REPUBLIC OF CAMEROON

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REPUBLIQUE DU CAMEROUN

Paix-Travail-Patrie

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# INTERNSHIP REPORT

# CONCEPTION AND IMPLEMENTATION OF A WEB APPLICATION FOR DONATION AND VOLUNTEERING FOR ORPHANAGES IN CAMEROON

Internship carried out from the 1$^{st}$ of July to the 30$^{th}$ of September 2024 at the company **WELLDONE PLANET** as part of my pursuit to obtain a

**Higher Technician Diploma (HTD) in Computer Science**

Option: Software Engineering

**Submitted by:**

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**Mr TIOMELA JOU Daniel**
Technical Director

2023-2024
Academic Year

# DEDICATION

THIS WORK IS DEDICATED

**TO MY PARENTS MR AND MRS NDO FOR THEIR LOVE AND SUPPORT**

# ACKNOWLEDGMENTS

This work wouldn't have been a success without the important contributions of a good number of people who took it upon themselves to see this work accomplished. Our sincere gratitude goes to the following:

- **MR ARMAND CLAUDE ABANDA,** the resident representative of ACIS Cameroon, for his guidance and concealing throughout these years.
- The chief executive officer of WELLDONE Planet, **MR BOM GABRIEL,** and his staff for accepting us as interns in his institution.
- To our professional supervisor MR TIOMELA JOU DANIEL for his advice and follow-up.
- To our academic supervisor **Mrs TCHOUTOUO ISABELLE** for her time, advice, and follow-ups throughout this period.
- To all the professorial and administrative core members at AICS CAMEROON for their professionalism, love, support, and counselling.
- To MR MESSIO ULRICH and MR AGBOR ANDERSON, our Fathers for their love, support, and counselling throughout these years.
- To our parents (MR AND MISSES NDO), and brothers (CHRISPY NDO, DADDY NDO, WILLY NDO, IDRICE NDO) for their limitless love and support, their faith in us and our abilities, and their encouragement.
- To our late grandmother BENGONO OKONO MYRIAM for her love, patience, support, counselling, and faith in our abilities.
- To the ABONDO OKONO family for their patience, attention, and counselling.
- To Miss BENGONO PRYCIL for her love and support throughout these years.
- To our lovely friends, for their limitless love and support, their faith in our abilities, and their encouragement.
- To our classmates, for their collaborative work throughout this academic year.
- To all the interns of Welldone Planet, for their support and counselling.
- To Mrs IRENE, Mr Pelayah Epoupa, Miss Akamessa Zibi, Miss NKO'O MARIAMA, MR MBILA AROUN FADHIL for their support all through this year.

# ABSTRACT

The Sustainable Development Goals (SDGs) were established by the United Nations (UN) in 2016. Amongst these goals, we can denote No poverty (SDG 1), Zero hunger (SDG 2), and Good health and well-being (SDG 3). Cameroon being a member of the United Nations (UN) is equally expected to respect these goals. 49% of Cameroon's population are children and over 1,300,000 are orphans according to UNICEF. These goals are especially critical for children, particularly the orphans whose survival hinges on community aid. However, overcrowded orphanages and an alarming 37.5% poverty rate make it difficult to ensure even basic needs are met. We need a way to bridge this gap in resources. That is why we created a web application for donation and volunteering for orphanages in Cameroon, known as **JUSTGIVE**. JUSTGIVE is a web application that will enable donors to make donations online to orphanages, create and participate as volunteers in projects (either created by the orphanage(the orphanage manager of the orphanage) or by another donor), search orphanages according to their location, and make announcements. The orphanage (the orphanage manager of the orphanage) can create a list of their needs and can equally make announcements. This will help the effective distribution of the help given to orphanages, enable volunteering, and facilitate the donation process. To accomplish this project, we used UML (Unified Modelling Language) as our modelling language coupled with 2TUP (2 Track Unified Process) to form a method.

Keywords:

- Donation
- JUSTGIVE
- Orphanages
- Sustainable Development Goals (SDG)
- Volunteering

# RESUME

Les objectifs de développement durable (ODD) ont été établis par les Nations Unies (ONU) en 2016. Parmi ces objectifs figurent : Pas de pauvreté (ODD 1), Faim Zéro (ODD 2) et Bonne santé et bien-être (ODD 3). Le Cameroun, étant membre de l'ONU, est également tenu de respecter ces objectifs. Selon l'UNICEF, 49% de la population camerounaise est constituée d'enfants et plus de 1 300 000 sont orphelins. Ces objectifs sont particulièrement critiques pour les enfants, en particulier les orphelins dont la survie même dépend de l'aide de la communauté. Cependant, la surpopulation des orphelinats et un taux de pauvreté alarmant de 37,5% rendent difficile la satisfaction des besoins fondamentaux. Nous devons trouver un moyen de combler ce manque de ressources. C'est pourquoi nous avons décidé de créer une application web de don et de volontariat pour les orphelinats du Cameroun, appelée JUSTGIVE. JUSTGIVE est une application web qui permettra aux donateurs de faire des dons en ligne aux orphelinats, de créer des projets et de participer en tant que bénévoles à des projets (créés par l'orphelinat ou par un autre donateur), de rechercher des orphelinats en fonction de leur catégorie et de leur emplacement, et de consulter leur historique de dons et de volontariat. L'orphelinat peut également créer une liste de ses besoins, visualiser les statistiques des dons et des activités de volontariat. Cela permettra une distribution efficace de l'aide apportée aux orphelinats, favorisera le volontariat et facilitera le processus de don. Afin de réaliser ce projet, nous avons utilisé UML (Unified Modeling Language) comme langage de modélisation couplé à 2TUP (2 Track Unified Process) pour former une méthode de développement.

Mots-clés :

- Don
- JUSTGIVE
- Objectifs de développement durable (ODD)
- Orphelinats
- Volontariat

# SUMMARY

# LIST OF FIGURES

# LIST OF TABLES

# GLOSSARY

**2TUP**: Two Track Unified Process

**API**: Application programming Interface

**AICS**: African institute of Computer Science

**CD-ROM**: Compact Disk- Read Only Memory

**CRUD**: Create Read Update Delete

**NISC**: National Institute of The Statistics of Cameroon

**OCHA**:UN Office for the coordination of humanitarian affaires

**SDLC**: System Development Life Cycle

**UN**: United Nations

**UNICEF:** United Nations Children's Fund

# GENERAL INTRODUCTION

Cameroon, a nation rich in cultural diversity and natural resources, also bears the weight of societal challenges. Among these is the plight of orphans, a vulnerable population that relies heavily on external support for survival and development. While the spirit of philanthropy is evident in the Cameroonian society, the process of channeling aid to orphanages remains fraught with inefficiencies and inequities. Traditional methods of donation and volunteering often result in a misalignment between the needs of orphanages and the resources provided. Donors, acting on goodwill, frequently make contributions without a clear understanding of the specific requirements of these institutions. Moreover, the geographical dispersion of orphanages across the country poses challenges for potential donors and volunteers seeking to connect with these institutions. The lack of a centralized platform to facilitate transparent and efficient aid distribution exacerbates these issues. Orphanages, struggling with limited resources, find it arduous to reach out to potential donors and volunteers. As a result, some orphanages remain underserved, while others grapple with an overabundance of certain resources. To address these pressing challenges, we propose the development of a web application designed to permit the donation and volunteering process for orphanages in Cameroon. By creating a virtual hub that connects donors, volunteers, and orphanages, we aim to foster a more equitable and effective system of aid distribution. Our theme, "CONCEPTION AND IMPLEMENTATION OF A WEB APPLICATION FOR DONATION AND VOLUNTEERING FOR ORPHANAGES IN CAMEROON," encapsulates the core objective of this research. To ease understanding, we divided this document into eight (8)parts. These are:

1. Insertion Phase;
2. Existing System
3. The Specification Book;
4. The Analysis Phase;
5. The Conception Phase;
6. The Deployment phase or Realization phase
7. The Functionality Testing Phase;
8. The User Guide.

# PART 1: INSERTION PHASE

Preamble

The insertion phase document presents the internship environment, the structure of the organization, the conditions in which the interns were received, as well as the project to be realized during the internship.

Content :

# INTRODUCTION

Professional integration is a process, which allows an individual or a group of individuals to enter the labor market under conditions favorable to obtaining a job. As part of the academic internship, we were received as an intern within the premises of Welldone Planet for a period of three (03) months and the integration phase lasted 02 weeks. This part of the internship report which presents the structure or internship was carried out, its operation, the reception of the student and the research theme during this period.

# I. WELCOME AND INTERGATION

## a. WELCOME

On Monday, July 1st, 2024, at 8:00 am sharp, we were warmly welcomed within the Welldone Planet structure by its Technical Director, Mr. TIOMELA JOU Daniel. He generously took the time to introduce the organization.

## b. INTERGRATION

Mr. TIOMELA JOU Daniel organized a meeting during which he reminded us of the internal regulations of the establishment, its operation, its vision, its objectives, its services, its requirements and the development of the weekly schedule. Emphasis was placed on the conduct to be followed and compliance with the regulations. Several pieces of advice and recommendations were given to us during various online sessions with the Technical Director. Then we argued about our theme.

# II. GENERAL PRESENTATION OF WELLDONE PLANET

## a) HISTORY AND MISSION

Welldone Planet is a young and ambitious company founded in June 2023 by talented engineers from the African Institute of Computer Sciences Cameroon (AICS). Officially launched in June 2024, by the resident representative of AICS-CAMEROON, they aim to be a leader in digital services across Africa, focusing on Cameroon.

Welldone Planet is passionate about inspiring excellence and achieving innovation. Their motto, "Inspire Excellence, Achieve Innovation, Dream Big," perfectly captures their mission: empowering dreams through digital technology. They believe that anything is possible with the right digital tools, and they encourage everyone to dream big!

## b) VISION

Welldone Planet's vision is to become a major catalyst in the field of information and communication technologies in Africa and Cameroon in particular, with a focus on innovation and excellence. Firmly believing that digital can turn dreams into reality, Welldone Planet is committed to creating forward-thinking technology solutions that meet the growing needs of society. By inspiring excellence and achieving innovation, the company aspires to propel Cameroon, and by extension Africa, towards a future where new technologies are at the heart of economic and social development.

### c) GOALS

Welldone Planet sets ambitious goals to have a significant impact on the economic and social development in Africa. Based on a holistic approach and inclusive, the company aims to:

- **Reduce poverty:** By providing innovative technological solutions, Welldone Planet intends to create economic opportunities and improve the living conditions of local populations, thus contributing to the reduction of poverty.
- **Facilitate the development of human capacities**: Through programs of quality training in computer science and information technology, the company aspires to strengthen the skills of individuals and professionals, allowing them to remain competitive in the job market.
- **Holistic approach, including young people**: Welldone Planet adopts a global which includes young people, whether they are educated or have dropped out of school, to fully integrate them into technological development and economically, providing them with opportunities for learning and growth.
- **Support businesses in their growth**: By using techniques based on optimization, bottleneck management, and cost-effectiveness sustainability, Welldone Planet is committed to supporting businesses in their expansion, by improving their operational efficiency and economic performance.

## III. SERVICES PROPOSED AT WELLDONE PLANET

Welldone Planet offers a full range of technological services to meet the diverse needs of its customers. The company's main mission is to solve IT problems that businesses encounter daily in the areas following:

- IOT/IA BIG DATA
- Software Development
- IT maintenance
- Web and Mobile Development
- Infographics
- Cybersecurity
- Digital Marketing
- IT training

# IV. LOCALIZATION OF WELLDONE PLANET

Welldone planet is located at Carrefour Colombia at the DEMA's story building. The DEMA's story building is just 200meters away from the main road. Welldone planet is found at the fourth level more precisely at the door 403.This can be represented on the diagram bellow:

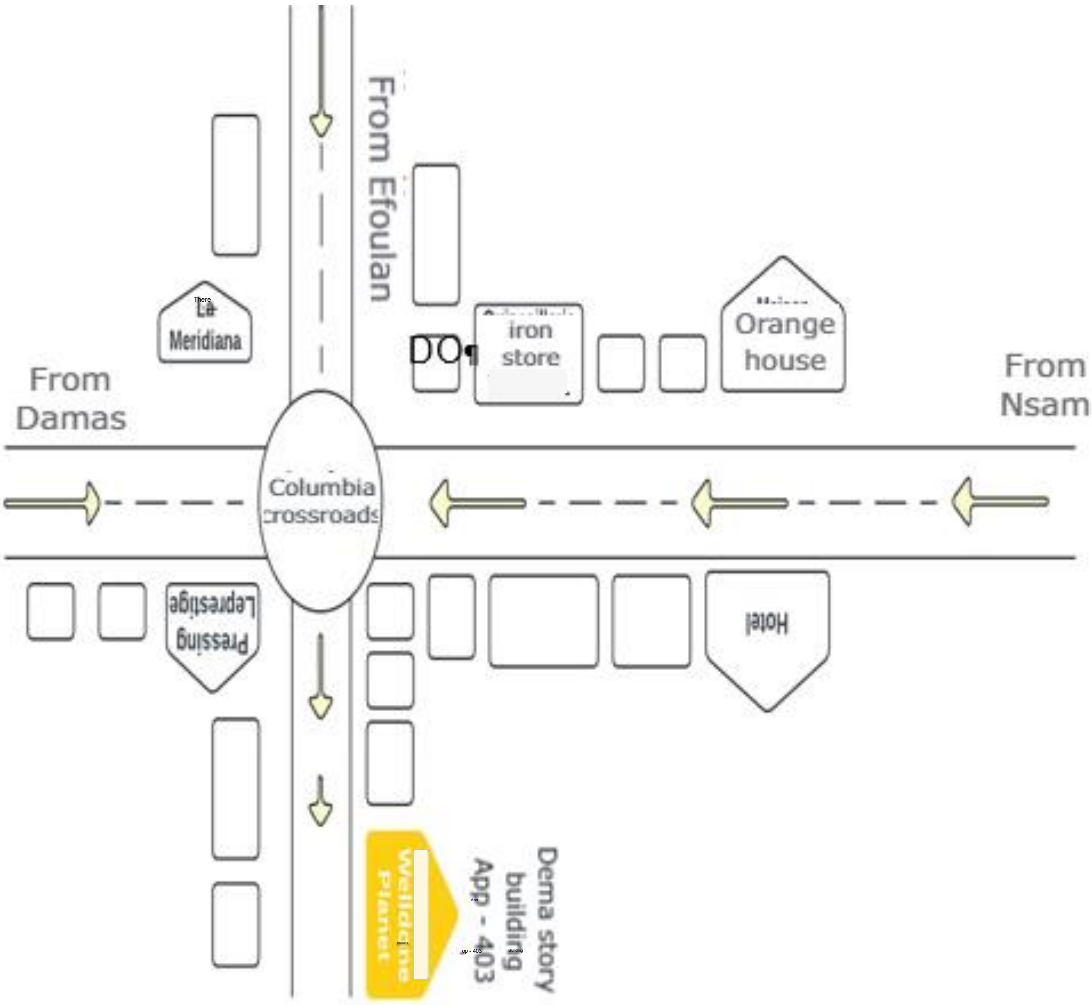



*Figure 1 localization plan of Welldone planet*

# V. THE ORGANIZATION OF WELLDONE PLANET

The organization chart is above all a communication tool intended to facilitate understanding of existing relationships and links within society. To this end, it allows a global view of the company in terms of services, divisions and more. **Welldone Planet** has six departments within it and we have been assigned to technical direction. This is the division colored in orange in the organization chart below:

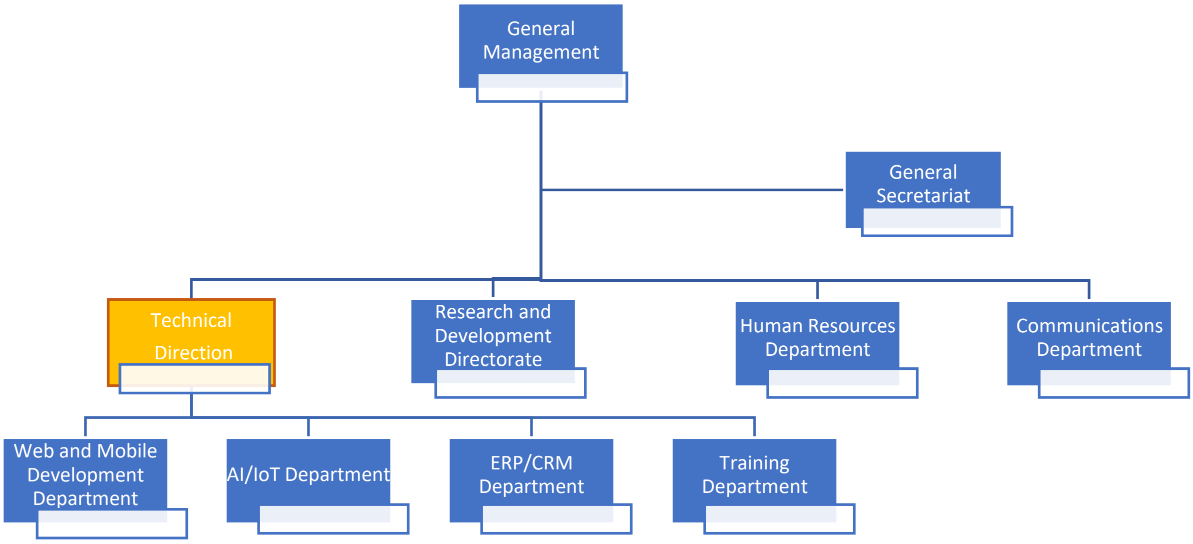



*Figure 2 Organization chart of Welldone planet*

# 1 ATTRIBUTION

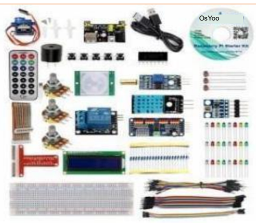
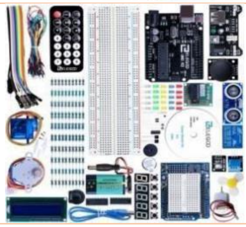
The Welldone Planet Company consists of executive-led divisions with competent individuals playing key roles. Namely:

- The Human Resources Division (HRD): headed by a director of human resources which is responsible for personnel management, training and skill development
- The Sales and Marketing Division (DCM): headed by a director sales and marketing who is responsible for sales, marketing and development commercial.
- The Communication Division (DCOM): headed by a director of communication of the company which is responsible for sales, marketing and Business Development**.**
- The Research and Development Division (DR&D): headed by a director of research and development who is responsible for innovation and development of new products.
- **The Financial Division (DRMF):** headed by a director of financial resources who is responsible for financial management, budgets and accounting.
- **The Technical Division (DT):** headed by a technical director who is responsible for the technical management, production and maintenance**..**
- **Secretariat** for invoicing, photocopying, customer registration

# VI. HARDWARE AND SOFTWARE RESOURCES OF WELLDONE PLANET

## i. HARDWARE RESOURCES

*Tableau 1 Hardware resources of Welldone planet*

| DESIGNATION | PICTURE |
|---|---|
| **3 HP desktops** |  |
| **3 Wifi Modem** |  |
| **3 plasma screens** |  |
| **1 fridge** |  |
| **1 canon printer** |  |
| **1 HP laptop** |  |
| **Ultiamte Kit for Rasberry** |  |
| **Super starter kit** |  |

| | |
|---|---|
| **Smart robots** |  |

### ii. SOFTWARE RESOURCES

*Tableau 2 Software resources of welldone planet*

| DESIGNATION | PICTURE |
|---|---|
| **Windows 10 operating system** |  |
| **Windows 11 operating system** |  |
| **Adobe Creative Cloud Suite 2024** |  |

## VII. BRIEF PRESENTATION OF THE PROJECT

Following our arrival as interns at Welldone planet, we were encouraged to dream big. Following the goals of Welldone planet, we were assigned an innovative project that will greatly improve the standard of living of orphans living in orphanages. This is a web application for donation and volunteering to orphanages. With this application we can donate monetary and material aid to orphanages and equally propose our selves as volunteers .

# CONCLUSION

During our internship at Welldone Planet, we underwent an insertion phase that provided us with a comprehensive understanding of the organization's missions, objectives, and activities while we pursued our academic internship. This phase allowed us to delve deeper into Welldone Planet, including its history, various departments, as well as the software and hardware resources utilized. As a culmination of this phase, we were assigned the theme "CONCEPTION AND IMPLEMENTATION OF A WEB APPLICATION FOR DONATION AND VOLUNTEERING FOR ORPHANAGES IN CAMEROON".

# PART 2:TECHNICAL PHASE

Preamble

The insertion phase helped us to establish a comprehensive understanding of the host organization. This phase emphasizes on the key characteristics, specificities, and objectives of the designated study subject.

Content :

# DOCUMENT 2: EXISTING

Preamble

The existing system refers to the established processes currently used. This documentation outlines how these processes function, including their limitations and any resulting problems. We aim to address these issues by proposing a new solution.

Content :

# INTRODUCTION

Engineers are society's problem-solvers, working to make life easier for everyone. Before proposing solutions, we need to understand the root causes of the problem and how it's currently addressed. Analyzing existing systems in our field is key to identifying problems and developing effective solutions. This ensures our proposals improve upon or replace the current system for the better.

# I. PRESENTATION OF THE THEME

Orphanages are the main institutions in charge of taking care of orphans. They greatly depend on community aid for their survival. Given the context of our country with a poverty level of 37.5%, we do not have the sufficient means to cover their needs completely. Implementing a web application for donation and volunteering for orphanages will facilitate the donation process and enable an efficient volunteering system for orphanages.

THEME: CONCEPTION AND IMPLIMENTATION OF A WEB APPLICATION FOR DONATION AND VOLUNTEERING FOR ORPHANAGES IN CAMEROON.

DEFINITION OF TERMS:

1. Web Application: This is a software program accessed and used through a web browser.
2. Donation: This is a voluntary gift of money or goods to a person or organization.
3. Volunteering: Volunteering is freely offering one's time and services to help others or a cause.
4. Orphanage: This is a residential institution caring for children who cannot be cared for by their families.

# II. DESCRIPTION OF THE EXISTING SYSTEM

We noticed that inefficient aid is given to these institutions(we mean by inefficient aid that, the donor's donators do not always take the time to ask about the orphanage's needs before donating hence, the orphanage receives help but it is not as efficient as it does not correspond to their need). In addition, there is no efficient volunteering system and finally, it is not easy to find orphanages. We decided to conduct a census from the 24 of July 2024 to 7 September 2024 to verify our hypotheses. Our hypotheses are as follows:

1. The donors do not take the time to go and as the orphanage's needs before doing a donation, which leads to an inefficient aid given to these orphanages.
2. Orphanages occasionally receive volunteers.
3. It is difficult to find an orphanage.

The result of our census is as follows:

**To the question:** How would you describe the accessibility to your orphanage?

The different options were:

1. Easily accessible from the main road.
2. Accessible with a four-wheel drive vehicle.
3. Not easily accessible and requires a long walk to reach.


*Figure 3 Donor questionnaire 1*

**To the question:** How many children are living at the orphanage?

The different options were:

1. Less than 20
2. In between 20 and 50
3. In between 51 and 100
4. More than 100

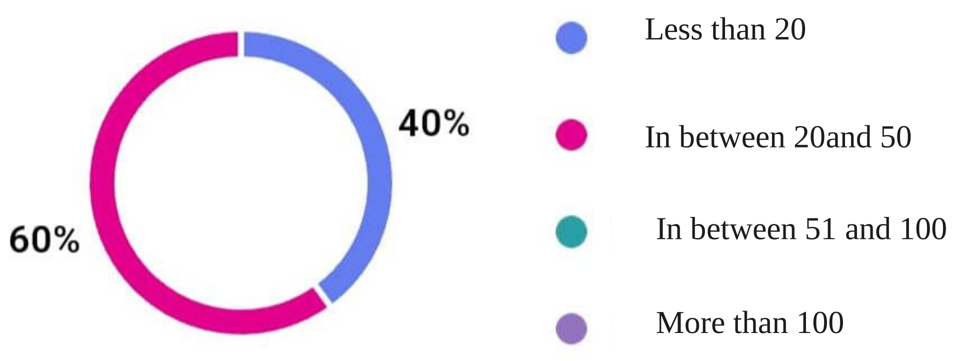



*Figure 4 Donor questionnaire 2*

**To the question:** How regularly do you receive donations ?

The different options were:

1 Less than one time a month

2 One time per month

3 Many times per month

4 One time per semester

5 Less than one time per month

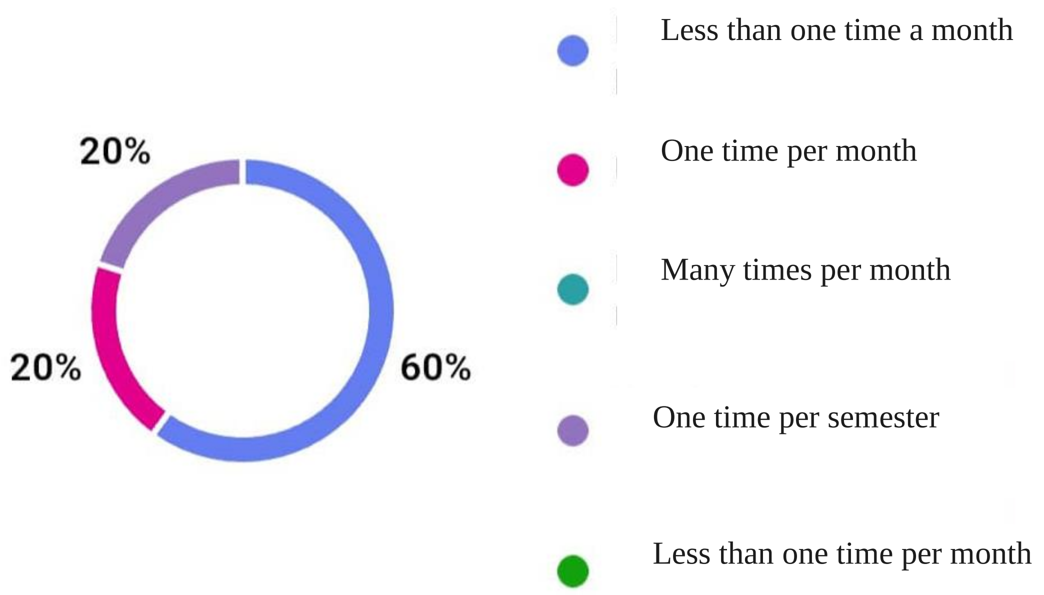



*Figure 5 Donor questionnaire 3*

**To the question:** Do you think the location of your orphanage influences the amount of donations you receive?

The different options were:

1 Yes, definitely

2 Yes ,to an extent

3 No

4 Other

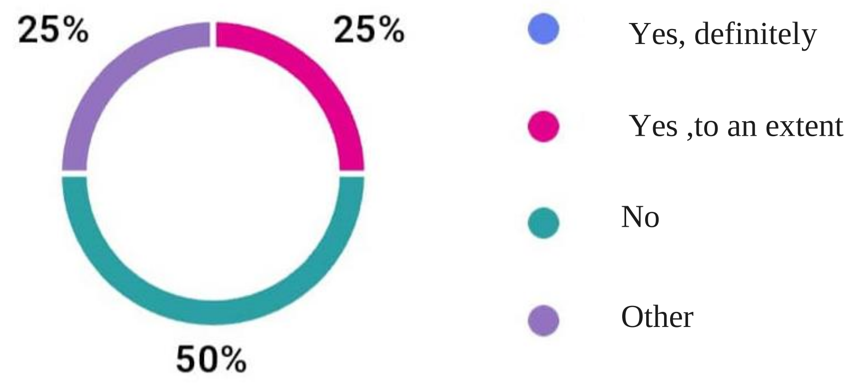



*Figure 6 Donor questionnaire4*

**To the question:** To What extent do donors investigate the specific needs of the orphanage before contributing?

The different options were:

1 Yes , regularly
2 Sometimes
3 Rarely
4 Never
5 Other

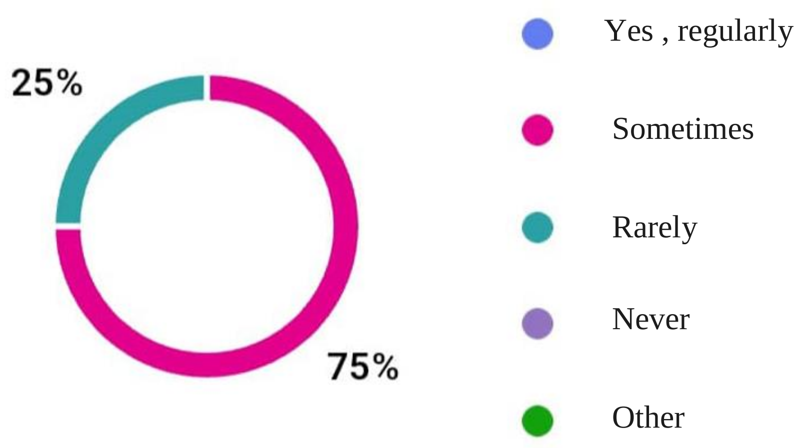



*Figure 7 Donor questionnaire 5*

**To the question:** Is your orphanage regularly funded by the government?

The different options were:

1 Yes
2 No
3 Occasionally
4 Other

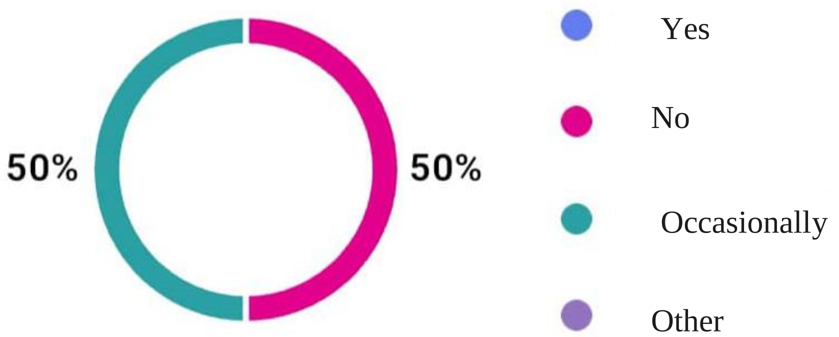



*Figure 8 Donor questionnaire 6*

**To the question:** How often do volunteers lend a hand at your orphanage?

The different options were:

1 Regularly(many times per week)
2 Occasionally (one time per month)
3 Rarely(few times a year)
4 Never

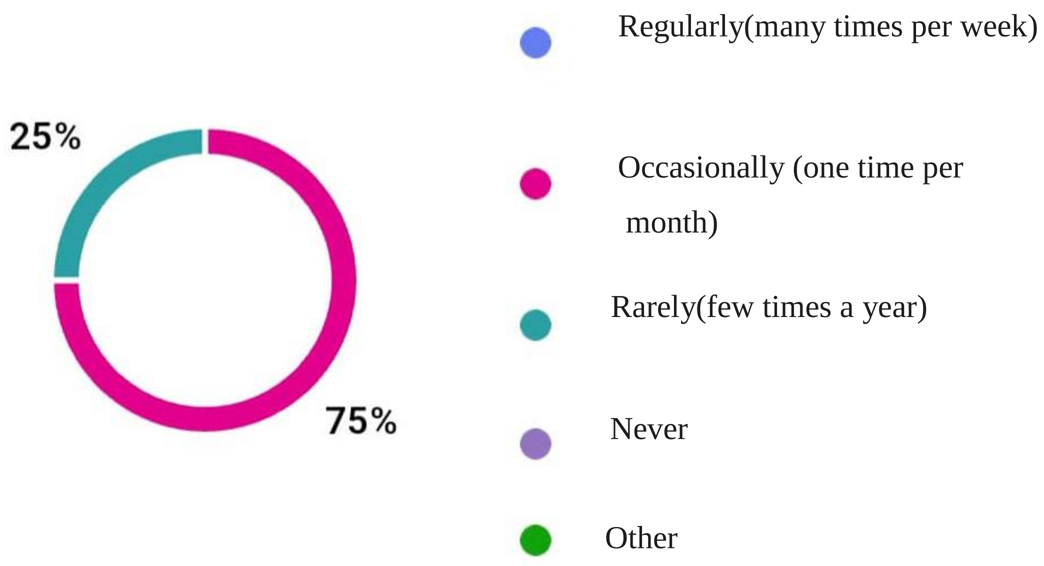



*Figure 9 Donor questionnaire 7*

From the above results, we came out with the following observations.

- Also, we can observe that the aid given to orphanages is inefficient. From our results, we can observe that donors only ask sometimes(75%) or rarely (20%) about the needs of the orphanage before donating.
- We can observe that, there is no real volunteering system in orphanages. From our results, we can observe that generally, orphanages receive a helping hand from the volunteers only once a month, which is not enough.
- We can observe that it is difficult to find an orphanage. The fact that orphanages are difficult to find contributes to the low quantity of donations.

In addition to these results, it is important to understand the procedure involved in making a donation to an orphanage . For a better understanding ,we will take the case of a non profit organization and an individual.

A non profit organization , is a legal entity organized and operated for a collective, public or social benefit, as opposed to an entity that operates as a business aiming to generate a profit for its owners. For example: Kidefind.

Kidefind in a non profit organization that fights against child abduction and helps underprivileged children.

When Kidefind does a donation, they follow these steps:

1. one most first of all find an orphanage respecting its criteria ,This step can take up to 2months .
2. contact the administration of the orphanage to now the needs of the orphanage. This step can take up to 3weeks.
3. Conduct the fundraising event to get the donations, this step can take 2months
4. Go to the orphanage to do donations to the orphanage
5. Go towards each donator to give feedback on the donations, how they have been used, and the ceremony. This can take up to 1 month in some cases.

When carrying out a donation through Kidefind, it may take up to about six months to complete the process of donating to an orphanage. As for individuals, the steps are generally the same, but many may skip steps 2 and 5. In some cases, an individual may simply visit an orphanage, make a donation, and leave. Individuals likely spend less time at steps one, two, and three

## III. CRITICISM OF THE EXISTING SYSTEM

The current donation and volunteering procedures have many shortcomings which lead to an increase in the number of orphanages living in critical conditions among these include :

- Little or no communication between the donors and the orphanage managers. A good amount of donors do not take the time to ask the orphanage managers about the needs of the orphanage and go forward to make donations that will not resolve the orphanage's present problems.
- It is very difficult to find orphanages, this is because most of them are found far into the quarters to ensure the protection of the children . As a result, these orphanages do not receive the help they need to respond to the children's necessities.
- There is no real volunteering system. Most of the volunteers are the friends of the orphanage managers, family members of the members of the orphanage, associations, and NGOs. It was somehow a miracle to meet another group of people volunteering at the orphanage. This results in a low human resource to take care of the children.
- It is time-consuming and tedious.

# IV. LIMITATIONS OF THE EXISTING SYSTEM

At the end of our collection of information concerning the donations and volunteering in orphanages, we find that this system has weaknesses that should not be overlooked. To these we proposed the following solutions:

*Tableau 3 Limitations of the existing sytem*

| LIMITATION | CONSEQUENCE | SOLUTION |
|---|---|---|
| **Inefficient aid** | Orphanages find themselves with a lot of material that is not of primordial aid to them. | The system gives the possibility to orphanage managers a list of their needs. This list must be obligatorily consulted by the donors before donating. |
| **No volunteering system** | The orphanages do not receive the sufficient human aid they need to take care of the children | The system gives the possibility to propose as a volunteer for a specific project |
| **It is difficult to find an orphanage** | This makes it even more difficult to help orphanages hence they do not receive the aid they need | The system enables a search of orphanages according to their location and category |
| **It is time-consuming** | It discourages people to donate to orphanages. Hence, they receive less donations. | The system provides the possibility of making online donations. |

# V. PROBLEMATIC

How can we improve the support for donation and volunteering to benefit orphanages?

# VI. PROPOSED SOLUTION

After our study and criticism of the existing system, we propose to design a web application for donation and volunteering in orphanages .This application will allow the orphanage manager , donors , orphanage manager of Cameroon to:

Orphanage manager:

- Do wish list
- Create orphanage
- Delete orphanage
- Update orphanage info
- Upload orphanage document

- Create a project
- Update a project
- Delete a project
- View project
- Do announcement

Donors :

- Make a donation
- Participate to a project(volunteer, donate)
- Create a project
- Update a project
- Delete a project
- Do announcement
- Search orphanage and project

System Admin:

- Validate orphanage
- Manage user account

# CONCLUSION

The study of the existing system is a very complex and challenging task. How ever it is an essential step in the SDLC (System Development Life Cycle). Taking the time to study an existing system permits system analyst lay a proper foundation for the new system. In this document we got to know the weaknesses, challenges, and the problematics of the existing system. Knowing these now permits us move further to implementing our new system

# DOCUMENT 3:THE SPECIFICATION BOOK

Preamble

This is a document that frames the student's academic internship, specifies the behavior of the student during his internship. This stage involves organization, planning, pedagogical standards and monitoring of work. It is considered as a communication and description tool which permits us to avoid inadequate results.

Content :

# INTRODUCTION

The specification book is a crucial part in software engineering process. For this reason, care must be taken when elaborating this document in order for the solutions to clearly satisfy the client's needs. This document permits us specify the expectations of the client(s), there by establishing a certain agreement between the client(s) and the person(s) to realize or develop the project. This specification book is to avoid inadequate production result.

# I. CONTEXT AND JUSTIFICATION

## a. CONTEXT

According to UNICEF, children make up forty-nine percent (49%) of Cameroon's population, with over 1,300,000 of these children being orphans. Due to poverty, the majority of these children are placed in orphanages. However, there are not enough orphanages in Cameroon to accommodate them, leading to overcrowding, as mentioned in an article published by the ffca-asso. This overcrowding makes it very difficult for orphanage managers to properly care for the children. To address this issue, orphanages receive assistance from the government, non-governmental organizations, associations, enterprises, and individuals. While these donations are used to meet the primary needs of the children, they are still insufficient. Research conducted by doctors at the General Hospital of Douala (Mrs. Calixte Ida Penda, Mrs. Cecile Oalla Ebongue, Mr. Nda Mefoo, and 9 others) has shown that children in orphanages suffer from malnutrition. Additionally, an article published by OCHA (UN Office for the Coordination of Humanitarian Affairs) in May 2023 revealed that 37.5% of the population lived in poverty, with rates exceeding 60% in some regions. For example, the poverty rate in the Far North of Cameroon is 69.2%, according to the National Institute of Statistics of Cameroon. These statistics highlight the urgent need for assistance in orphanages.

## b. JUSTIFICATION

The establishment of a web application for donation and volunteering for orphanages in Cameroon is a game changer. It would permit the following: Enable an efficient volunteering system in orphanages, Improve the efficiency of the aid given to orphanages, and facilitate the finding of orphanages.

# II. PROJECT OBJECTIVES

## a. GENERAL OBJECTIVE

Facilitate the donation process to orphanages and establish an online volunteering system.

## b. SPECIFIC OBJECTIVES

- ✓ Enable an online donation system to orphanages.
- ✓ Permit the creation of projects for orphanages.
- ✓ Permit donors to participate to a project either through donations or volunteering
- ✓ Enable orphanage managers to express the needs of the orphanage
- ✓ Easily find orphanages though the platform

The project has the following characteristics:

✓ Name of Project: JUSTGIVE

✓ Project Target: Facilitate the donation procedure to orphanages and enable an online volunteering system.

✓ Technical Specification: Web Application

# III. EXPRESSION OF NEEDS

## a) FUNCTIONAL NEEDS

These are the requirements that a system must meet to fulfill its purpose, typically expressed in terms of systems' output, input, and behaviors. They are as follows:

The System Administrators (Super Admin) should be able to:

- ❖ Validate orphanage
- ❖ Manage user account

The Orphanage manager should be able to:

- ❖ Do a wish list
- ❖ Create orphanage
- ❖ Delete orphanage
- ❖ Update orphanage info
- ❖ Upload orphanage document
- ❖ Create a project
- ❖ Update a project
- ❖ Delete a project
- ❖ View project
- ❖ Do announcement

The Donor should be able to:

- ❖ Make a donation
- ❖ Participate to a project(volunteer, donate)
- ❖ Create a project
- ❖ Update a project
- ❖ Delete a project
- ❖ Do announcement
- ❖ Search orphanage and project

## b) NON-FUNCTIONAL NEEDS

It specifies the quality attributes of a software system. They judge the software system application based on the system's requirements, performance, usability, scalability, portability and other non-functional standards that are critical to its success. Failing to meet the non-functional needs can result in the system not fulfilling the user's needs.

- Performance and Scalability:

**Performance** defines how fast a software system can respond to a user's actions under a certain workload. In most cases, this metric explains how long a user must wait before the targeted operation happens or responds given that other users are using that application at that same moment. Performance may also describe the background processes invisible to users. Our goal will be to provide users with the best performance as it affects the overall user experience.

**Scalability** accesses the highest workload under which the system will still meet the performance requirements. In this project, we will make it as flexible as possible for users to feel comfortable and secure when using the application.

- The application should have a friendly user interface (UI) and be easy to use and understand;
- The code should be clear and facilitate future development and improvements;
- The application should implement strong security measures;
- It should be accessible on all platforms as long as it has a browser and an internet connection.

# IV. PROJECT PLANNING

We began our internship on the 1st of July, till the 30th of September a total of 3 months. To this, it was an obligation for us to manage our time properly so as to finish on time. This section presents how the work phases were scheduled throughout my internship period.

## 1 PROJECT PLAN

*Tableau 4 Project plan*

| Task | Duration(days) | Period |
|---|---|---|
| **Produce an insertion document** | 5days | 02$^{nd}$ July – 8$^{th}$ July |
| **The Existing System** | 5 days | 9$^{th}$ July - 15 July |
| **Production of Specification Book (with corrections)** | 12 days | 16$^{th}$ July – 31$^{st}$ July |

| | | |
|---|---|---|
| **Analysis of the solution and drafting of the analysis document(with corrections)** | 17days | 1th August – 23th August |
| **Conception** | 17days | 23th August – 16th September |
| **Realization** | 7days | 17th September – 25th September |
| **Functionality Test** | 2days | 26th September – 27rd September |
| **Writing user Guide** | 3days | 28th September – 30th September |

## 2 GANTT DIAGRAM

The Gantt diagram is a tool used in project management enabling the project manager to visualize on time the various tasks components if a project. It is all about a graphic representation that enables to graphically represent the project advancements. The Gantt Diagram for this project is as follows:

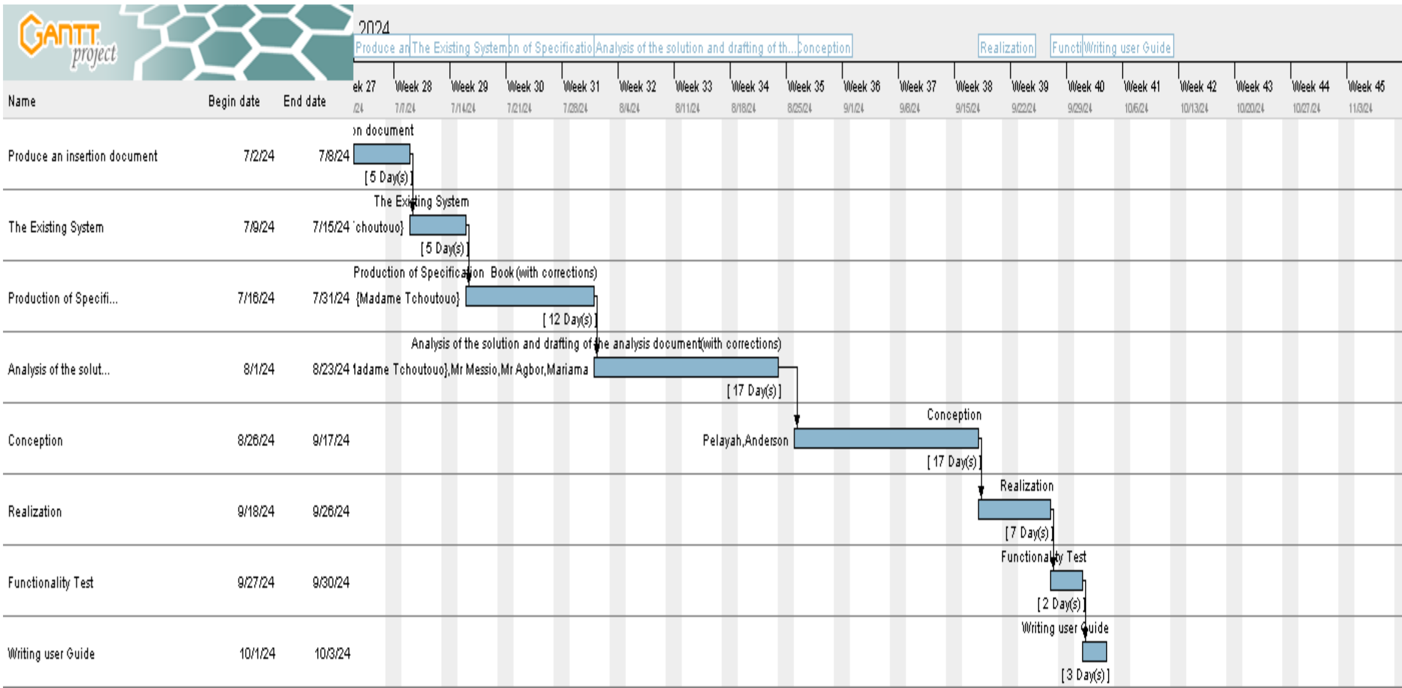



*Figure 10 Gantt chart of project*

# V. ESTIMATED COST OF PROJECT

## 1. SOFTWARE RESOURCE

For the realization of this project, we will need the following software applications:

Source: Mercurial 2024, https://www.microsoft.com/en-us/d/windows-11-home/dg7gmgf0krt0

*Tableau 5 Software resources cost*

| Software Resource | Description | Quantity | Total Price (FCFA) |
|---|---|---|---|
| **Microsoft Windows 11 pro** | Operating System | 01 | 120,174.45 |
| **Microsoft office 365** | It is a productivity suite that includes Excel, word, outlook etc. to create documents, spreadsheets, presentations and emails | 01 | 47,998 |
| **Visual Paradigm** | Modeling tool | 01 | Freemium |
| **Visual Studio** | Front-end and backend development environment | freemium | freemium |
| **POST MAN** | API testing software | freemium | freemium |
| **Total software resource cost:** | | | 168,172.45FCFA |

## 2. HARDWARE RESOURCE

This includes the material resources used to realize this project. The table below shows a representation of these different materials:

*Tableau 6 Hardware resource cost(source :Mercurial 2024)*

| Hardware Resource | Quantity | Total Price (FCFA) |
|---|---|---|
| **laptop 8G of RAM, SSD, corei5, processor 1.10GHz** | 01 | 200,000 |
| **Internet Connection(ORANGE)** | \ | 64,400 |
| **Total hardware resource cost** | | 264 ,400FCFA |

## 3. HUMAN RESOURCE COST

Source :Mercurial 2024

*Tableau 7 Human resource cost*

| Human resource | Quantity | Price (FCFA)/day | Duration | Total price(FCFA) |
|---|---|---|---|---|
| **Head of project** | 01 | 53,500 | 31days | 1,658,500 |
| **Analyst** | 01 | 100,000 | 21days | 2,100,000 |
| **Developer** | 01 | 80,000 | 30days | 2,400,000 |
| **Tester** | 01 | 50,000 | 10days | 500,000 |
| **Total human resource cost:** | | | | 6,658,500FCFA |

TOTAL ESTIMATED COST OF PROJECT:

*Tableau 8 Total cost of the project*

| | |
|---|---|
| **SOFTWARE RESOURCE** | 168,172.45 FCFA |
| **HARDWARE RESOURCE** | 264,400FCFA |
| **HUMAN RESOURCE** | 6,658,500FCFA |
| **Unexpected charges (10%) {(Total cost) * 10% }** | 709,107.245FCFA |
| **TOTAL COST** | 7,091,072.45FCFA |
| **Total** | 7,800,179.70FCFA |

The total cost of the project:7,800,179.70FCFA

# VI. PARTICIPANTS

The table below presence the different persons who took part in the accomplishment of this project. They include:

*Tableau 9 Project participants*

| Names | Functions | Role and Tasks |
|---|---|---|
| **Mr TIOMELA JOU DANIEL** | Technical Director | Professional Supervisor |
| **Mrs. TCHOUTOUO** | Lecturer at AICS-Cameroon | Academic Supervisor |
| **Miss.NYANGONO NDO EMMANUELLA DORETTE EMIE GRACE** | Second year Student at AICS-Cameroon | Project head, analyst, conception and coding of project, testing |

# VII. PROJECT CONSTRAINTS

To put in place this application, we must respect three main constraints which includes:

- ❖ Budget Constraint: the fixed budget for the realization of this project must be strictly respected;
- ❖ The Dateline Constraint: the realization of this system must respect the dateline attributed to it and should be completed in a duration of two months (2) including respecting the fixed objectives of the project;
- ❖ Quality Constraint: after fixing the budget and a dateline, we must produce a flexible, standalone, and reusable application;

# VIII. DELIVERABLES

In this section. We will present the different elements that will be delivered at the end of this project. They include:

- ❖ A CD-ROM composed of the web application;
- ❖ The installation guide and user manual of the application;
- ❖ PowerPoint Presentation of the application;
- ❖ A video to show a brief demonstration of how the application works

# CONCLUSION

To conclude the specification book has permitted us to present the general and specific objectives of the project, the project constraints, the functional and nonfunctional needs of the project, lay out the different resources of the project, and finally the project planning. The end of this document will enable us to easily dive into drafting the analysis document which will bring out the technical aspects of the project as a whole.

# DOCUMENT 4 : THE ANALYSIS PHASE

Preamble

The analysis file following the specifications represents a more detailed explanation of the problem and the solution with the help of diagrams, the chosen analysis method to be used through out the project and the reason for this choice. All these globs to having a better understanding of the information system to be modeled.

Content :

# INTRODUCTION

Analysis is the decomposition of a problem into its various elements for computer processing. Therefore, for the successful completion of a project, it is very important to carry out a good analysis in order to achieve the objectives specified in the specification book. The realization of a good project is mainly based on its analysis. Thus, a poorly analyzed project cannot give good results.

# I. METHODOLOGY

Analysis is the fundamental step in the conception of a software. It is the base to the realization of every information system. An information system is an organization of resources (human, software and hardware), data and feedbacks to attain a specific objective. Several methods and languages have been put in place to facilitate the analysis and conceptions of information systems which we principally have UML and MERISE.

MERISE (Méthode d'Etude et de Realization Informatique pour les Systèmes d'Entreprise) it is a method of analysis, conception and realization of information systems
mostly used in French Speaking Companies or Enterprises. It is based on the separation of data and processing them using several conceptual and physical models. Its principal objective is to conceive an information system. Merise proposes to consider the real system from two points of view i.e.

The Static View (data);

❖ The Dynamic View (treatments).

It is a systematic and conceptual method of analysis.

UML (Unified Modeling Language) is a modeling language that gives a standardized visualized view of an information system. It uses diagrams to represent each aspect of a system in an object-oriented way.

UML proposes an approach different from MERISE as its readability and reusability make it an ideal choice for programmers.

UML is currently at its 2.5 version and presents several advantages which include:

❖ It is a formal and standardized language.

❖ It provides a visual representation of the structure and behavior of the system which can help to communicate and understand the design of the system.

❖ It is flexible and customizable to suit different domains and technologies.

❖ It is readable and re-useable i.e. you can easily see the relationships and interactions between classes and objects in UML.

❖ It is a planning tool that can be used to help design and document a system before the implementation phase.

We can present the UML hierarchy as follows:

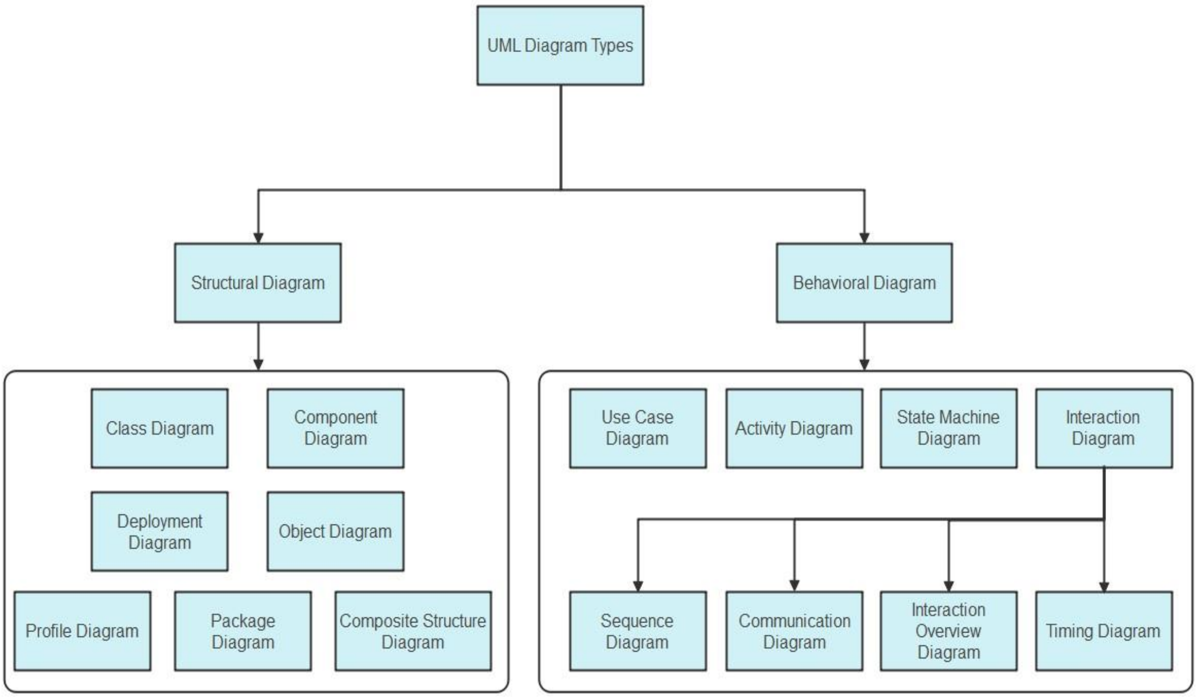



*Figure 11 UML architecture*

Source: https://images.edrawsoft.com/articles/what-is-uml/uml-diagram-types.png

## a) COMPARATIVE ANALYSIS BETWEEN MERISE AND UML

*Tableau 10 Comparative analysis between MERISE and UML*

| MERISE | UML |
|---|---|
| **Méthode d'Etude et de Realization Informatique pour les Systèmes d'Entreprise** | Unified Modeling Language |
| **Merise is a method of analysis** | UML is modeling language |
| **Merise is a systematic method of analysis and design for information systems i.e. it uses a system approach** | UML is however not a method but rather an object-oriented language, uses the 2TUP approach to model a system |

| | |
|---|---|
| **It is based on procedural approach** | Based on object approach |
| **Merise proposes to consider the real system from two points of view:**<br>**The Static view(data)**<br>**The Dynamic view(treatments)**<br>**Which means in Merise we have a separate study of data and treatments** | UML offers a different approach from that of MERISE in that it combines data and process |

## b) PRESENTATION OF THE 2TUP

2-TUP (Two Track Unified Process) is Y in shape and is a unified process that has as its objective to bring a solution to constraints of functional changes imposed on information systems. It proposes a development cycle that separates the functional aspects from the technical aspects. 2-TUP therefore distinguishes two main branches :

(The functional and technical branch) which their different results are being merged to realize our system.

Any unified process UP (Unified Process) process meets the following characteristics:

It is iterative and incremental. Some of the advantages of the 2TUP include:

- It allows the addition of modules.
- It allows flexibility in the analysis of a project
- It is easy to design and understand
- It allows to have a preliminary view

Below is the representation of 2-TUP:

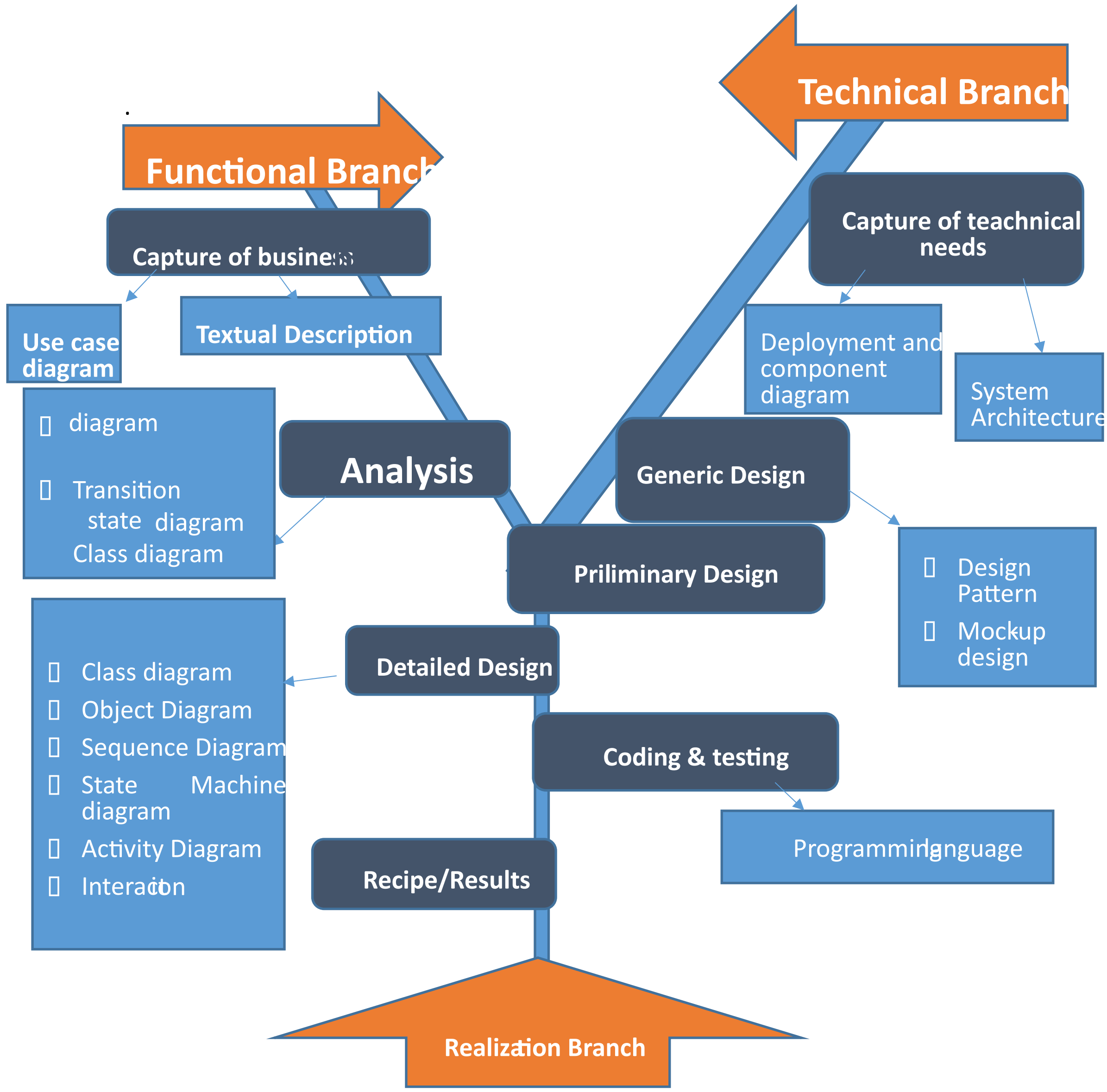



*Figure 12 2TUP diagram*

**Source: Miss. CHAYIE FOUSSOUM HERMANNE LUCRESE'S Report (session 2023)**

i.) **The Functional Branch:** this branch captures business needs, resulting in a model focused on the end-user's requirements.

ii.) **The Technical Branch:** this branch proposes a generic design from the technical needs, identifies the materials, tools and technologies to be used.

iii.) **The Implementation:** The implementation phase consists of bringing the two branches together, providing the application design and finally the delivery of a solution adapted to the needs.

a.) Functional Branch (left branch)

- **Capture of Business Needs:** this part minimizes the risk of producing an inadequate system with the needs of users and also verifies it consistency.
- **Analysis**: this is the study of the specifications to find out what the system will be made of in terms of the business rules. We have the following diagrams: Use Case, Activity Diagram, State Machine Diagram, Communication diagram

b.) Technical Branch

- **Capture of Technical Needs**: this stage consists of the identification of tool, materials and technologies to use in developing the system. This technical architecture will be presented in this stage.
- **Generic Designs**: the technical architecture will be presented in this stage. We have the following diagram: mockUp7 Design.

c.) Implementation Branch (middle branch):

- **Preliminary Branch:** this is the stage where the analysis model is integrated into the technical architecture. The goal here is to know what technical component will be used depending on the features of the analysis. We have the following diagrams: interaction overview, Component, Deployment, package diagram and Composite Diagram.
- **Detailed Design:** this is the detailed design of each feature of the system. We have the following diagrams: class, object, sequence and timing Diagrams.
- **Coding and Testing:** This is the programming phase of the designed features alongside testing of the coded features.
- **Recipe (Result):** this is the validation phase of the functions of the system developed.

## c) CHARACTERISTICS OF 2TUP

The two-track unified process was built to be used with the Unified Modeling Language to become a method and has the following characteristics:

- **It is user oriented**: 2TUP is built from user's expectation
- **It is component oriented**: it offers flexibility to the model by supporting the re-use of components.
- **It is an iterative process**: processes are being done in iterations and each iteration shows a precise level of abstraction.

- **It is an incremental process:** allowing a better functional and technical risk management and thus constituting the deadlines and the cost control.

## d) JUSTIFICATION ON THE CHOICE OF THE ANALYSIS

Our choice was based on the modeling language UML associated to it 2TUP due to certain criterion which are as follows:

- UML is based on object-oriented approach;
- UML associate's data and treats them;
- It is a modeling language which focus mainly on users' needs and expectations;
- 2TUP is a process built from object-oriented approach and is built from UML;
- 2TUP facilitates the modeling of complex systems which can further be readjusted.

# II. MODELLING

## a) USECASE DIAGRAM

### i.) DEFINITION

The use case diagram simply shows the relationship between the actors of the system and the system .It permits to easily visualize the different actions that can be done by the users. It also clearly defines the boundaries of the system.

### ii.) FORMALISM

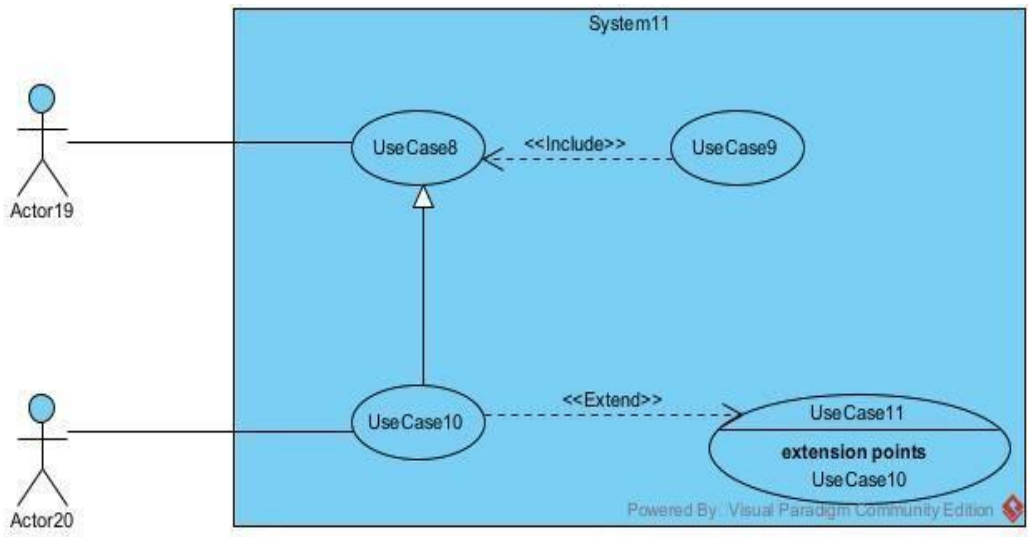
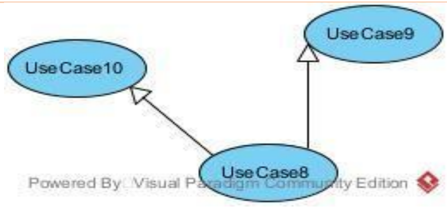
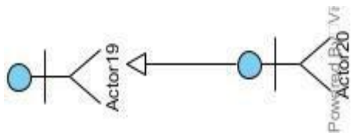
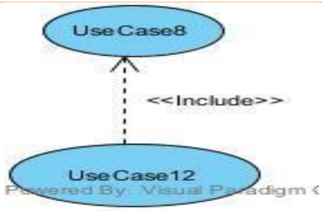
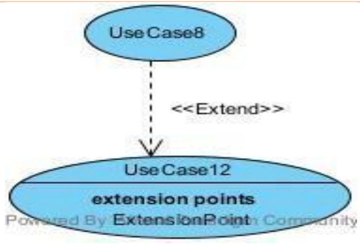



*Figure 13 formalism of use case diagram*

## COMPONENTS OF USECASE DIAGRAM

Components of usecase diagram

*Tableau 11 Components of usecase diagram*

| Elements | Description and main properties | Representation |
|---|---|---|
| **Actor** | Represents the entity who directly interacts with the system. The actor is one who performs the different tasks or actions of the system. |   |
| **Use Case** | it represents the functionality of the system. It is an action that an actor can perform |   |
| **Generalization** | Generalization is a relationsl between a general use case (the pare and a more specific use case (the chi that inherits and can add to or overri the behavior of the parent use case. |   |
| **Inheritance** | this is the only possible relationship that can exist between actors whereby an actor can inherit from another |   |
| **Inclusion** | It denotes that an included action must be performed before another action can be done. |   |
| **Extension** | It denotes that an action may be performed while another action is going on e.g. a use case B extends A implies B is an optional part of A. |   |

iii.) THE ACTORS OF OUR SYSTEM

- The primary actors

*Tableau 12 Primary actors of our system*

| Actor | Role |
|---|---|
| **visitor** | He can give financial aid to projects and orphanages |
| **Donor** | He can give both financial and material aid to projects a orphanages. Equally, he can participate in projects and create projects. |
| **Orphanage manager** | He can validate the proposed projects, create projects, and validate material donations. |
| **System administrator** | He can equally, validate accounts, manage user accoun |

- The secondary actors

*Tableau 13 Secondary actors of our system*

| Actor | Role |
|---|---|
| **Payment API** | Permits us to make financial donations and participation |

## iv.) TEXTUAL DESCRIPTION OF SOME USECASES

The Textual description is a simple text that describes how a use case can be performed in a more precise and understandable way following a chronological order. We will present in the following lines the textual description of some use cases.

Textual description of <<authenticate>>

*Tableau 14 Textual description of <<authenticate>>*

| **Title** | **Use Case « Authenticate »** |
|---|---|
| **Summary** | The user authenticates |
| **Actor** | Donor, orphanage manager, system admin |
| **Stakeholder** | Justgive |

| | |
|---|---|
| **Version** | 1.0 |
| **Precondition** | The application is launched<br>The actor has an account on the platform |
| **Trigger** | User clicks on login button. |
| **Nominal scenario** | 1. System displays login form.<br>2. User fill and submit form.<br>3. The System verifies form conformity<br>4. The system sends a login query to the database<br>5. The Database executes the login query and sends the response<br>6. The system displays the error message |
| **Alternative Scenario (invalid credentials)** | **7.** The system displays the error message.<br>**8.** Returns to step(1) |
| **Alternative Scenario(if the user does not exist)** | 9. The system displays the error message<br>10. Returns to step(1) |
| **Exceptional scenario** | The system stops unexpectedly |
| **Post-condition of success** | Display home page |
| **Postcondition of failure** | Display home page |

Textual description of << Make financial donation >>

*Tableau 15 Textual description of << Make financial donation >>*

| | |
|---|---|
| **Title** | **Make financial donation** |

| | |
|---|---|
| **Summary** | The user makes a financial donation |
| **Actor** | Donor |
| **Stake holder** | Justgive |
| **Version** | 1.0 |
| **Precondition** | The application is launched<br>The actor is authenticated<br>The actor is on the orphanage's page |
| **Trigger** | The user clicks on make a financial donation button |
| **Nominal scenario** | 1 The system displays the financial donation form<br>2 The user fills and submits the form<br>3 The system checks conformity<br>4 The system sends payment query to API<br>5 The API executes the query<br>6 The System sends a save query to the Database<br>7 The system displays a success message |
| **Alternative Scenario (invalid credentials)** | 8 System displays the error message.<br>9 Returns to step(2) |
| **Alternative Scenario(if payment)** | 10 System displays the error message<br>11 Returns to step(2) |
| **Exceptional Scenario** | The system stops unexpectedly |
| **Post-condition of success** | Display orphanage detail interface |

| | |
|---|---|
| **Postcondition of failure** | Display orphanage detail page |

Textual description of << Apply for material donation >>

*Tableau 16 Textual description of << Apply for material donation >>*

| **Title** | **Apply for a material donation** |
|---|---|
| **Summary** | The user makes a material donation. |
| **Actor** | Donor |
| **Stakeholder** | Justgive |
| **Version** | 1.0 |
| **Precondition** | The application is launched.<br>The actor is authenticated<br>The actor is on the orphanage's page |
| **Trigger** | The user clicks on making a material donation. |
| **Nominal scenario** | 1 The system displays the material donation form<br>2 The user fills and submits the form<br>3 The system checks conformity<br>4 The system sends a save query to the Database<br>5 The system displays success message |
| **Alternative Scenario (invalid credentials)** | 6 System displays error message.<br>7 Returns to step(2) |

| | |
|---|---|
| **Exceptional scenario** | The system stops unexpectedly |
| **Post-condition of success** | Display orphanage detail interface |
| **Post condition of failure** | Display orphanage detail page |

Textual description of <<participate in a project>>

Textual description of << Participate materially in a project >>

*Tableau 17 Textual description of << Participate materially in a project >>*

| **Title** | **Make material participation** |
|---|---|
| **Summary** | The user makes a material participation |
| **Actor** | Donor |
| **Stakeholder** | Justgive |
| **Version** | 1.0 |
| **Precondition** | The application is launched<br>The actor is authenticated<br>The actor has searched for a project |
| **Trigger** | The user clicks on participate materially |
| **Nominal scenario** | 1 The system displays material participation form<br>2 The user fills and submits the form<br>3 The system checks conformity<br>4 The System sends a save query to the Database<br>5 The system displays success message |
| **Alternative Scenario (invalid credentials)** | 6 System displays error message.<br>7 Returns to step(2) |
| **Exceptional scenario** | The system stops unexpectedly |
| **Post-condition of success** | Display project detail page |
| **Post condition of failure** | Display project detail page |

Textual description of << Volunteer in a project >>

*Tableau 18 Textual description of << Volunteer in a project >>*

| **Title** | **Volunteer to a project** |
|---|---|
| **Summary** | The user volunteer to a project |
| **Actor** | Donor |
| **Stake holder** | Justgive |
| **Version** | 1.0 |
| **Precondition** | The application is launched<br>The actor is authenticated<br>The actor searched a project |
| **Trigger** | User clicks on choose participation button |
| **Nominal scenario** | 1 The user clicks on volunteer<br>2 The system displays volunteer form<br>3 The user clicks on confirmation button<br>4 The System sends save query to the Database<br>5 The system displays success message |
| **Post-condition of success** | Display project detail page |
| **Exceptional scenario:** | The system stops un volunteerily |

Textual description of << Participate in a fundraise >>

*Tableau 19 Textual description of << Participate in a fundraise >>*

| **Title** | **Participate in a fundraise** |
|---|---|

| | |
|---|---|
| **Summary** | The user participates financially to a project |
| **Actor** | Donor |
| **Stake holder** | Justgive |
| **Version** | 1.0 |
| **Precondition** | The app is launched<br>The actor is authenticated<br>The actor searched a project |
| **Trigger** | The user clicks on make financial participation |
| **Nominal scenario** | 1 The system displays financial participation form<br>2 The user fills and submit form<br>3 The system checks conformity<br>4 The system sends payment query to API<br>5 The API executes the query<br>6 The System sends save query to the Database<br>7 The system displays success message |
| **Alternative Scenario (invalid credentials)** | 8 System displays error message.<br>9 Returns to step(2) |
| **Alternative Scenario(if payment)** | 10 System displays error message<br>11 Returns to step(2) |
| **Exceptional scenario** | The system stops unexpectedly |
| **Post-condition of success** | Display home page |
| **Post condition of failure** | Display home page |

## v.) GENERAL USECASE OF JUSTGIVE

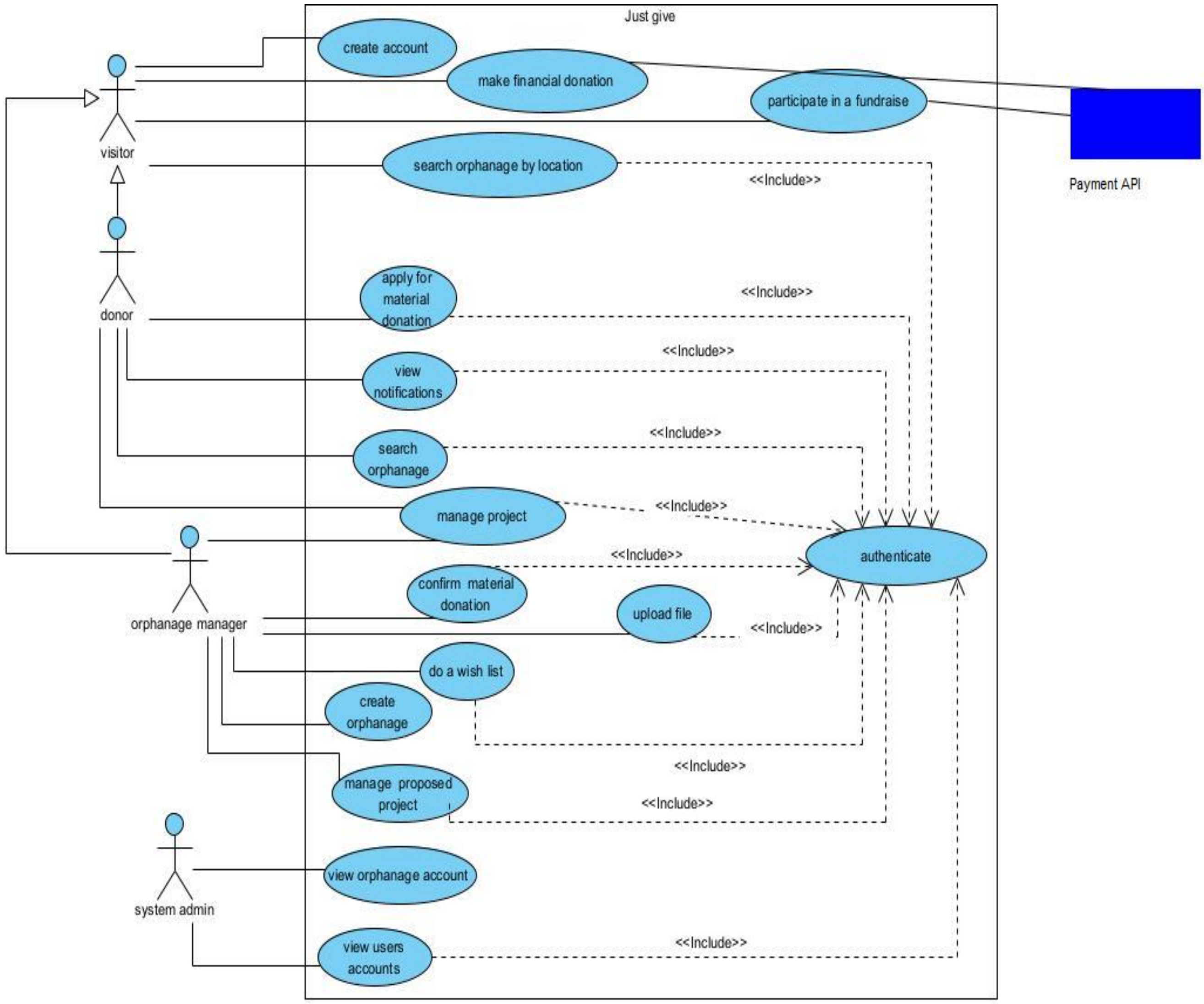



*Figure 14 general usecase of justgive*

## vi.) SPECIFIC USECASE DIAGRAMS

Specific usecase of manage project

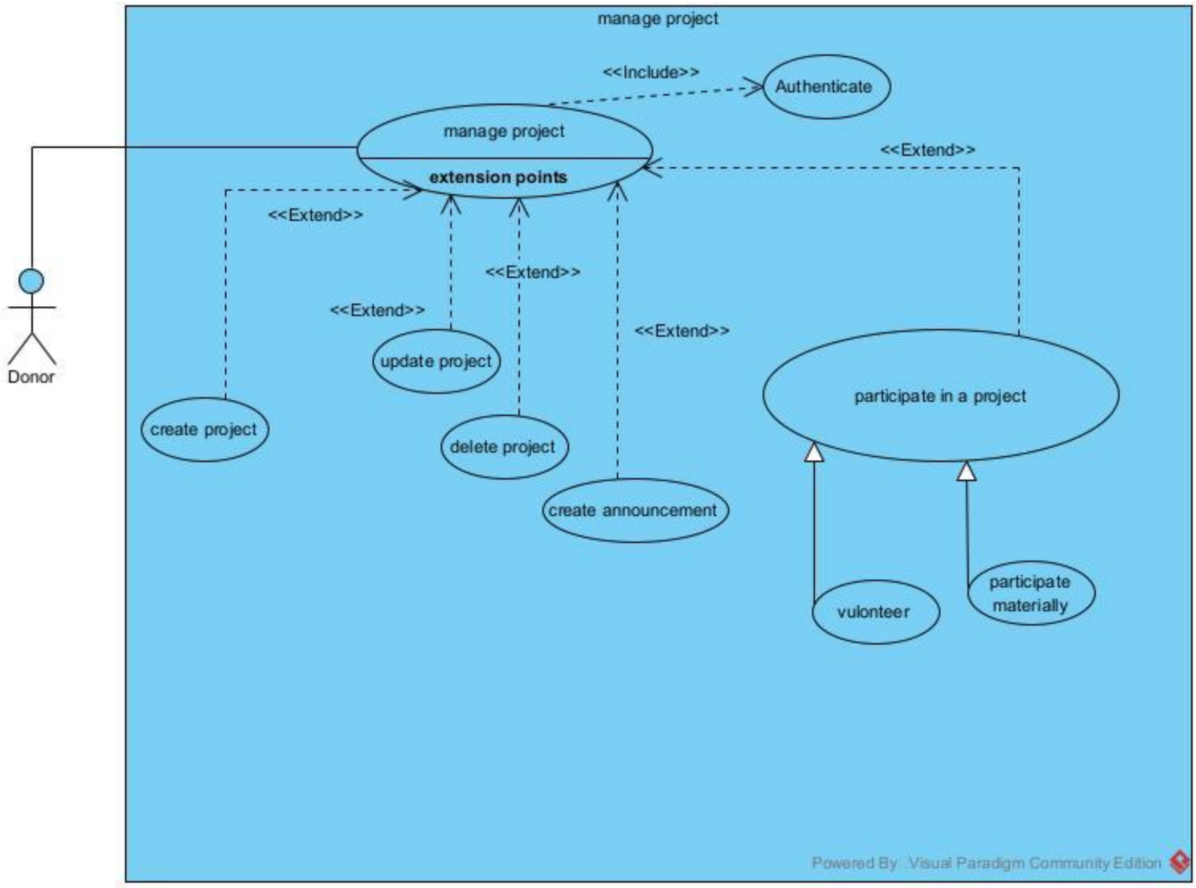



*Figure 15 Specific usecase of manage project*

Specific usecase of make financial donation

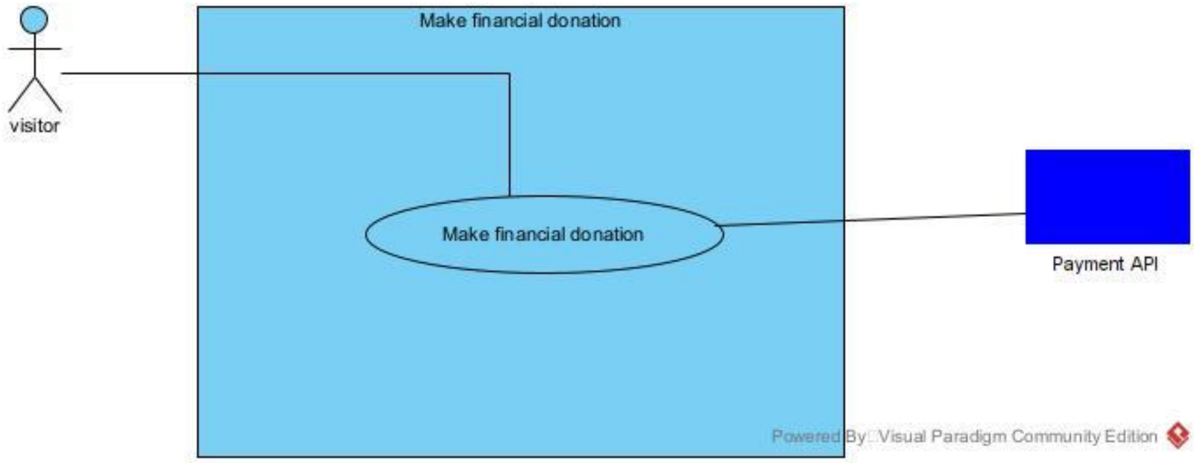



*Figure 16 Specific usecase of make financial donation*

Specific usecase for apply material donation

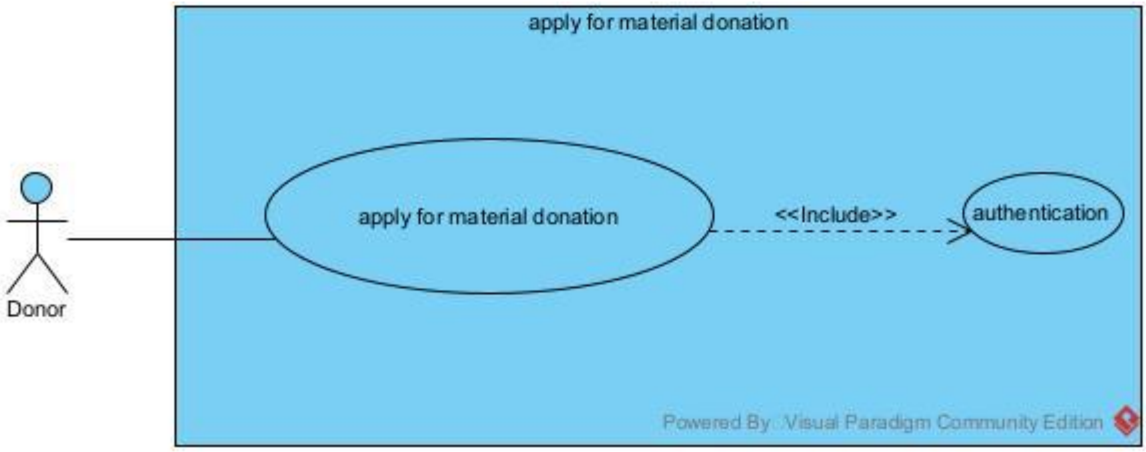



*Figure 17Specific usecase for apply material donation*

Specific usecase of participate in a fundraise

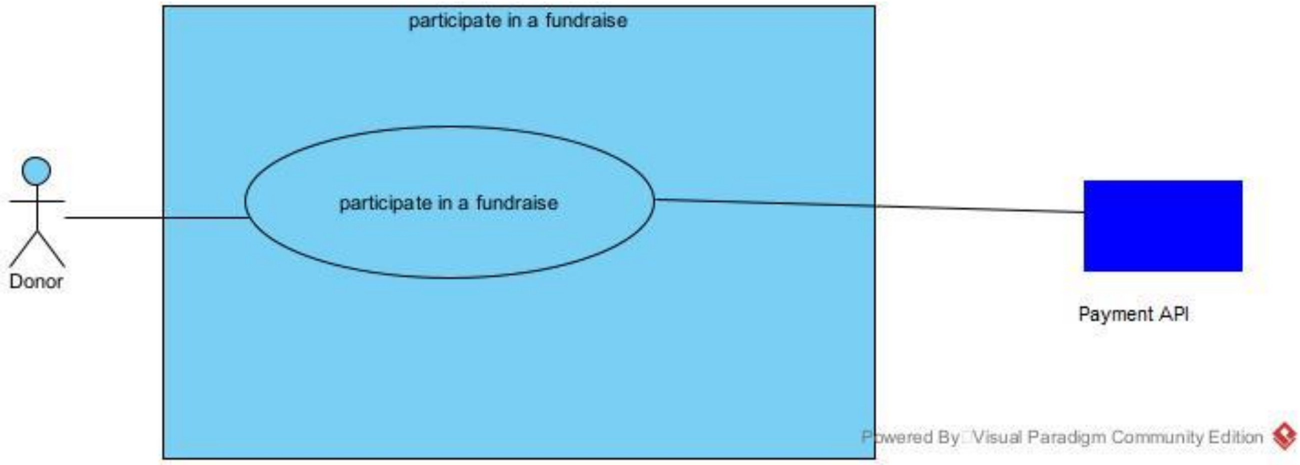



*Figure 18 Specific usecase of participate in a fundraise*

## b) COMMUNICATION DIAGRAM

### 1 DEFINITION

The communication diagram is used to model the specific interaction between the participants of the system. It describes both the static and dynamic behavior of the system. It shows the chronological succession of the actions performed by an actor to realize an operation.

## 2 FORMALISM

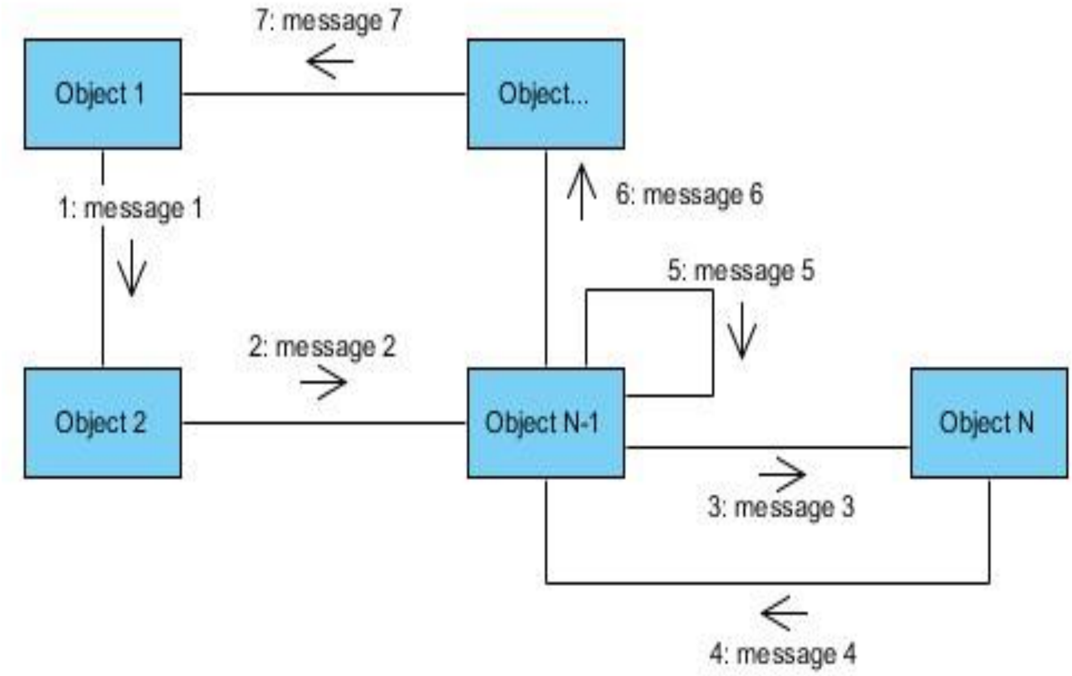
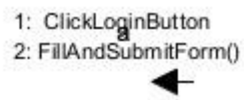
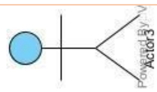



*Figure 19 Formalism of communication diagram*

## 3 COMPONENTS OF COMMUNICATION DIAGRAM

*Tableau 20 Components ofthe communication diagram*

| Element | Description and main properties | Representation |
|---|---|---|
| **Object** | Represents a role or object instance that participated in the interaction model. |  |
| **Message** | Defines a particular communication between lifelines in an interaction. |  |
| **Actor** | It models the type of role played by an entity that interact with the subject |  |
| **Link** | It initiates an association it connects two objects together for them to communicate. | |

Communication diagram of <<participate in a project>>

Communication diagram of <<volunteer in project>>

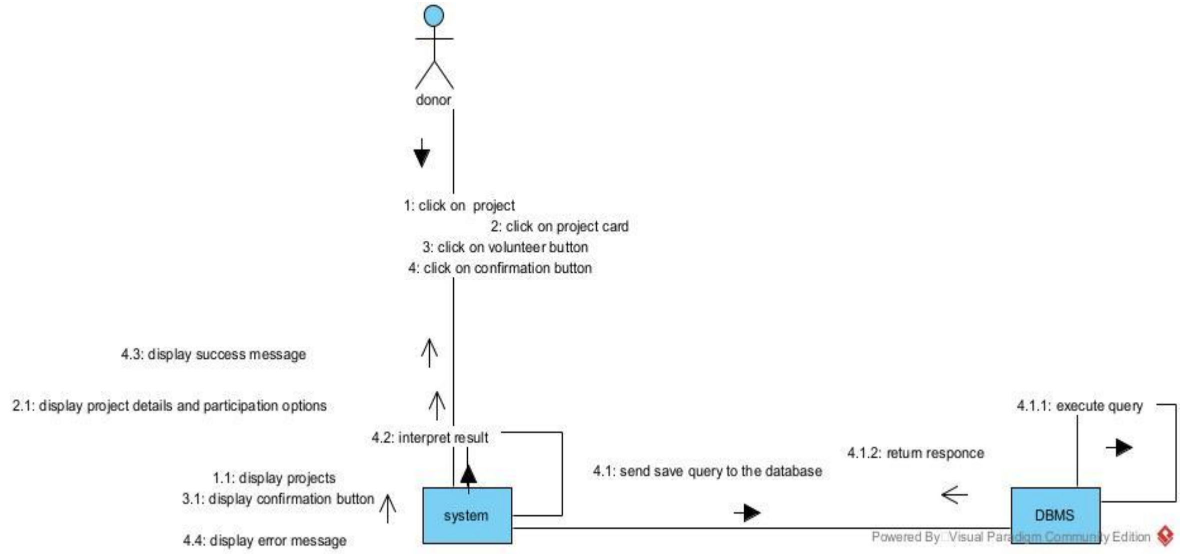



*Figure 20 Communication diagram of <<volunteer in project>>*

Communication diagram of <<participate materially in a project>>

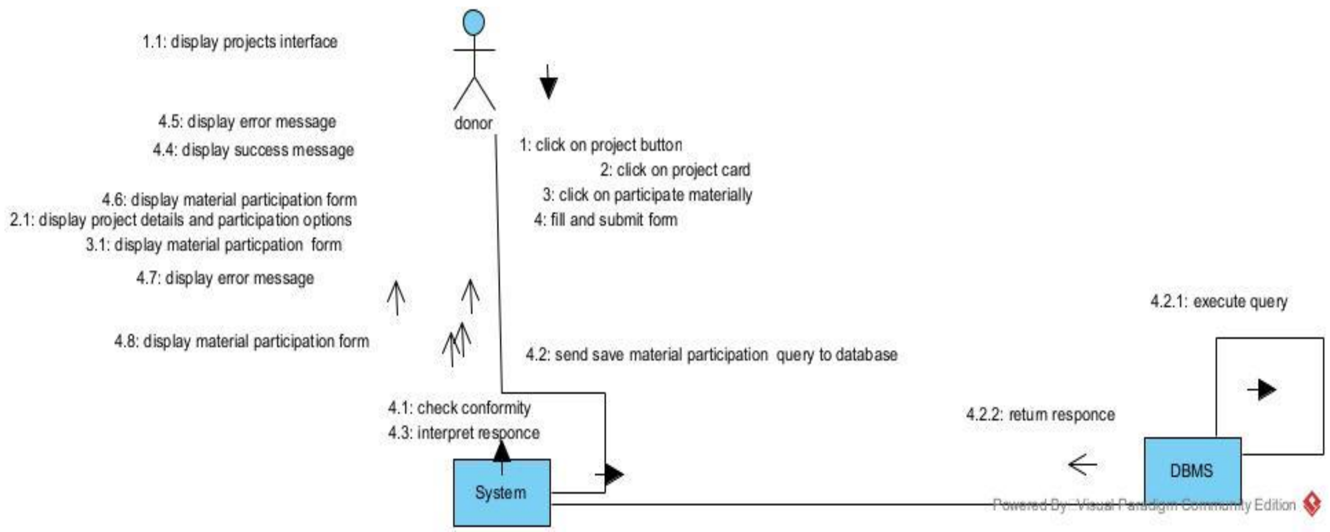



*Figure 21 Communication diagram of <<participate materially in a project>>*

Communication diagram of << Apply for material donation>>

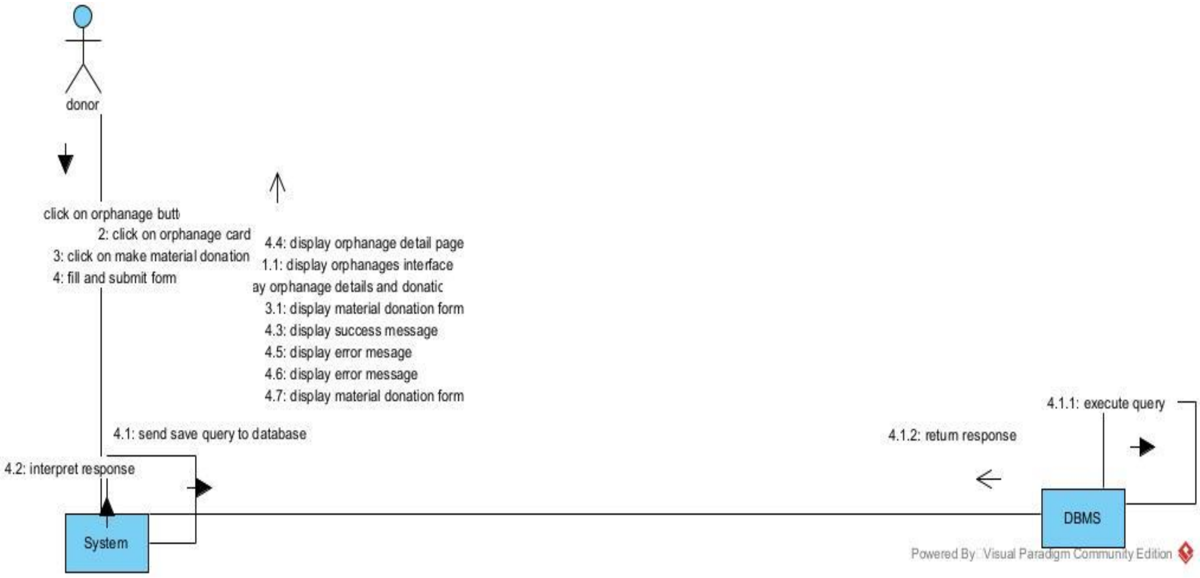



*Figure 22 Communication diagram of << Apply for material donation>>*

Communication diagram of <<make financial donation>>

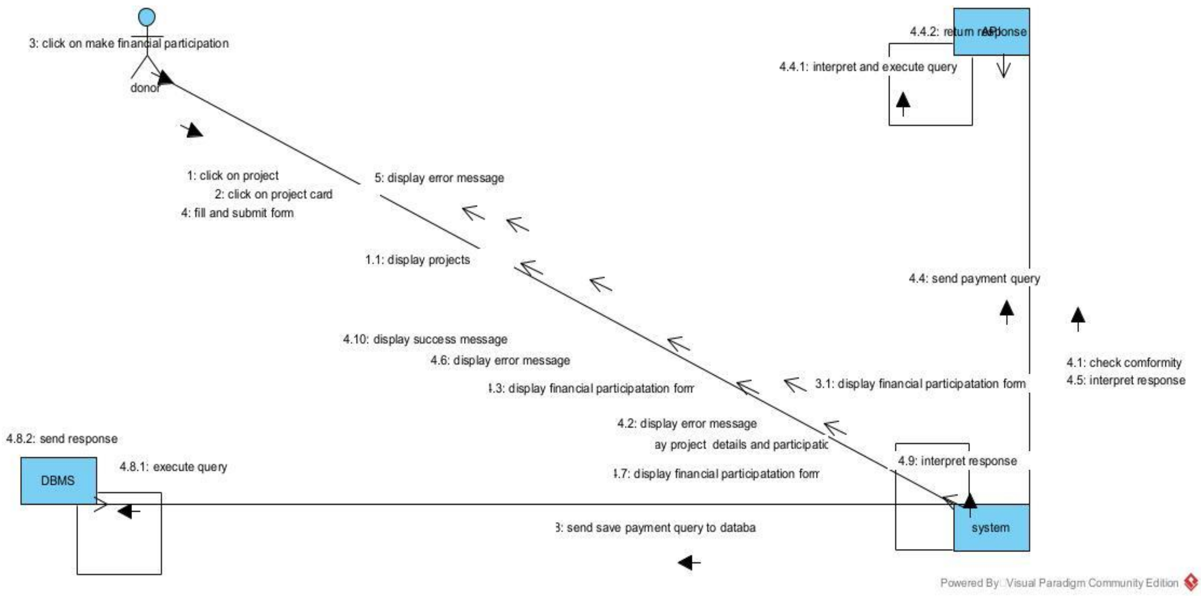



*Figure 23 Communication diagram of <<make financial donation>>*

Communication diagram of <<participate to fundraise>>

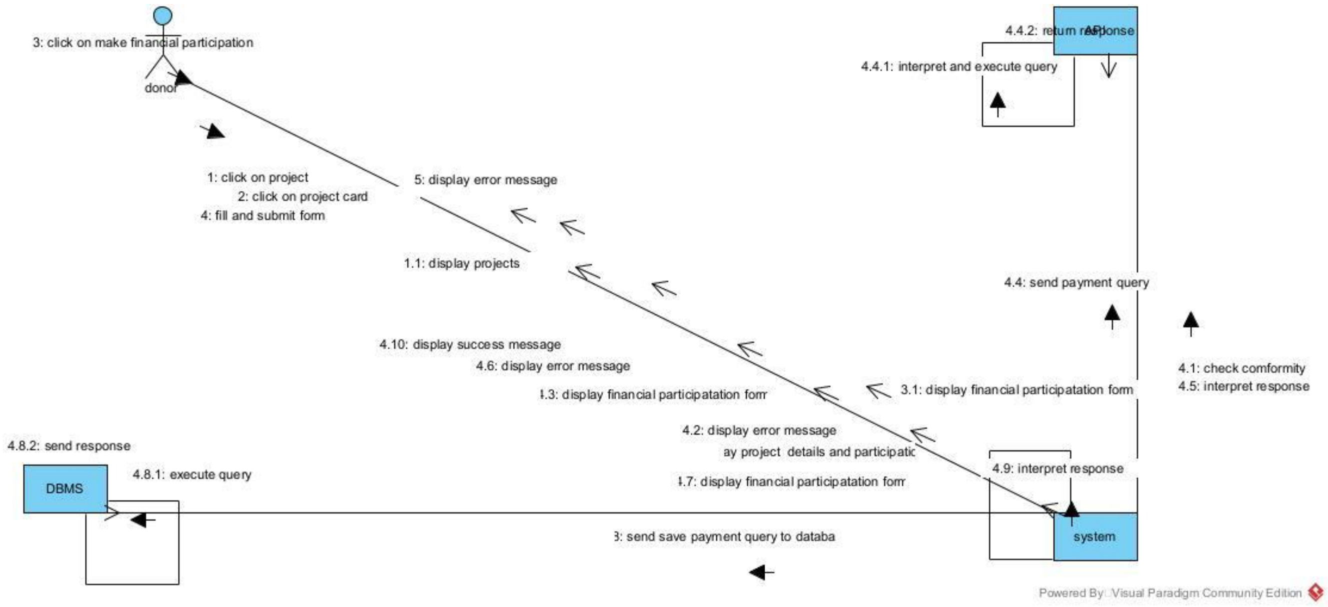



*Figure 24 Communication diagram of <<participate to fundraise>>*

## c) SEQUENCE DIAGRAM

### 1 DEFINITION

A sequence diagram is a UML diagram that illustrates the sequence of messages between objects in an interaction. It consists of a group of objects that are presented by lifelines and the messages that they exchange over time during the interaction. These interactions are arranged from top to bottom following their order of occurrence.

# 2 FORMALISM

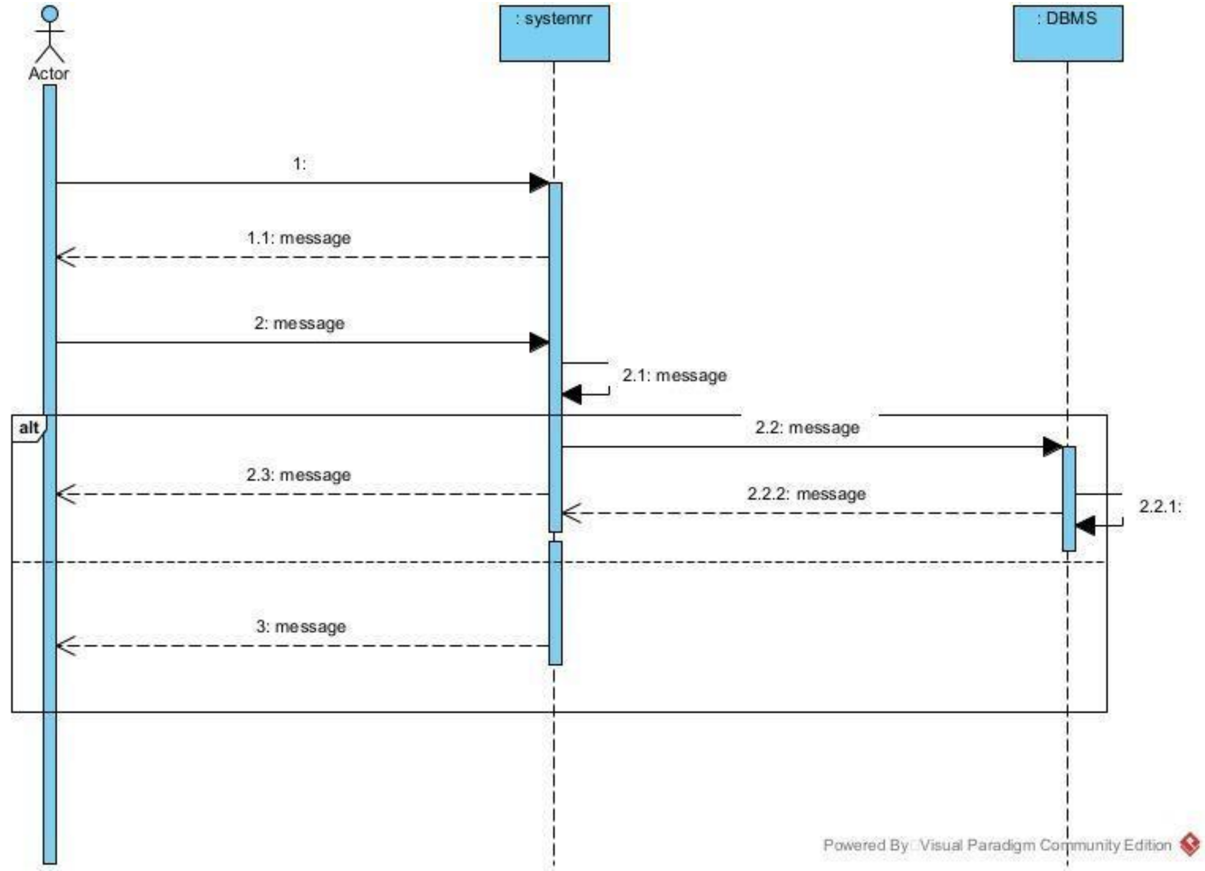
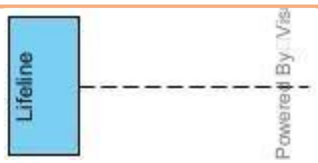
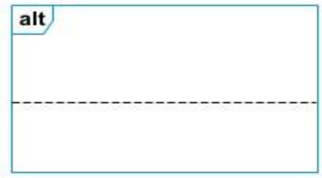
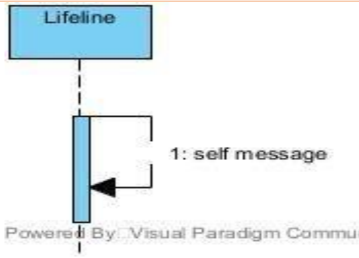
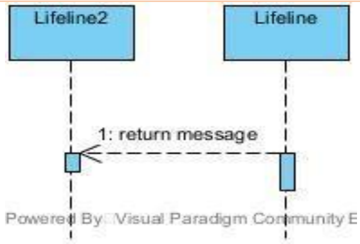



*Figure 25 formalism of  sequencce diagram*

# 3 COMPONENTS OF SEQUENCE DIAGRAM

*Tableau 21 components of the sequence diagram*

| Element | Description and main properties | Representation |
|---|---|---|
| **Lifeline** | They represent rows or object instances that participates in a sequence being modeled. |  |
| **Messages** | These are arrows which shows the direction of message flow. We have the synchronous, the asynchronous and the self-messages. |  |

| | | |
|---|---|---|
| **Combined Fragment** | It represents a choice of behavior in which at most one operand will be chosen. |  |
| **Self Message** | Represents the execution or operation call on thesame object |  |
| **Return Message** | Represents a response to a message |  |

Sequence diagram of <<authenticate>>

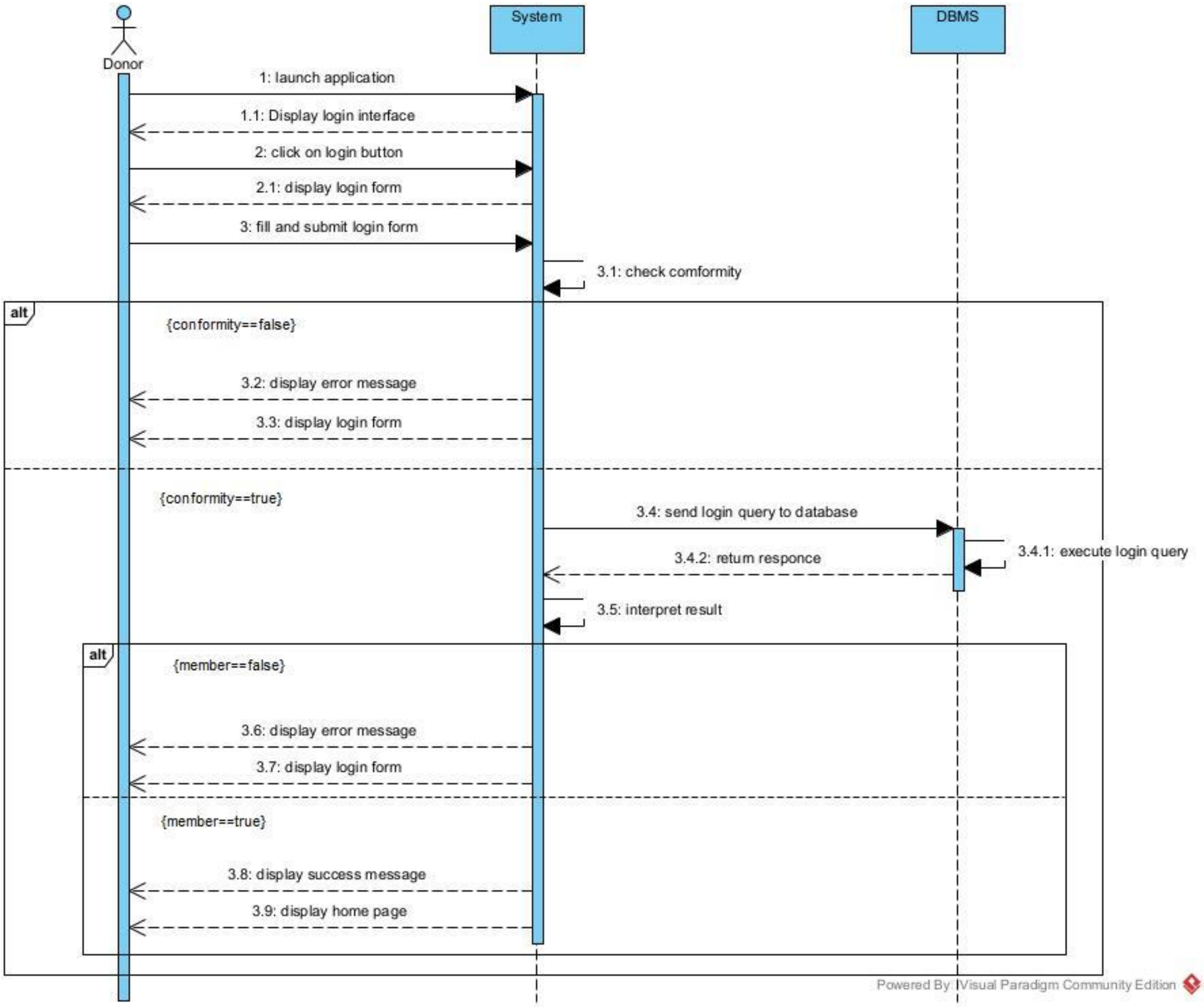



*Figure 26 Sequence diagram of <<authenticate>>*

Sequence diagram of <<Apply for material donation>>

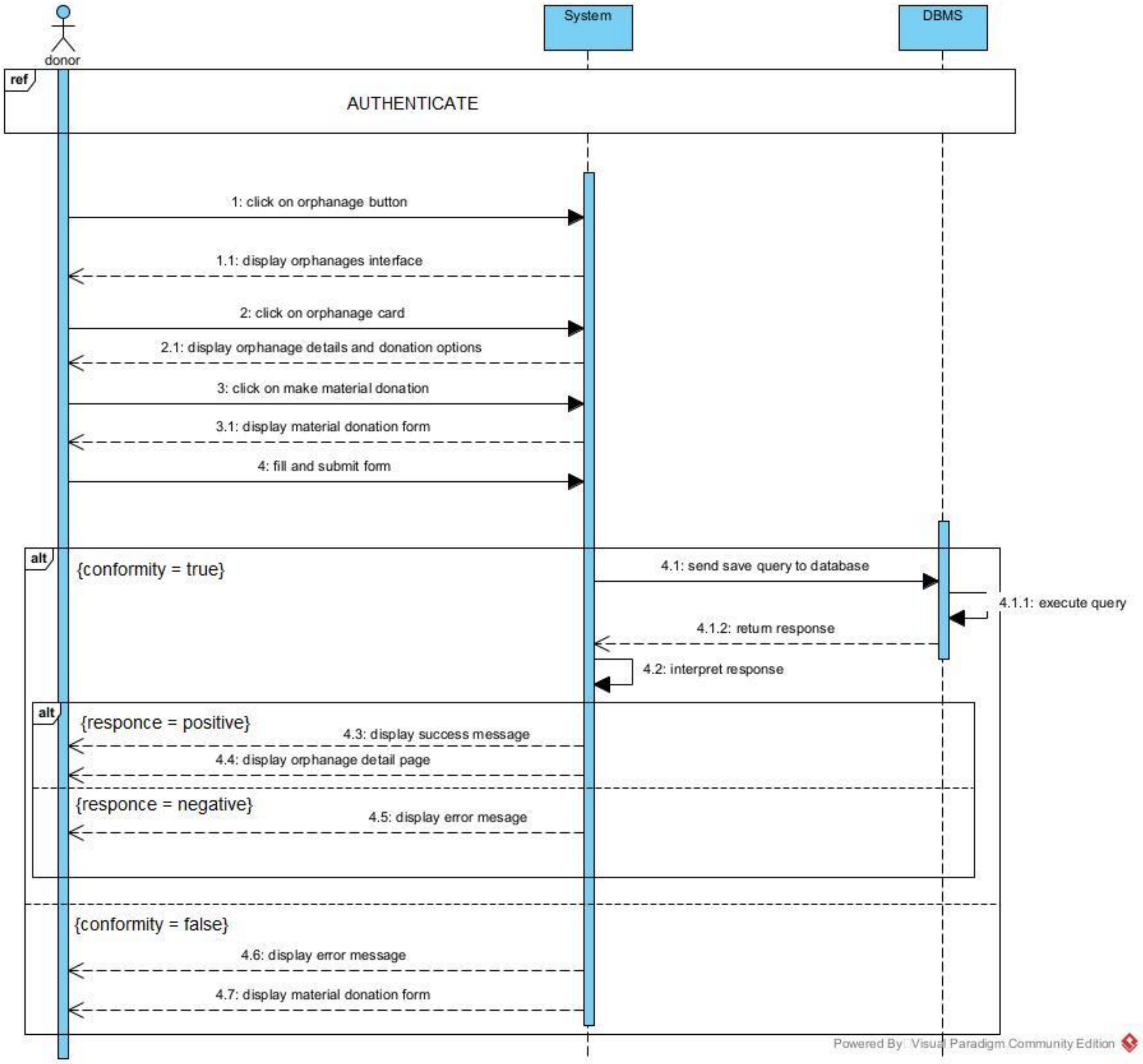



*Figure 27 Sequence diagram of <<Apply for material donation>>*

Sequence diagram of << participate in a project>>

Sequence diagram of << participate materially to a project>>

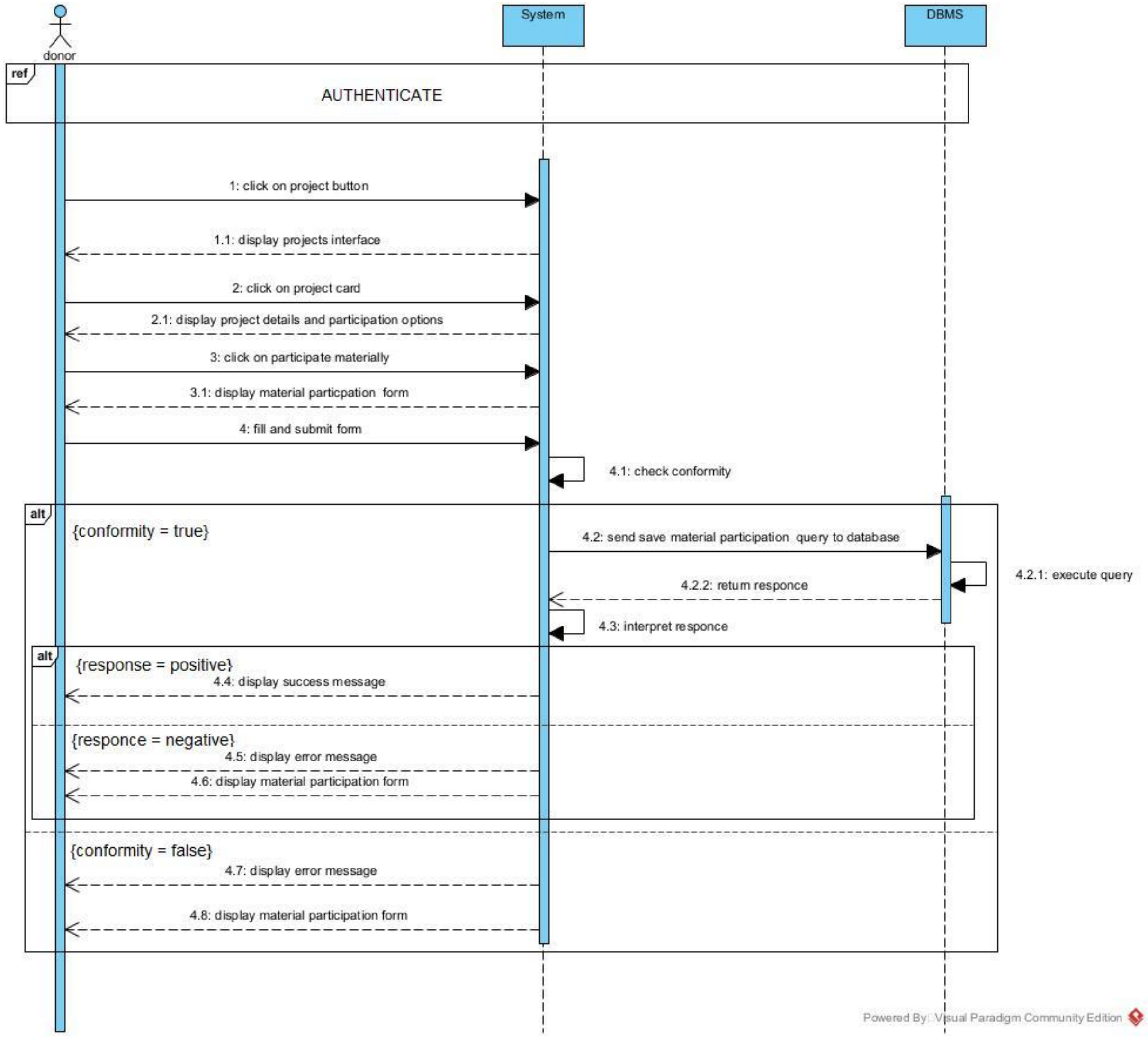



*Figure 28 Sequence diagram of << participate materially to a project>>*

Sequence diagram of <<volunteer in a project>>

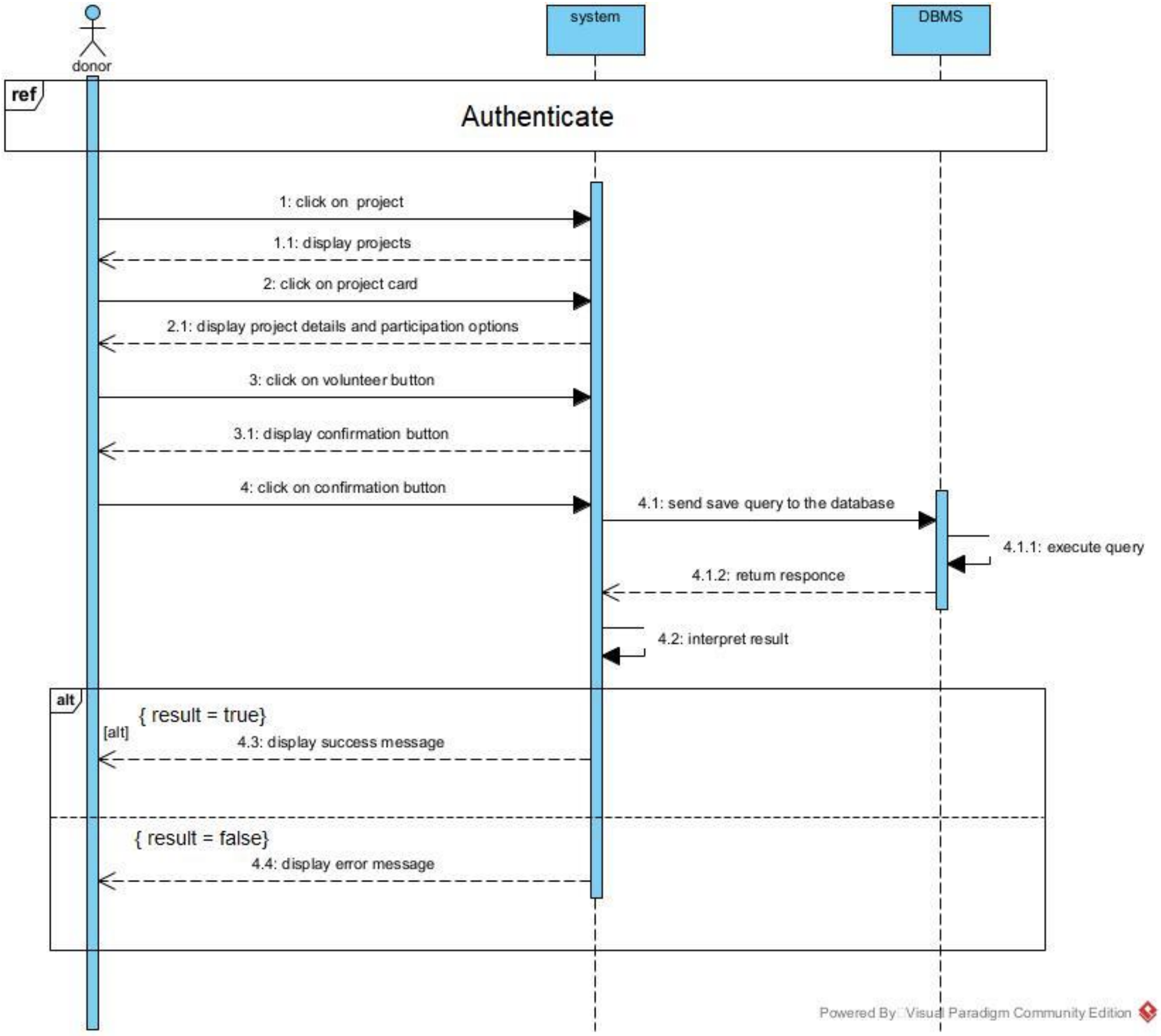



*Figure 29 Sequence diagram of <<volunteer in a project>>*

Sequence diagram of <<participate in fundraise>>

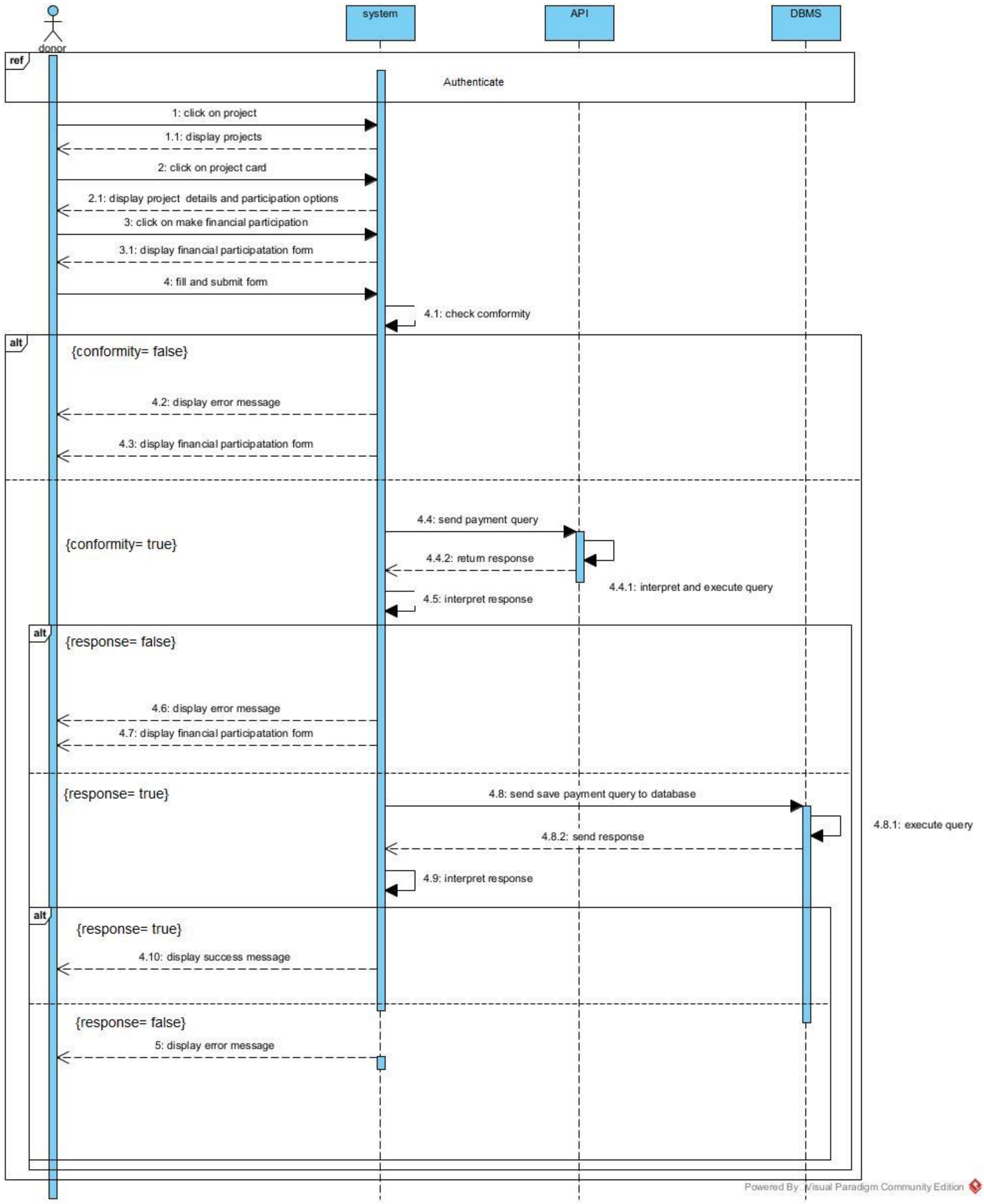



*Figure 30 Sequence diagram of <<participate in fundraise>>*

# d) ACTIVITY DIAGRAM

## 1 DEFINITION

An activity diagram provides a view of the behavior of a system by describing a sequence of actions in the process. The activity diagram makes it possible to emphasize the processing and to graphically formalize the sequence of actors carried out in a use case.

## 2 FORMALISM

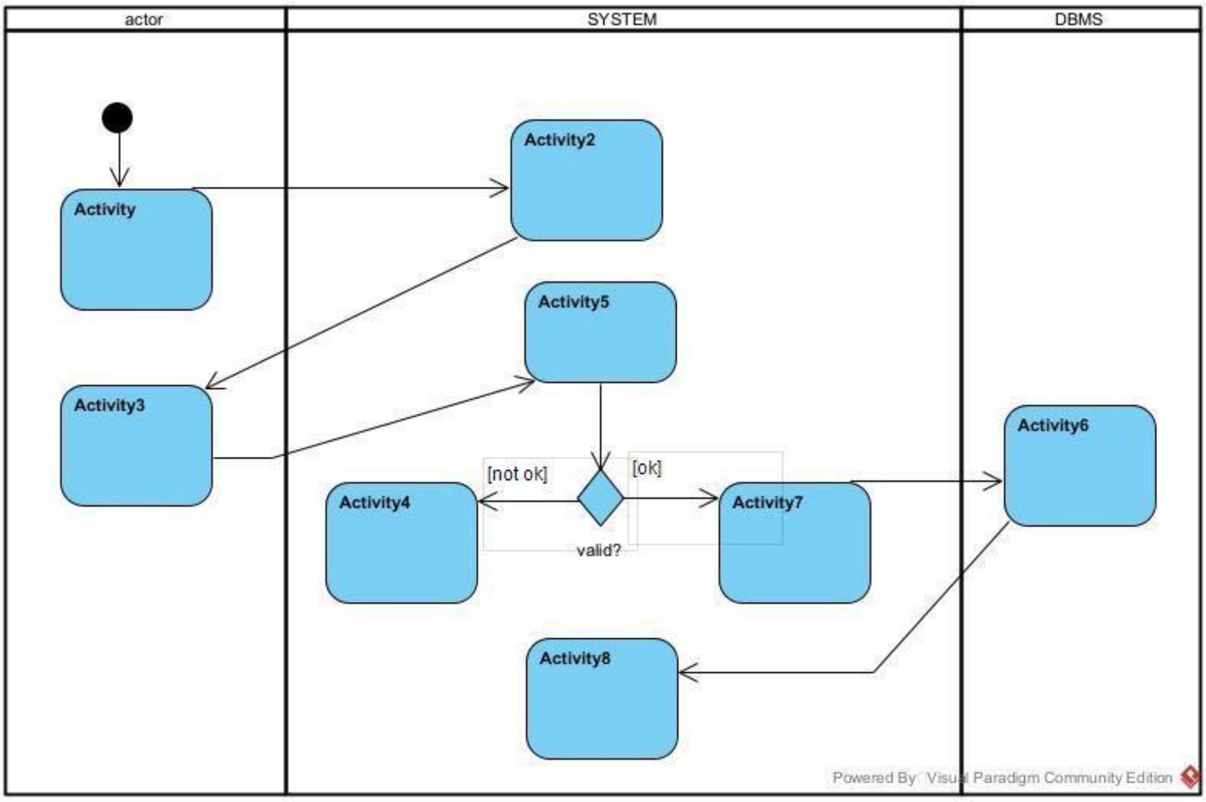
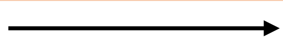
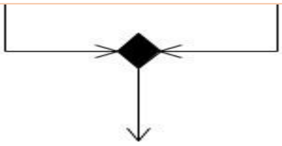
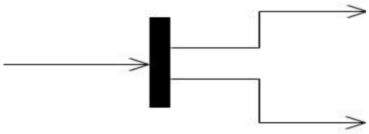
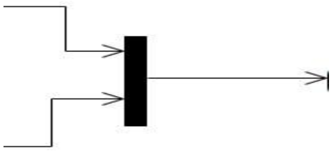
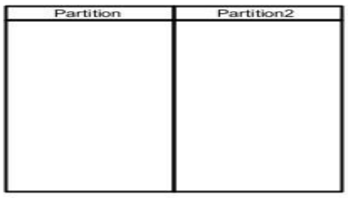



*Figure 31 Formalism of sequence diagram*

## 3 COMPONENTS OF ACTIVITY DIAGRAM

*Tableau 22 Components of activity diagram*

| Element | Description | Representation |
|---|---|---|
| **Activity** | It describes the action of an object and it is represented by rectangle with |  |
| **Activity edge** | A directed connection between two activity nodes through which tokens may flow. |  |

| | | |
|---|---|---|
| **Decision node** | It is a decision node that takes token slight round borders (condition) from one edge and |   |
| **Initial Node** | It is a control node at which flow starts when an activity is invoked |  |
| **Final Node** | A control node that stops all flows in an activity. It is represented by circle with a filled circle inside |  |
| **Action** | Represents a single step in an activity which cannot be further decomposed |   |
| **Merge node** | Reunite different decision paths created using a decision node. |  |
| **Fork node** | Slits behavior into parallel or concurrent flows of activities (or actions) |  |
| **Join node** | Unites a set of parallel or concurrent flows of activities or actions. |  |
| **Swimlane and partition** | A way of grouping activities performed by the same actor in an activity diagram or to group actions in the same thread. |   |

Sequence diagram of <<authenticate>>

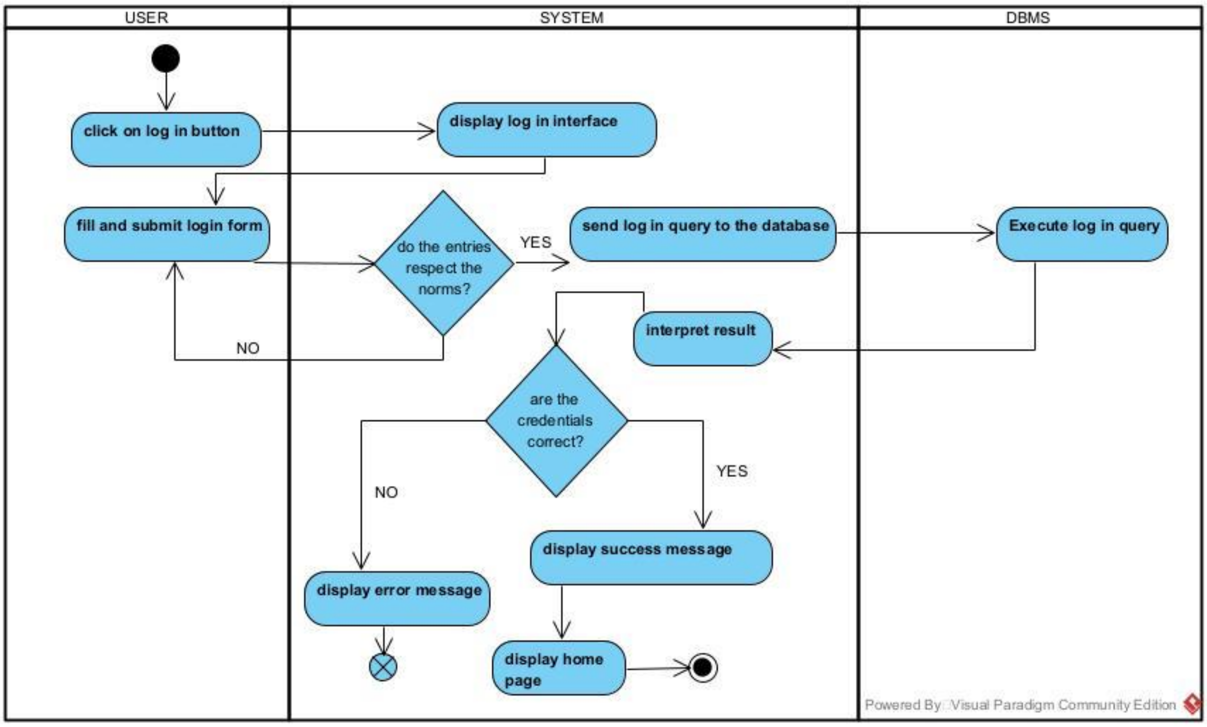



*Figure 32 Sequence diagram of <<authenticate>>*

Activity diagram of <<Make Financial Donation>>

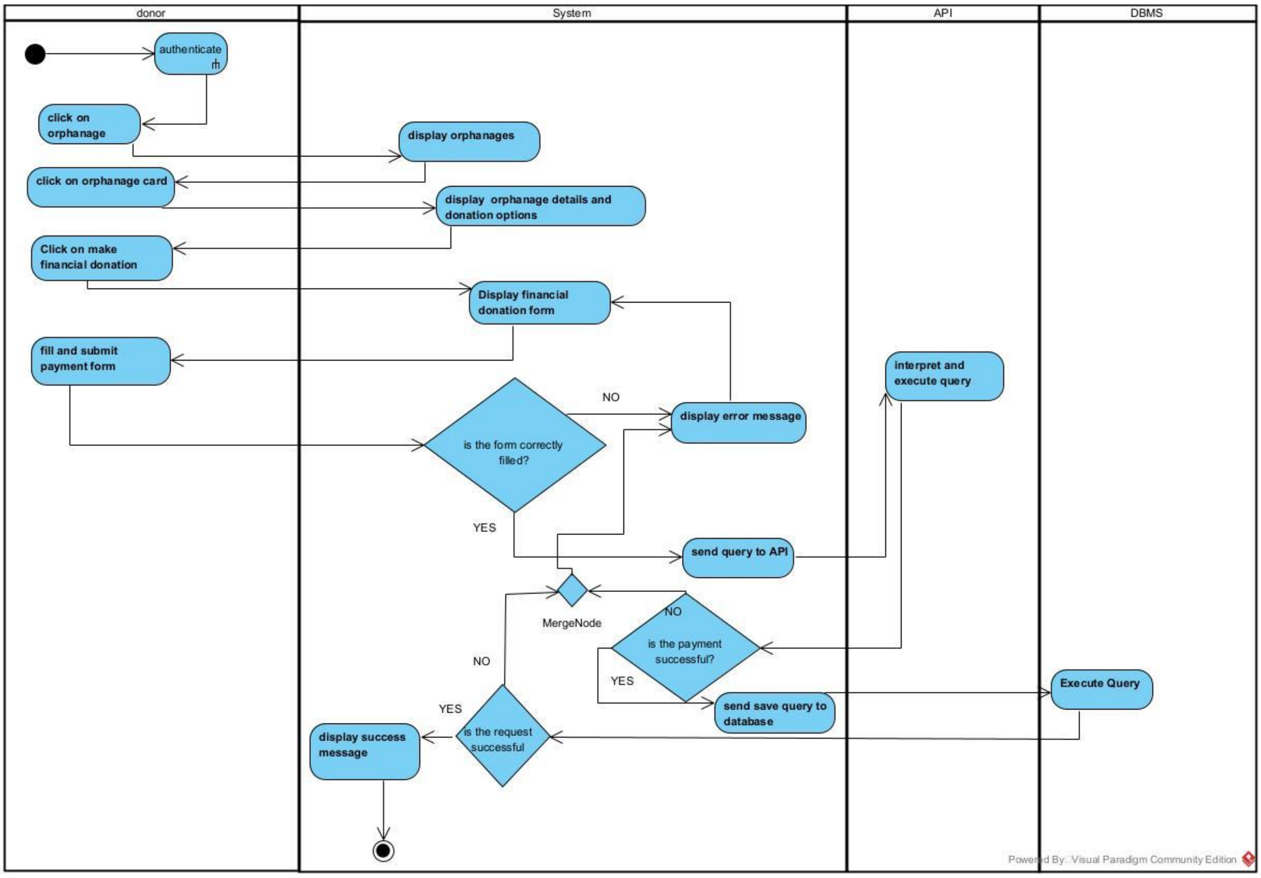



*Figure 33 Activity diagram of <<Make Financial Donation>>*

Activity diagram of <<Apply for material donation>>

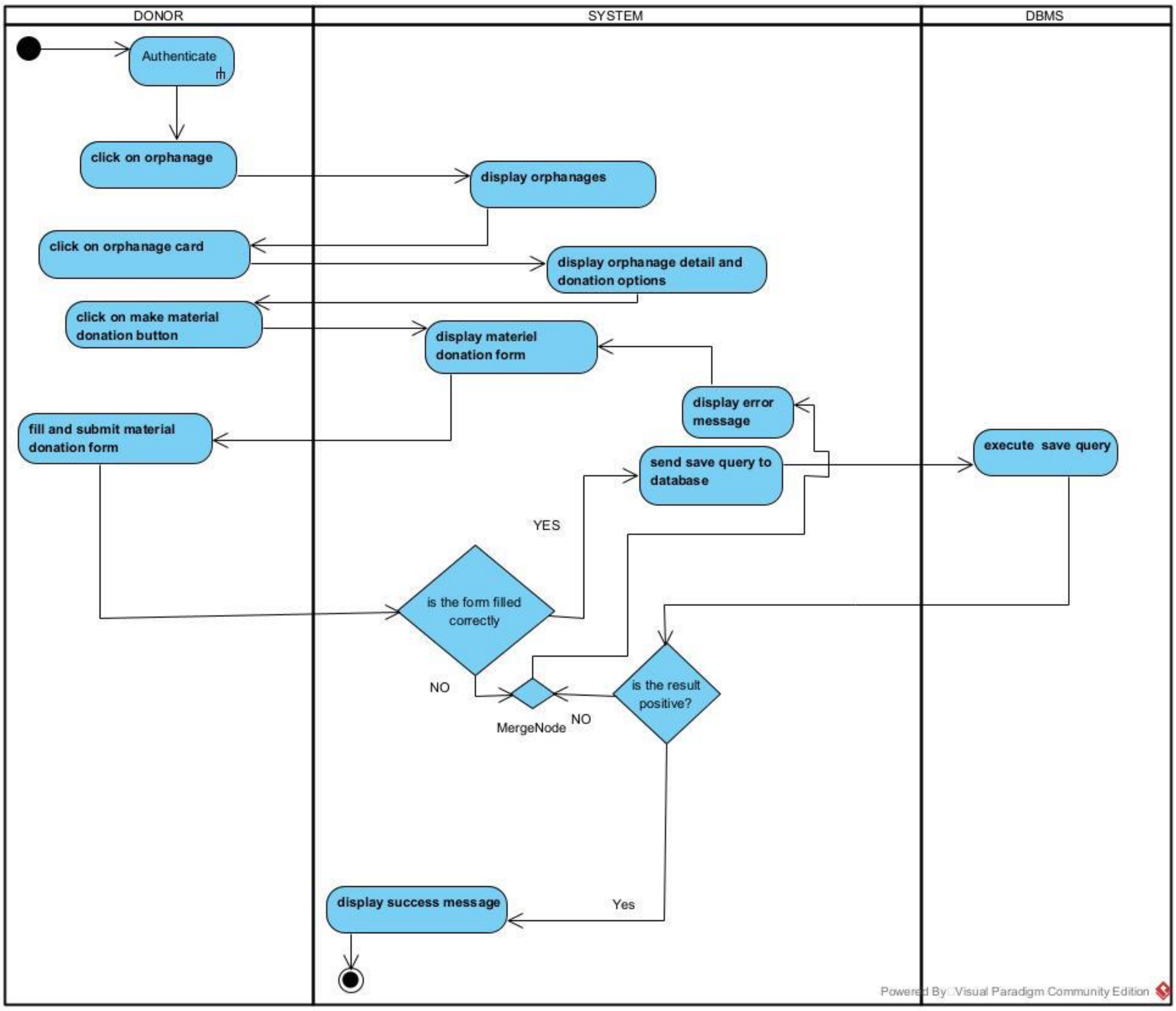



*Figure 34 Activity diagram of <<Apply for material donation>>*

Activity diagram of << participate in a project>>

Activity diagram of <<Participate materially in a project>>

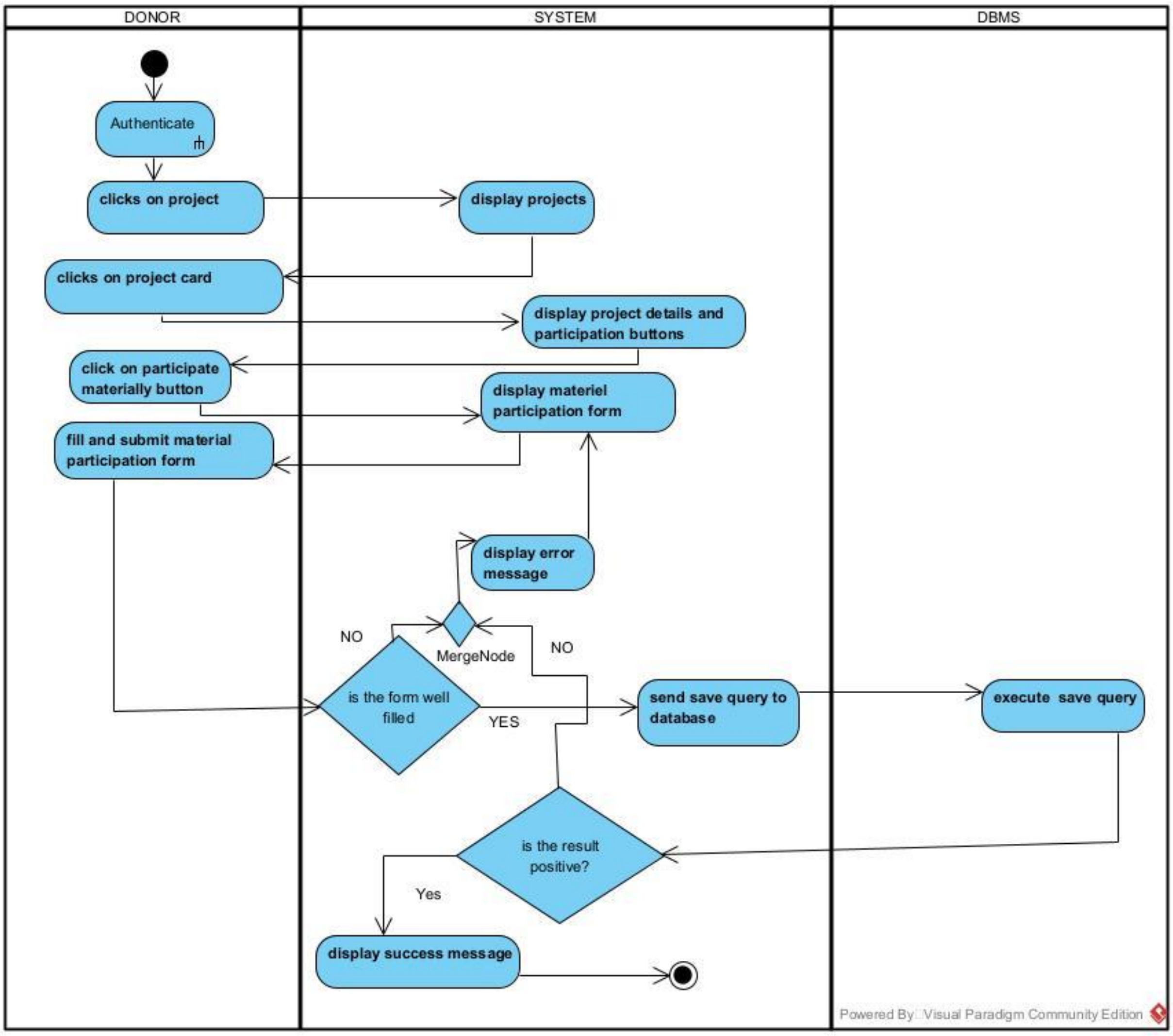



*Figure 35 Activity diagram of <<Participate materially in a project>>*

Activity diagram of <<Volunteer in a project>>

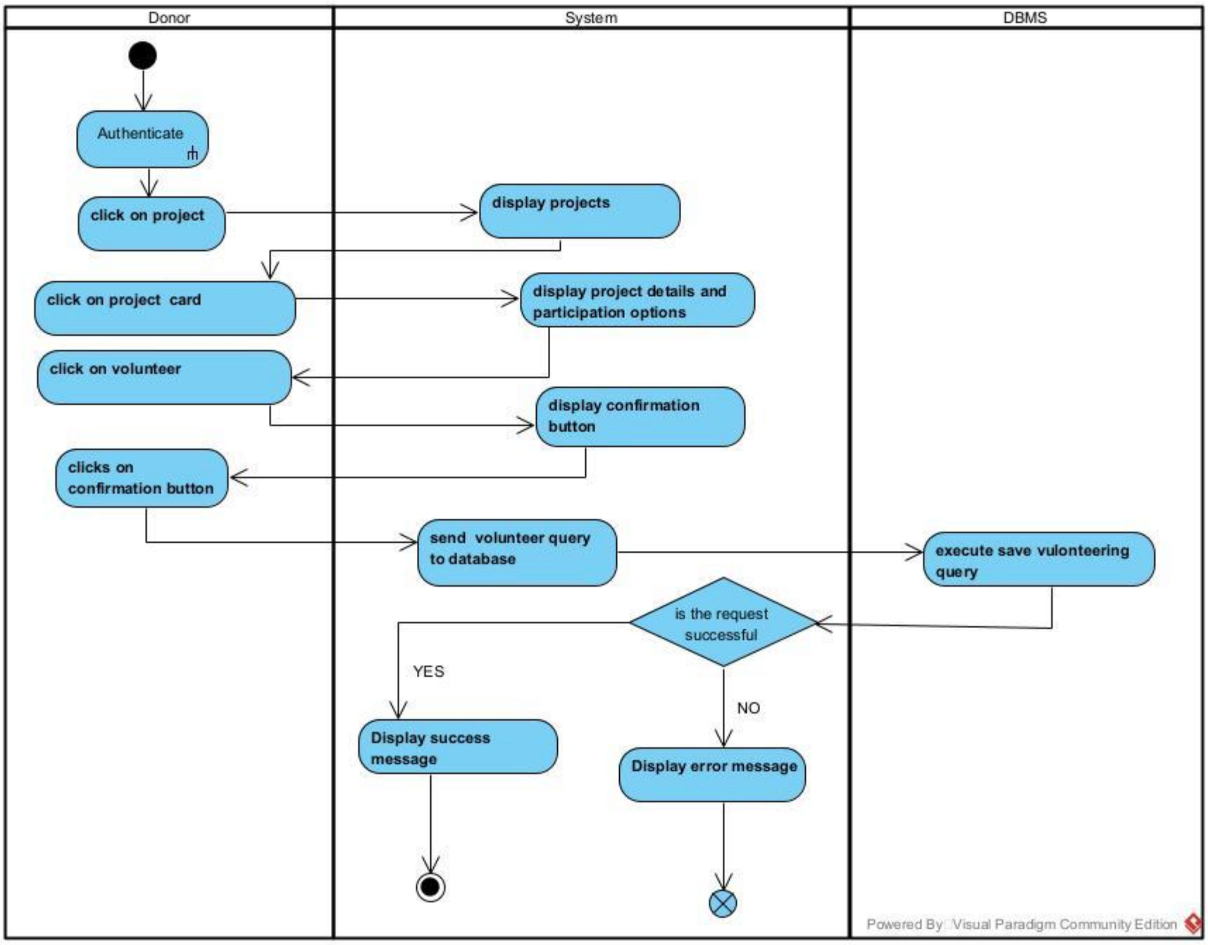



*Figure 36 Activity diagram of <<Volunteer in a project>>*

Activity diagram of <<participate in fundraise>>

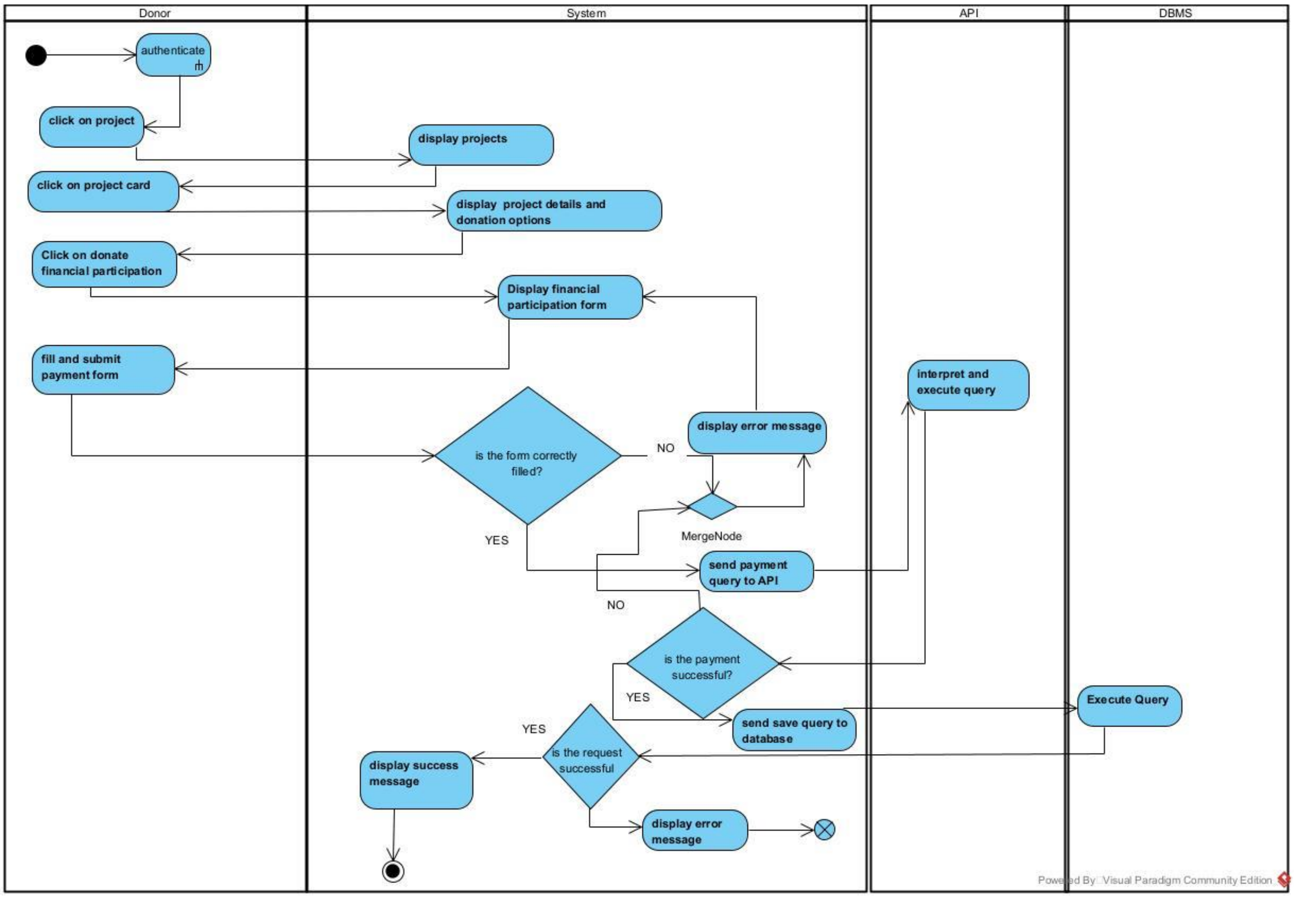



*Figure 37 Activity diagram of <<participate in fundraise>>*

# CONCLUSION

Having reached the end of our analysis, we were asked to illustrate the language and method used during this phase, present the different diagrams of the functional branch. we can then say that this analysis file has allowed us delimit the functional needs of the web application that will be designed and to have a detailed overview of the new system to be set up without notwithstanding we quickly dive into the conception phase.

# DOCUMENT 5: CONCEPTION PHASE

Preamble

This document makes it possible to model the proposed solution as a whole and to gather the information necessary to set up a complex and efficient database. It therefore foresees the future system.

Content :

# INTRODUCTION

The realization of an analysis file is very important for the advancement of the application, it guarantees the reliability of maintenance and use of a software. That said, after analyzing our system, we will move on to the design of our future system. To do this we will follow the steps of the technical branch of the 2TUP process, making the diagrams associated with these steps namely the class diagram and the state machine diagram.

# I. CAPTURE OF TECHNICAL NEEDS

## 1. PHYSICAL ARCHITECTURE

When designing an application, it is important before going any further to know the architecture of this application. The choice of your architecture can be based on: the amount of users, the data volume, transaction volume, performance requirements and security needs.

The architecture of our project is an n tier architecture .We made this choice due to the fact that we are making use of external APIs. In our case a payment API.

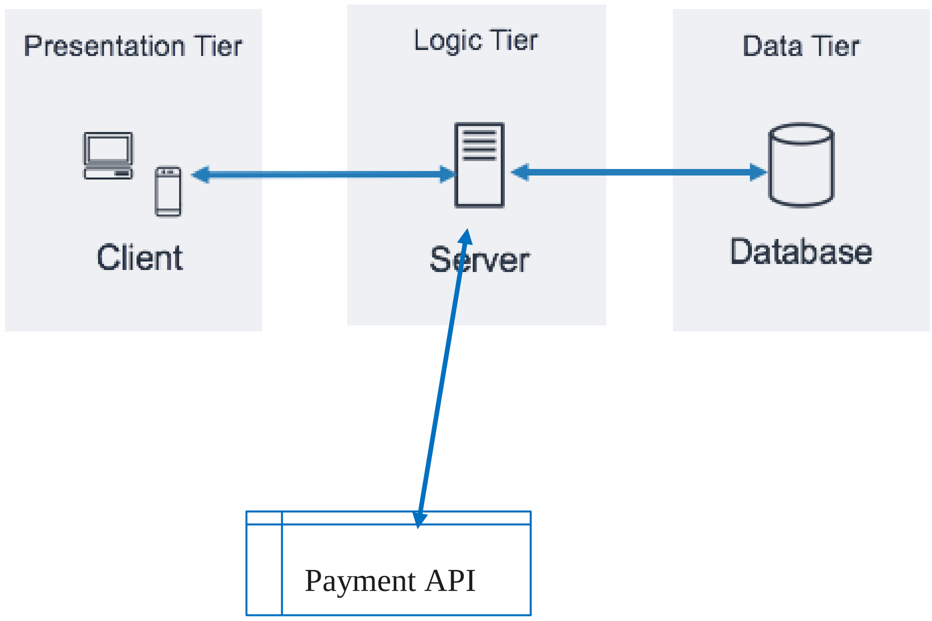



*Figure 38 Physical architecture*

Source:https://docs.aws.amazon.com/whitepapers/latest/serverless-multi-tier-architectures-api-gateway-lambda/three-tier-architecture-overview.html

## 2. LOGICAL ARCHITECTURE

The application logic of our system is based on MVC design pattern which is a very practical way to organize the different classes in an application development project facilitating error detection, collaboration, and code reusability.

MVC (Model View Controller) design pattern is an architecture and a design method that organizes the human-machine interface (HMI), a view (user Interface) and a controller (Control Logic, event management, synchronization), each having a specific role in the interface.

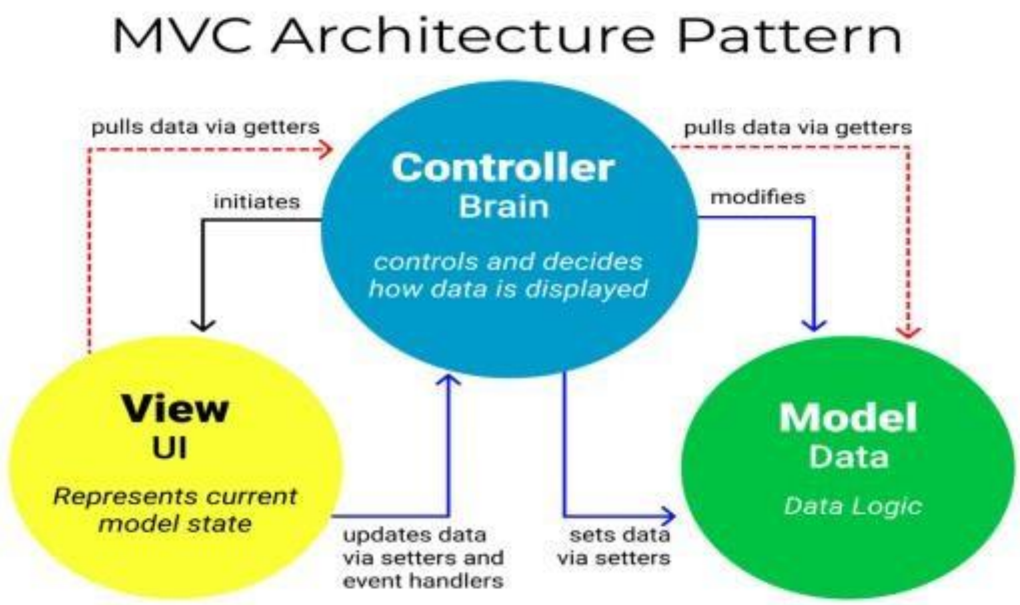



*Figure 39 Logical architecture of justgive*

*(Source: https://www.freecodecamp.org/news/the-model-view-controller-pattern-mvc architecture-and-frameworks-explained/)*

# II. CLASS DIAGRAM

## a) DEFINITION

The class diagram expresses the static structure of the system in terms of classes and the relationships between those classes. The interest of the class diagram is to model the entities of the information system. The class diagram is used to represent all the finalized information that is managed by the domain. This information is being structured, that is, it has been grouped into classes.

## b) FORMALISM

The diagram highlights possible relationships between these classes. A class is a set of elements of the same nature that can be uniquely identified. This diagram has several concepts: class, attribute, identifier, method, access modifier. (Public (+), protected (#), private (-), package (), derivative (/)), relationships and multiplicities. The table below shows the different components of a class.

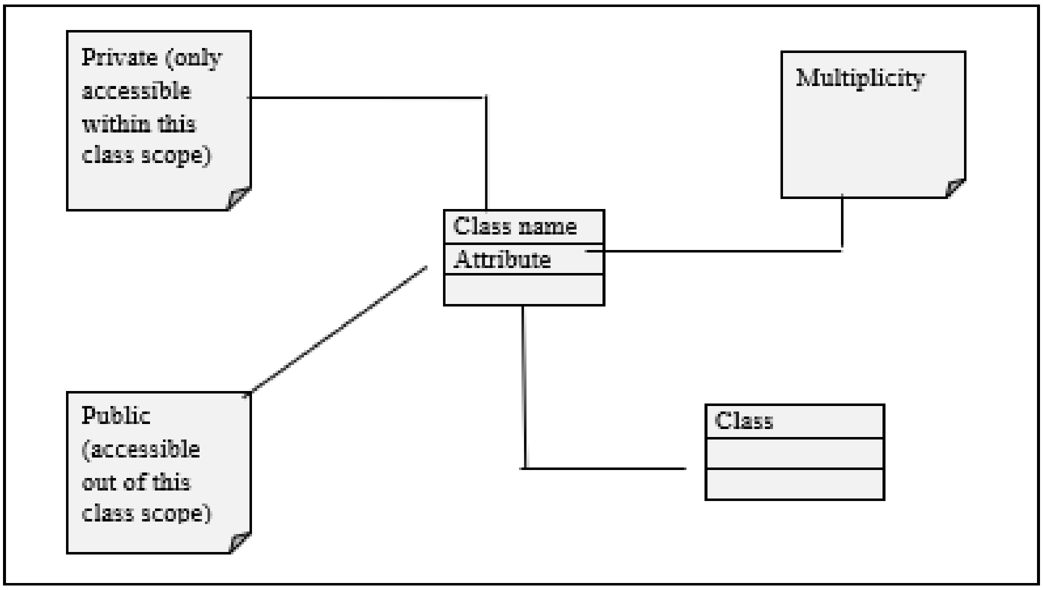
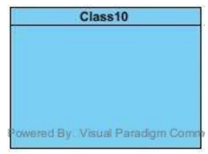
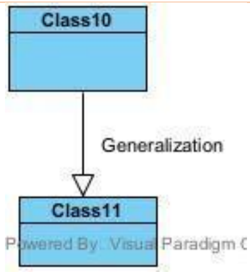
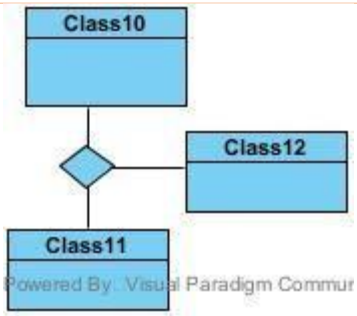
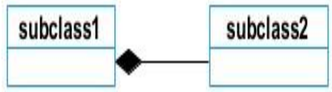
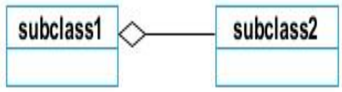
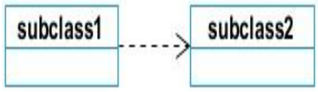
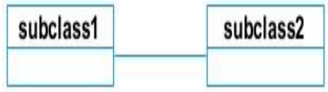
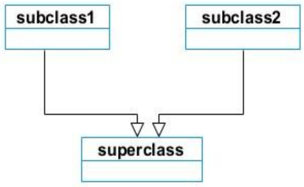



*Figure 40 Formalism of class diagram*

## c) COMPONENTS OF THE CLASS DIAGRAM

*Tableau 23 Components of the class diagram*

| Element | Description and main features | Representation |
|---|---|---|
| **Class** | Represents an entity within a system. It is an abstract representation or description of a set of objects in the application domain. |  |
| **Generalization** | Exist between two related classes where one is a specialized form of the other. |  |
| **N-ary relation** | It is a type of relationship used to model several classes |  |

| | | |
|---|---|---|
| **Composition** | If a parent of a composite is deleted, usually, all of its parts are deleted with it. |   |
| **Aggregation** | If the parent of the aggregate is deleted, usually the children are not deleted. |   |
| **Dependency** | It existed between two classes, if one changes it may cause the change in the order, but the other way around. |   |
| **Association** | It is a general type of relationship between elements, it may include cardinality, roles etc. |   |
| **Generalization** | it a relationship between a whole thing (called superclass) and a more specific thing (called subclass) |   |

Class diagram of justgive

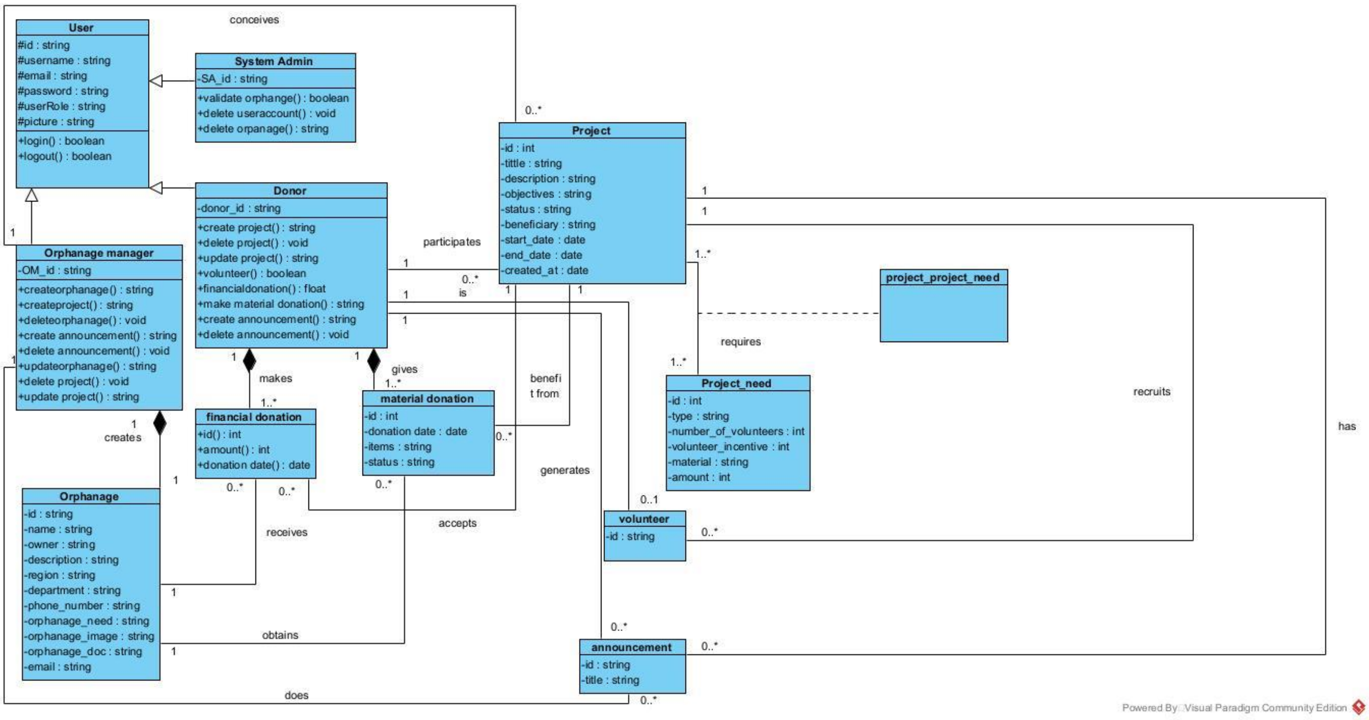



*Figure 41 Class diagram of justgive*

# III. STATE MACHINE DIAGRAM

## 1. DEFINITION

State diagrams are used to document the various modes (states) that a class can take, and the events that cause a state transition. Their role is to present the treatments (operations) that will manage the field studied. They define the sequence of class states, also known as the state chart diagram, which models the dynamic flow of control from the state of a particular object within a system.

## 2. FORMALISM

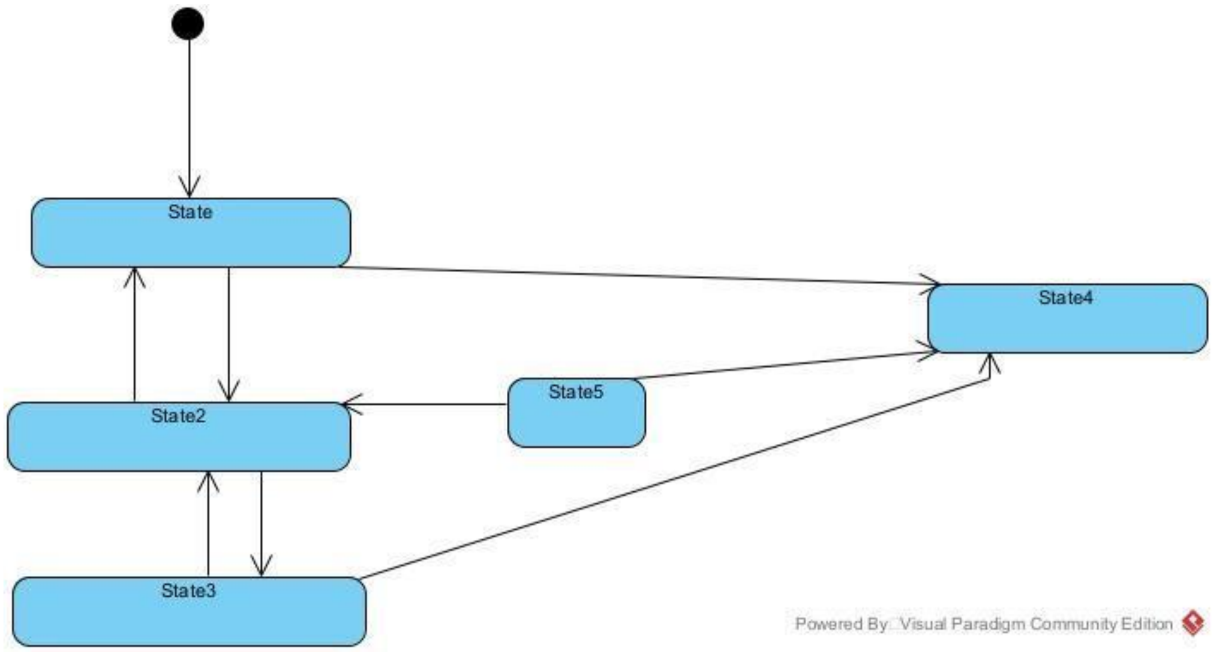
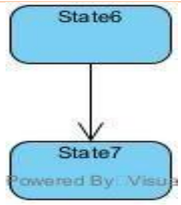



*Figure 42 formalism of state machine diagram*

## 3. COMPONENTS OF STATE MACHINE DIAGRAM

*Tableau 24 Components of the statemachine diagram*

| Element | Description and features | Representation |
|---|---|---|
| **Initial State** | It represents the final or beginning of state and it is represented by a block filled circle |  |
| **Final State** | It represents the final state or end of a system. It is represented by a filled circle within another circle |  |

| | | |
|---|---|---|
| **Transition** | it is a change of control from one state to another due to the occurrence of some events. It is represented by an arrow labelled with the event that causes it's change of state. |  |
| **State Box** | It depicts the condition or circumstances of particular object of a class at a specific point of time. |  |
| **Choice pseudo-State** | A diamond symbol that indicates a dynamic condition with branched potential results |  |

State machine diagram of Justgive

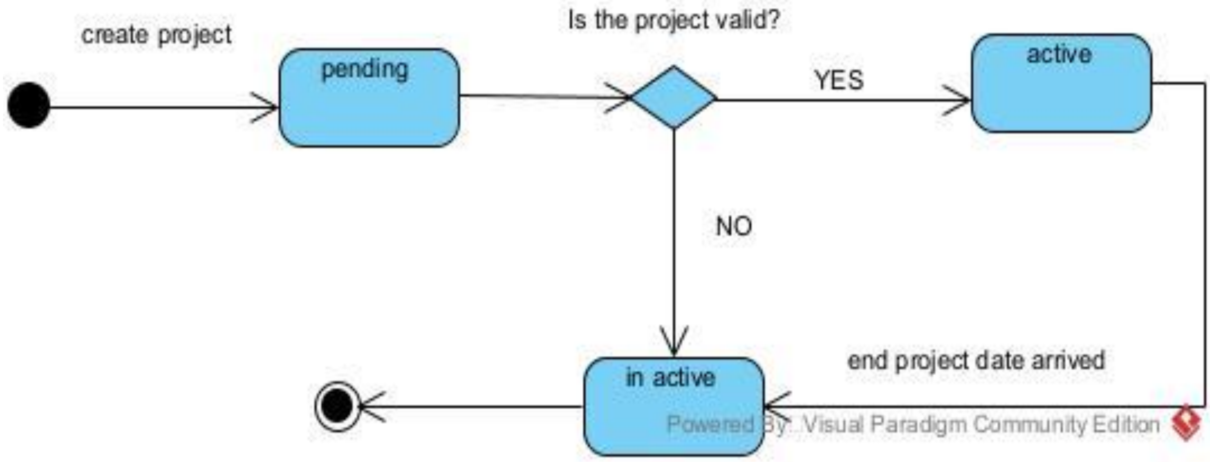



*Figure 43 State machine of justgive*

# IV. PACKAGE DIAGRAM

## 1. DEFNITION

They are structural diagrams used to show the organization and arrangement of various model elements in the form of a package. A package diagram is group of related UML elements such as diagrams, documents, classes, or even other packages. Package diagrams are commonly used to provide a visual organization of the architectural layout within any UML classifier such as software systems.

## 2. FORMALISM

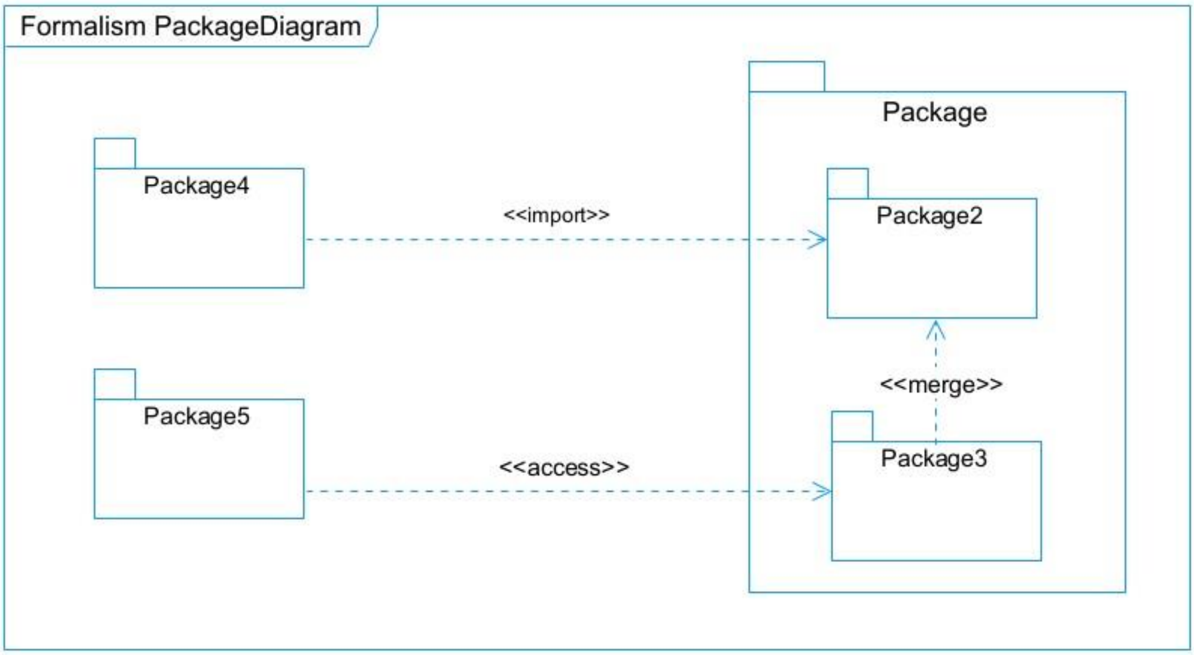
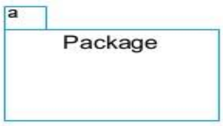
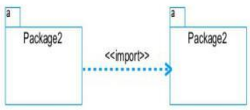
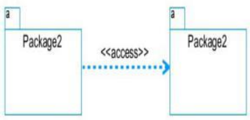
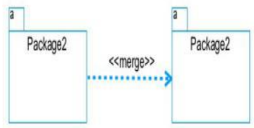



*Figure 44 Formalism of package diagram*

## 3. COMPONENTS OF THE PACKAGE DIAGRAM

*Tableau 25 Components of the package diagram*

| Element | Description and main features | Representation |
|---|---|---|
| **Package** | It is namespace used to group together logically related elements within a system. Each element contains within the package should be a packageable element and have a unique name |  |
| **Package import** | A relationship Indicate that, functionality has been imported from one package to another. |  |
| **Package access** | A relationship Indicates that one package requires assistance from the function of another package. |  |

| | | |
|---|---|---|
| **Package merge** | It is a relationship which shows that, the functionality of two packages are combines to a single function. |   |

Package diagram of Justgive

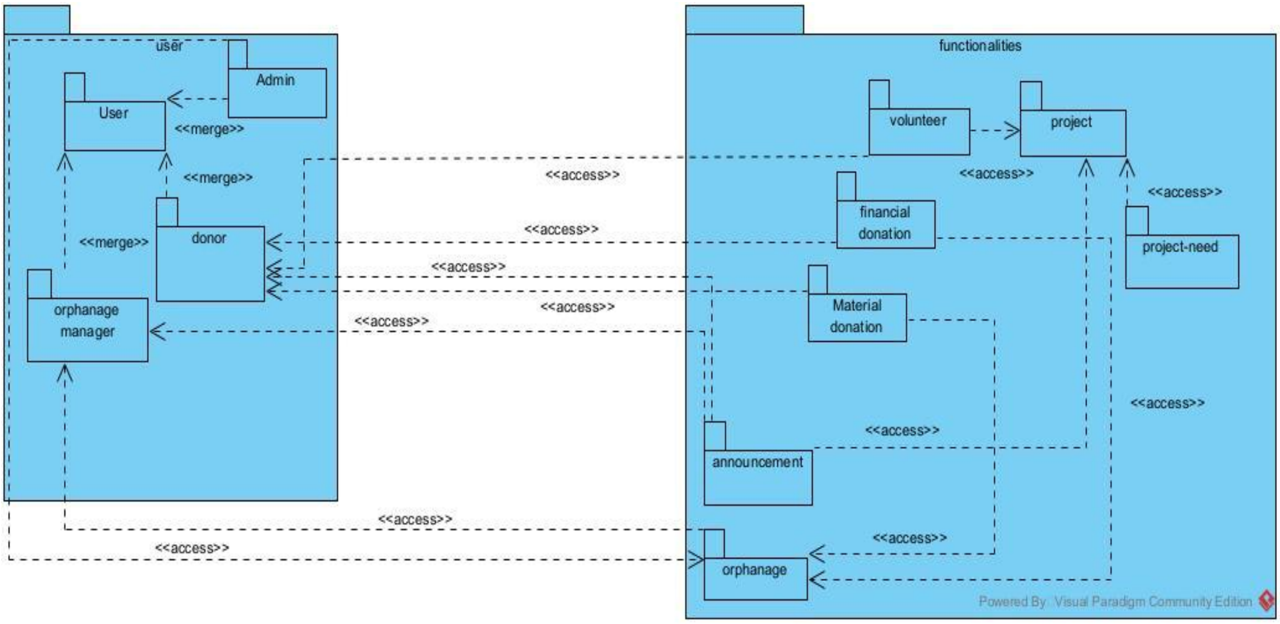



*Figure 45 Package diagram of Jusstgive*

# CONCLUSION

The conception phase allowed us to bring out the data necessary for the creation of our database and the implementation of our application. The different elements modeled in this part allowed us to have a global view of the different modules of our application; therefore, the next step of our project will be the drafting of the realization file taking into account the different elements modeled above.

# DOCUMENT 6:REALIZATION

Preamble

The realization document consists of the implementation of the project in a programming language according to the specifications defined in the previous file. It contains component and deployment diagrams; the testing and development phases. At the end of this part, programming documentation will be produced explaining the architecture of the database and the architecture of our code.

Content :

# INTRODUCTION

Design and modeling are important links in the development process of an application because they make it possible to define and specify the different elements constituting the application or system studied. Realization is the culmination of the work carried out upstream because it makes it possible to physically create the various objects that were designed, and to bring the computer system to life. Throughout this phase, we will present the software, physical, and hardware architecture of our system, through diagrams such as the component diagram, deployment diagram, physical model, and code architecture.

# I. DEPLOYMENT DIAGRAM

## 1 DEFINITION

The deployment diagram models the hardware components used to implement a system and the association between those components. Deployment diagrams can be implemented as early as the design phase to document the system's physical architecture. As elements used in deployment diagrams, we can have, components, as in component diagrams, and nodes, which represent the physical processing resources of the system, and their associations.

## 2 FORMALISM

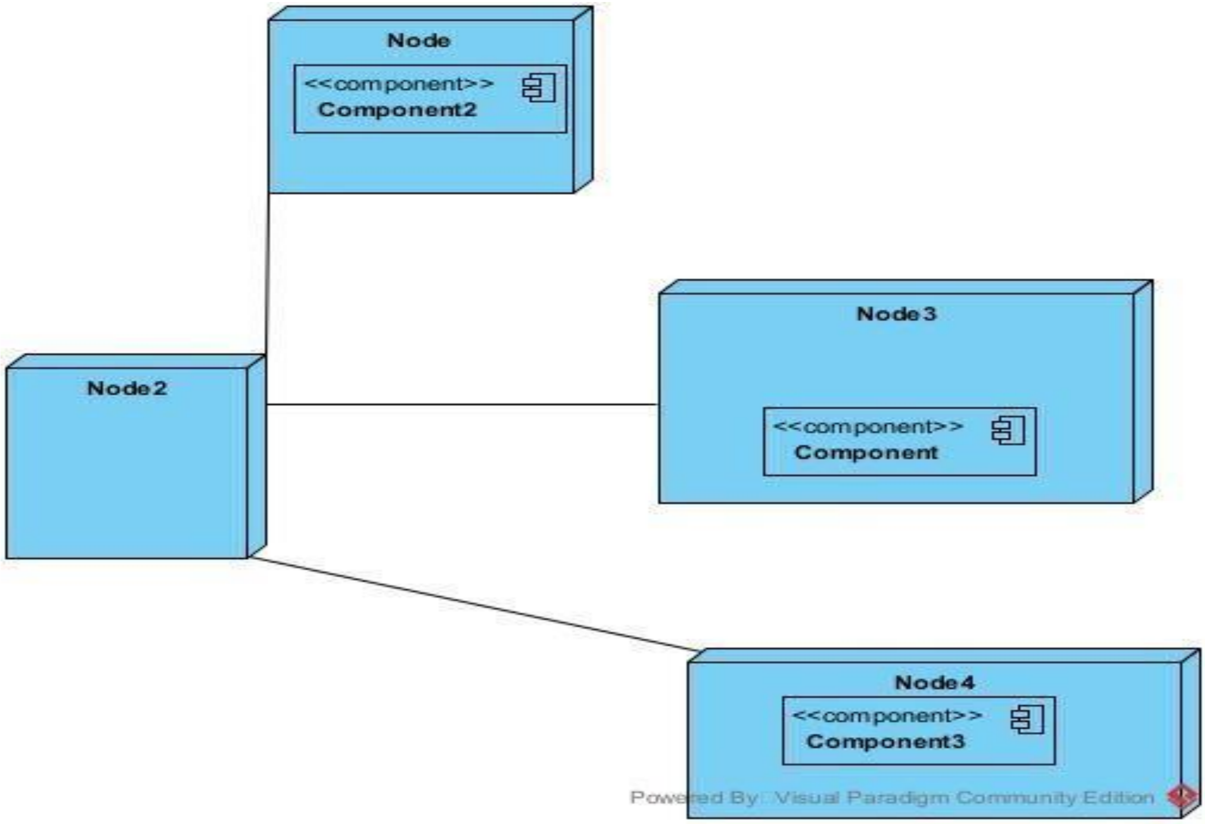
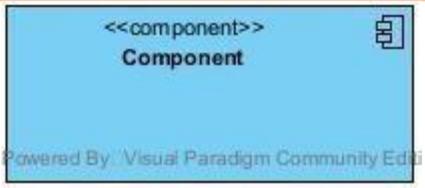
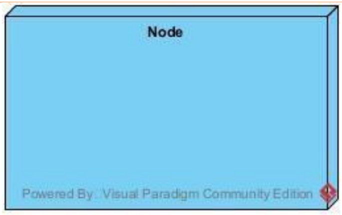
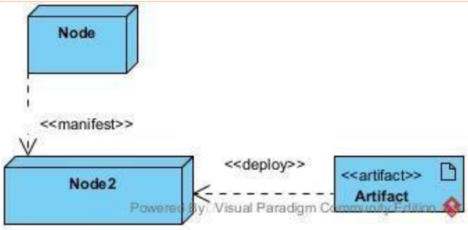



*Figure 46 Formalism of deployment diagram*

## 3 COMPONENTS OF THE DEPLOYMENT DIAGRAM

*Tableau 26 Components of the deployment diagram*

| Element | Description and main | Representation features |
|---|---|---|

| | | |
|---|---|---|
| **Component** | A component represents a software entity of the system. |  |
| **Node** | A node represents a set of hardware elements of the system. A three-dimensional cube represents this entity or node. |  . |
| **Interface** | Represents a declaration of a set of coherent public features and obligations |  |
| **Dependency (manifestation deployment)** | A relationship between two model elements where changes in one of them will similarly affect the other |  |

DEPLOYMENT DIAGRAM OF JUSTGIVE

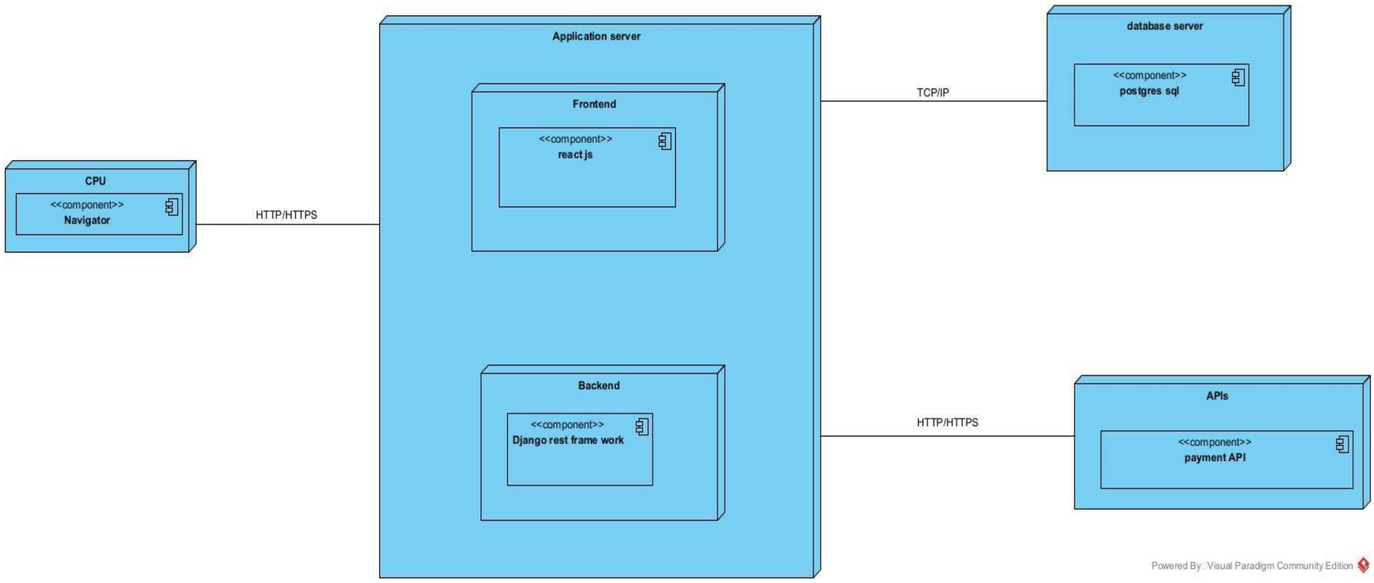



*Figure 47 Deployment diagramof justgive*

# II. COMPONENT DIAGRAM

## 1 DEFINITION

The component diagram is a UML diagram that provides a graphical representation of dependencies and generalizations between software components, including source code components, binary code components, and executable components.

## 2 FORMALISM

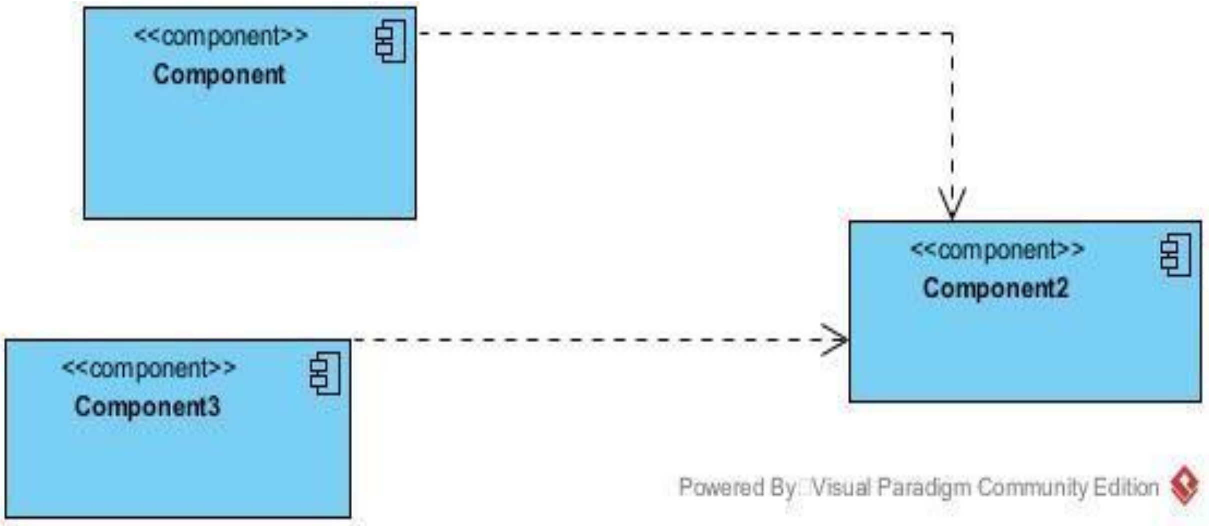
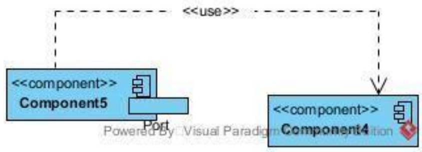



*Figure 48 formalism of component diagram*

## 3 COMPONENTS OF THE COMPONENT DIAGRAM

*Tableau 27 Components of the component diagram*

| Element | Description and main featu | Representation |
|---|---|---|
| **Component** | Represents the modular part the system that encapsulates content. In other words, it is part of a system or applicatio |   |
| **Interface** | Represents a declaration of a set of coherent public feature and obligations |   |
| **Port** | Specifies a distinct interactior point between that classifier element and environment |   |

| | | |
|---|---|---|
| **Usage** | A relationship in which one element requires another element for its full implementation or operation |   |
| **Instance Specificatio** | It has the capability of being a deployment target in a deployment relationship |   |

Component diagram of JustGive

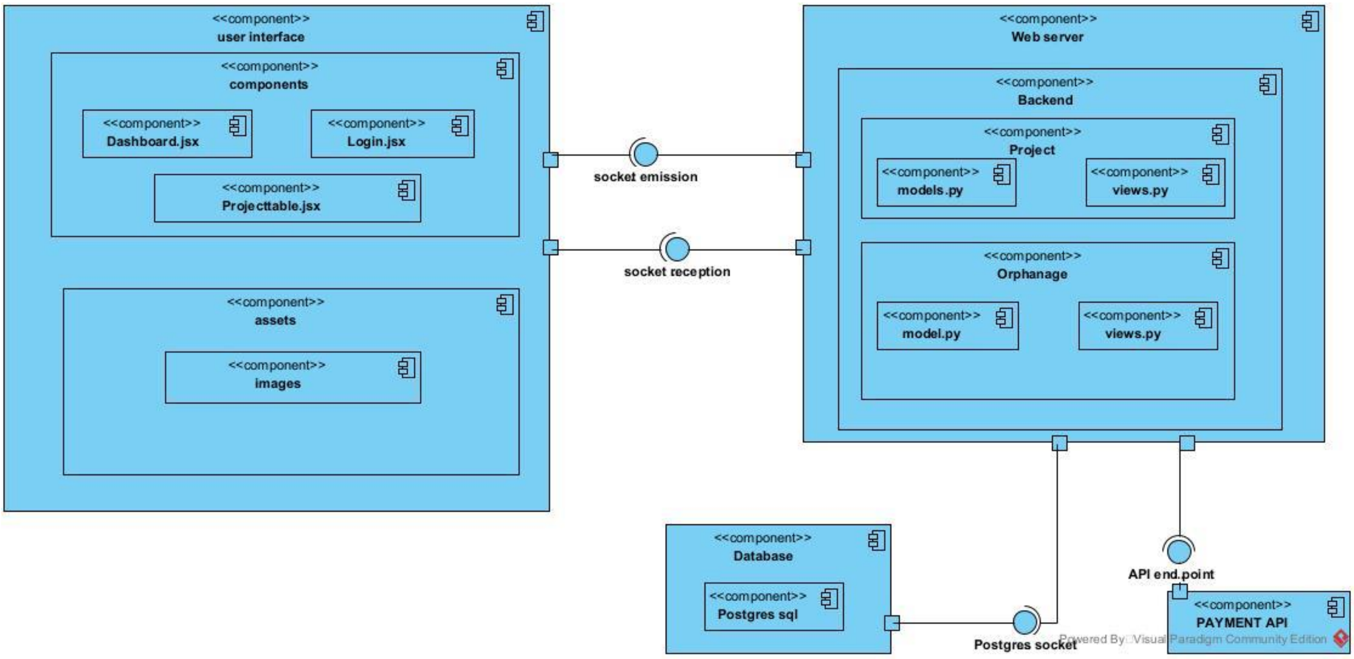



*Figure 49 Justgive component diagram*

# III. TECHNICAL CHOICES

## SOFTWARE TOOLS USED FOR THE DEVELOPMENT OF JUST GIVE

*Tableau 28 Tools used to develop Justgive*

| Software | Role | Reason for use | Image |
|---|---|---|---|
| **OS WINDOWS 10** | The operating system required to run the software is the one on which we have worked. | Windows 10 has a user-friendly interface that is familiar to many users, making it easy to navigate and use. | |
| **Visual Paradigm** | The software engineering workshop was used for solution modeling. We utilized this tool for modeling the various diagrams of our system | . Supports multiple modeling languages and methodologies, making it suitable for various project requirements and enabling teams to adapt to different development processes. | |
| **Visual studio code** | Visual Studio Code is a lightweight, open-source code editor developed by Microsoft, supporting multiple programming languages. Using for the writing of application code. | . Combines tools for version control, debugging, and terminal access in one interface, improving efficiency and reducing the need for multiple applications. | |
| **Github** | It is a website and cloud service that helps developers | Git allows developers to track changes, revert to previous versions, and maintain a history | |

| | | | |
|---|---|---|---|
| | store and manage their code. | of modifications, ensuring that no work is lost and enabling easy recovery. | |
| **Framework react** | A popular JavaScript library for building UI. It provides a component-based approach and allows for efficient rendering and state management. Used for building the front | Promotes the creation of reusable components, which enhances code maintainability and allows for easier updates and testing. | |
| **DJango** | Django is a popular high-level Python web framework. | Django enables a rapid development,scalability,security, and has a large community . | |
| **MS Word** | MS Word is a widely used word-processing application that allows users to create, edit, and format text documents. Using in-application report creation. | Offers extensive options for formatting text, images, and layouts, enabling the creation of professional quality documents. | |

| | | | |
|---|---|---|---|
| **MS PowerPoint** | MS PowerPoint is a presentation software that allows users to create dynamic slideshows comprising text, images, videos, and animations. Able us to create slideshow presentations for application. | Its intuitive interface and templates make it easy for users to create professional presentations quickly, regardless of their technical skill level. | |
| **Chrome** | Google Chrome is a fast, secure, and user-friendly web browser developed by Google. Used to test our application UI | Chrome is optimized for fast page loading and efficient resource management, providing a smooth browsing experience. | |
| **GanttProject** | GanttProject is a free, open-source project management tool that allows users to create Gantt charts for planning and scheduling projects. | Allows users to assign tasks, set deadlines, and manage resources effectively, facilitating better project coordination and efficiency. | |

# CONCLUSION

In this phase, the presentation of the application was the subject. It began with the presentation of the various architectures, which has permitted us to; present in a general manner the components attached to the DBMS and its deployment.

# DOCUMENT 7:FUNCIONALITY TESTING

Preamble

This phase aims at testing the software or application to verify if the system is working as expected. It involves testing individual functionalities and capabilities of the software to ensure that they work correctly and meet the specified needs.

Content :

# INTRODUCTION

It is a systematic process that involves evaluating the functionality of a software application or system in different phases or stages of its development lifecycle. This testing approach ensures that each phase of the software development process is thoroughly tested for its intended functionalities and overall performance. Its main goal is to identify and rectify any defects, and bugs during the process.

# I. APPLICATION MODULES

i. User registration and login

Secure signup and login with authentication.

ii. Orphanage

Creation

Orphanage creation is made only by the orphanage managers

View

All projects can be viewed on the project page. To see its details, the user has to select an orphanage in particular.

Detail view:

View orphanage details

iii. Projects:

Creation

Projects can be created by both the orphanage managers and the donors

View

All projects can be viewed on the project page.To see its details, the user has to select a project in particular.

Detail view:

View project details

iv. Donation

It is possible to make a financial and material donation to either a project or an orphanage.

v. Volunteer

It is possible to volunteer for a project

vi. Announcement

The project creator can create an announcement that will be visible to the project volunteers

vii. Validate material donation

The orphanage manager can confirm that he received a material donation

viii. Validate project

The orphanage manager can validate the proposed project

ix. Orphanage validation

The orphanage must be validated by the System administrator

x. Account management

The system administrator must be able to delete an account and modify some user information

xi. User dashboard

On the user dashboard he can view his profile, and the projects he created, and if he is an orphanage manager, can view the proposed projects.

# II. TESTING

User registration and login

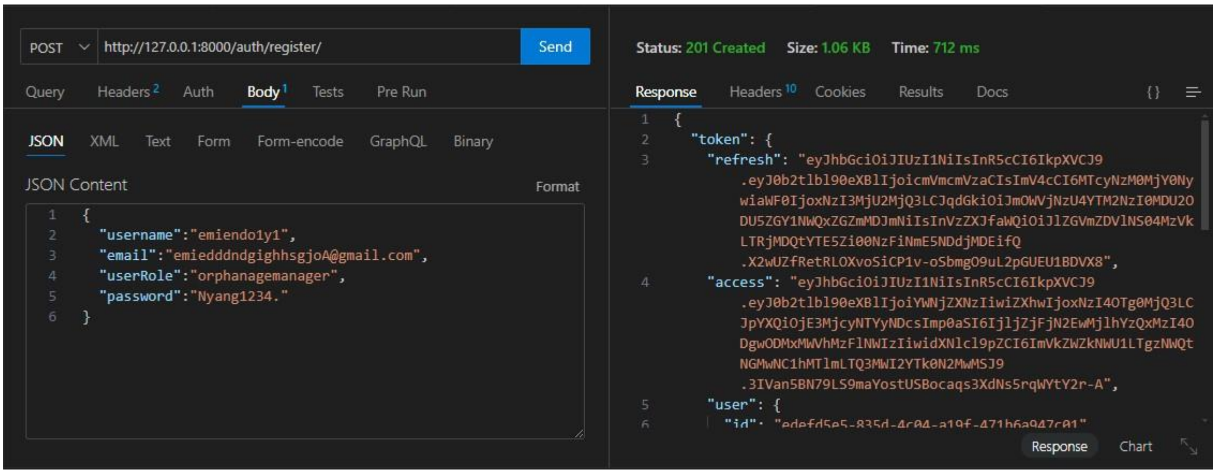



*Figure 50 User registration test*

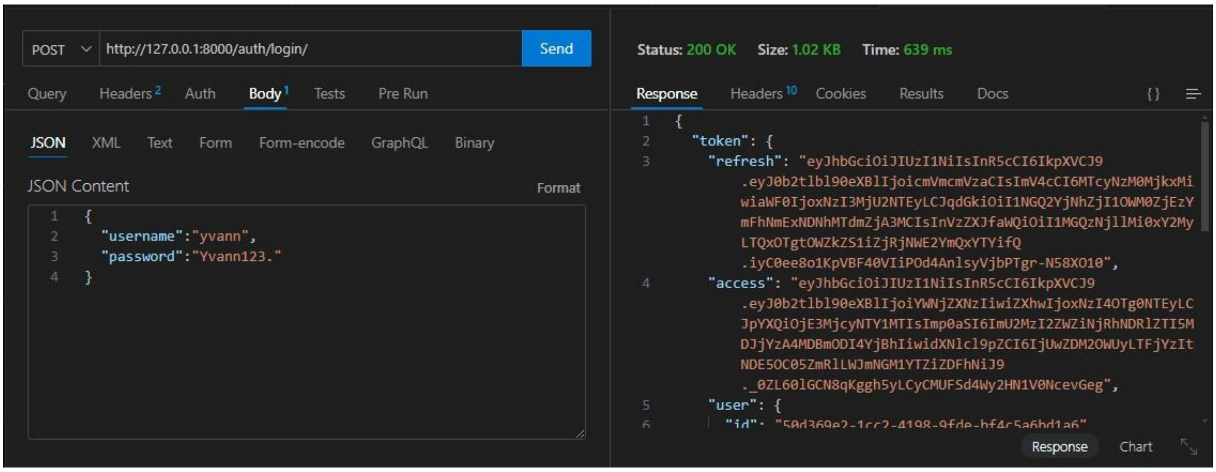



*Figure 51 Login test*

Project creation

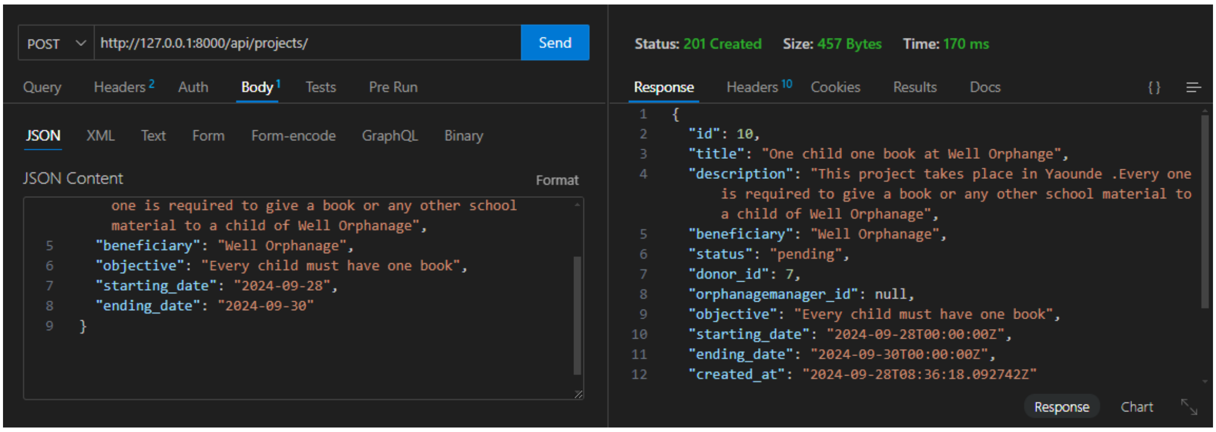



*Figure 52 Project creation test*

Announcement creation unit testing

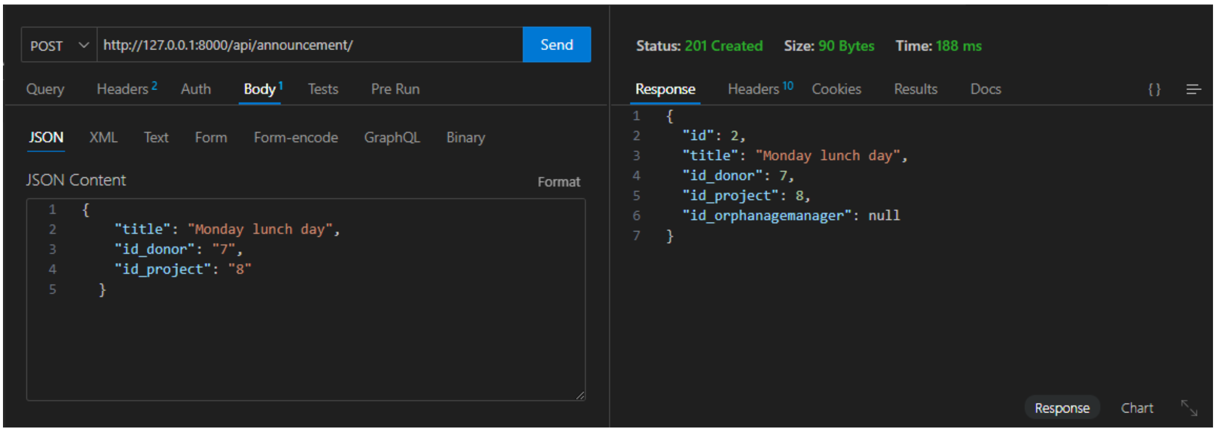



*Figure 53 Announcemt creation test*

Volunteer

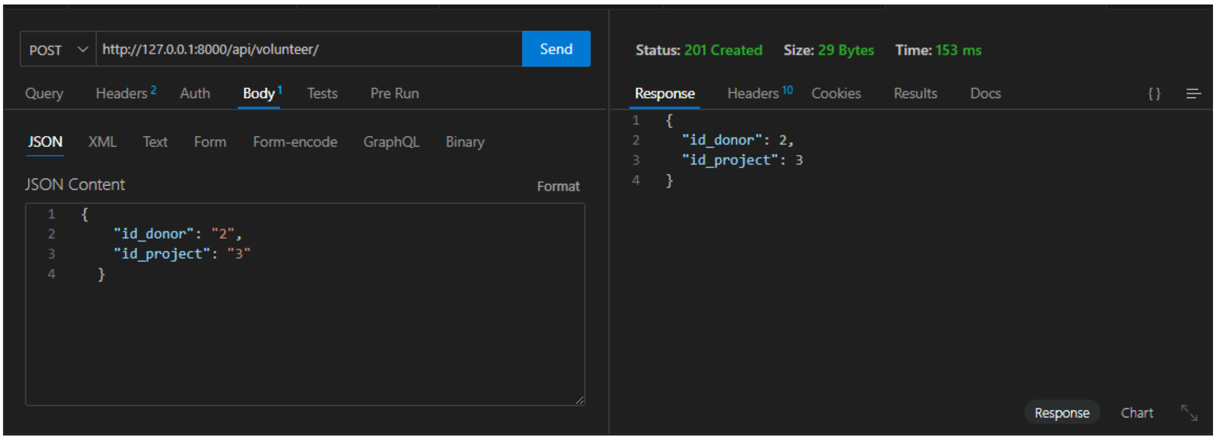



*Figure 54 Volunteer test*

Make material donation unit testing.

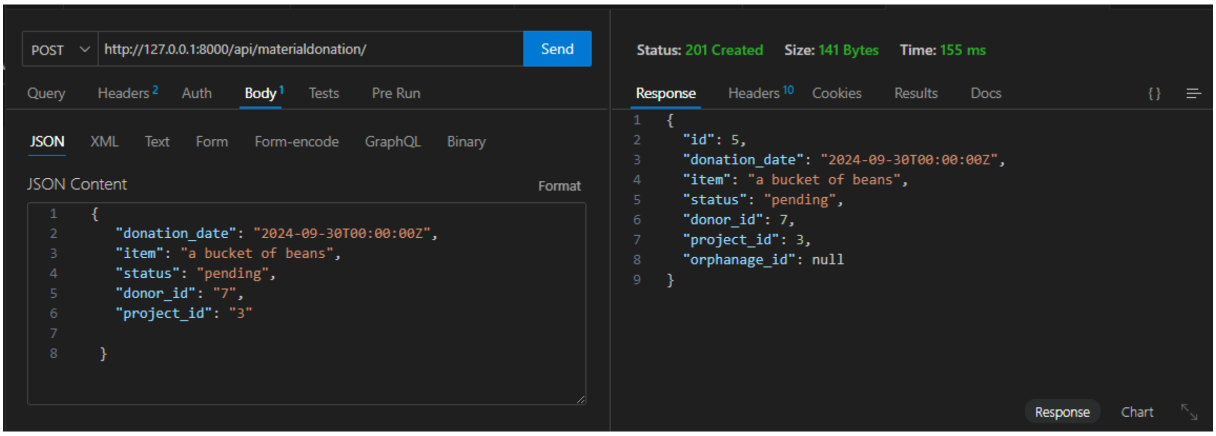



*Figure 55 Apply for material donation test.*

# CONCLUSION

Testing is an integral part of the software development life cycle and is performed to verify the correctness, reliability, and robustness of the implemented functionality. The functionality and testing phase is therefore a critical stage in software development where the implemented features are validated, and any issues are identified and resolved.

# DOCUMENT 8: USER GUIDE

Preamble

A user guide explains how to use a software application in a language that a non-
a technical person can understand. Thus, it enables the user to easily use the application to familiarize themselves with the software and discover all its functionalities.

Content :

# INTRODUCTION

The user manual is the document established after the realization of an application, software or platform. It informs on the question "how to use the software, the application or the platform that you have in front of you? ". It is therefore crucial for us to offer the various users of our platform a guide allowing them to easily perform various operations on our platform.

# I. DESCRIPTION OF THE APPLICATION

JustGive is a powerful web platform dedicated to enabling donations and volunteer work in orphanages. It empowers users to create and support projects for registered orphanages, fostering a seamless and impactful avenue for contributing to a worthy cause. JustGive stands as a transformative force for orphanages in Cameroon, embodying the ethos of "Give today, save tomorrow."

# II. USER GUIDE

In this section, I will explain the steps for carrying out the operations in detail. I will accompany the explanations with tests and screenshots to help users understand how to perform the tasks and provide them with an overview of the expected results.

Connecting to the platform is not complicated at all. However, it is essential to meet a

a number of prerequisites, namely:

Have a terminal (computer, tablet, or smartphone);

Have node installed as well as python

Have Access to the Internet

Have installed a web browser (preferably the most recent version) in your terminal:

Go to the project's directory.

Type "code .'' to open that directory in vs code.

Install the dependencies with 'npm install'

To launch the server run 'npm run dev'

Click on the link.

For the backend, you will need to install what's found in your requirements.txt file using the command.

Lunch your virtual environment with '[the name of your virtual environment]\Script\activate'

Lunch the server with 'python manage.py runserver'

Here is the result you will see:

Landing page:

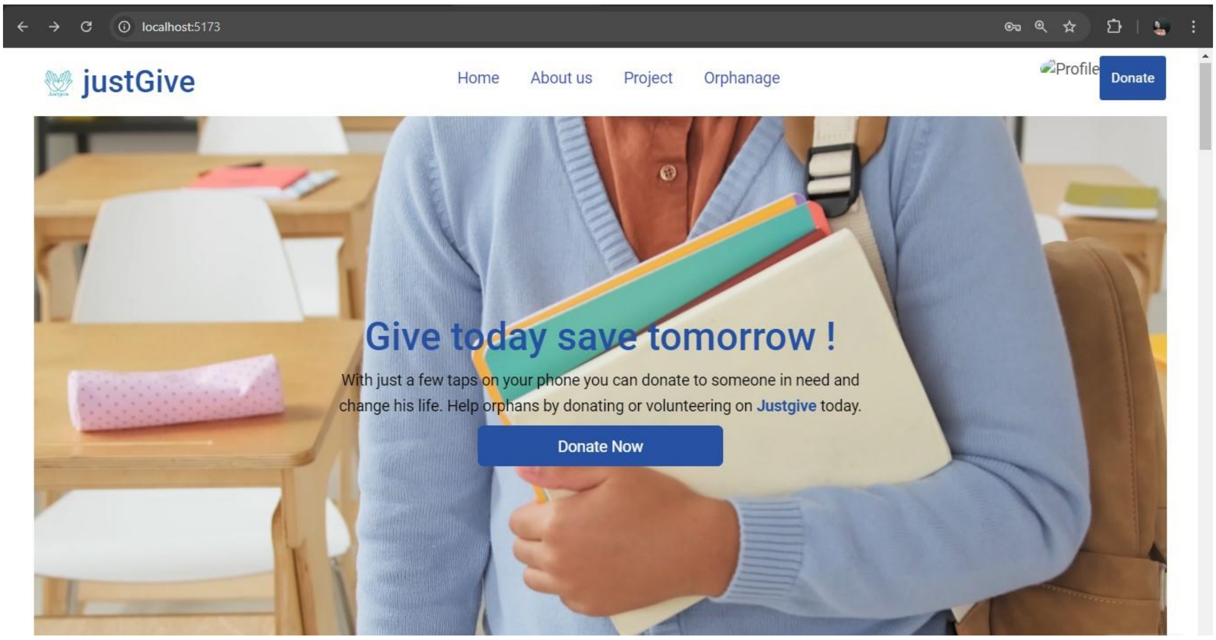



*Figure 56 Justgive landing page*

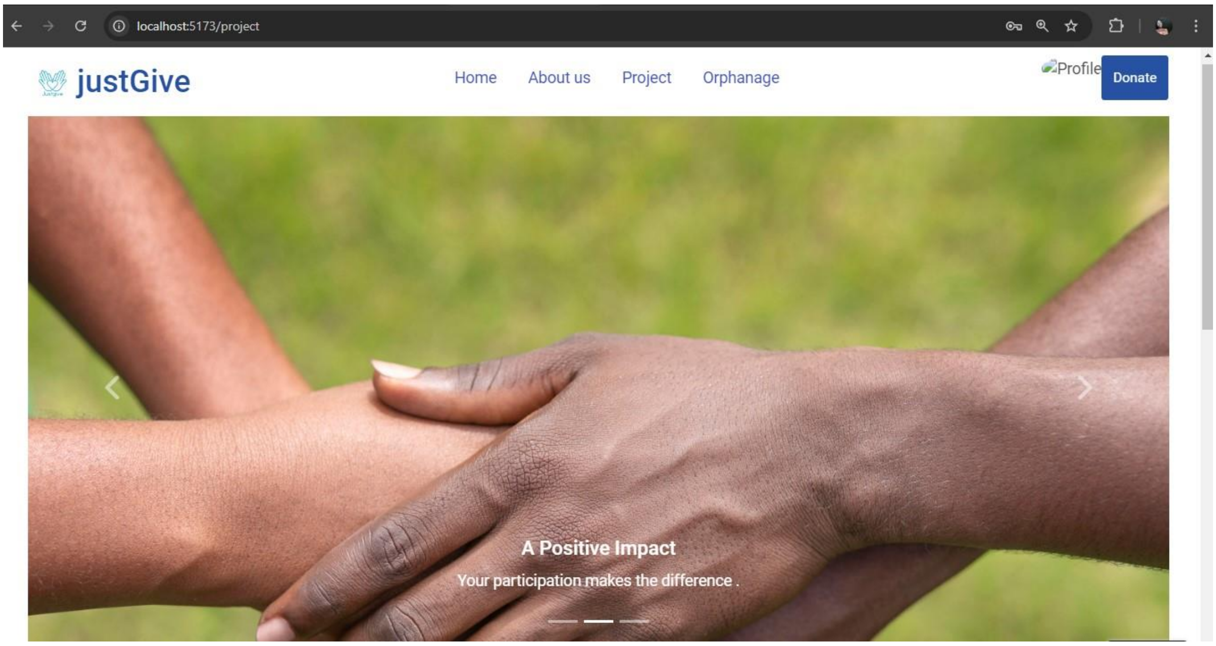



*Figure 57 Project page*

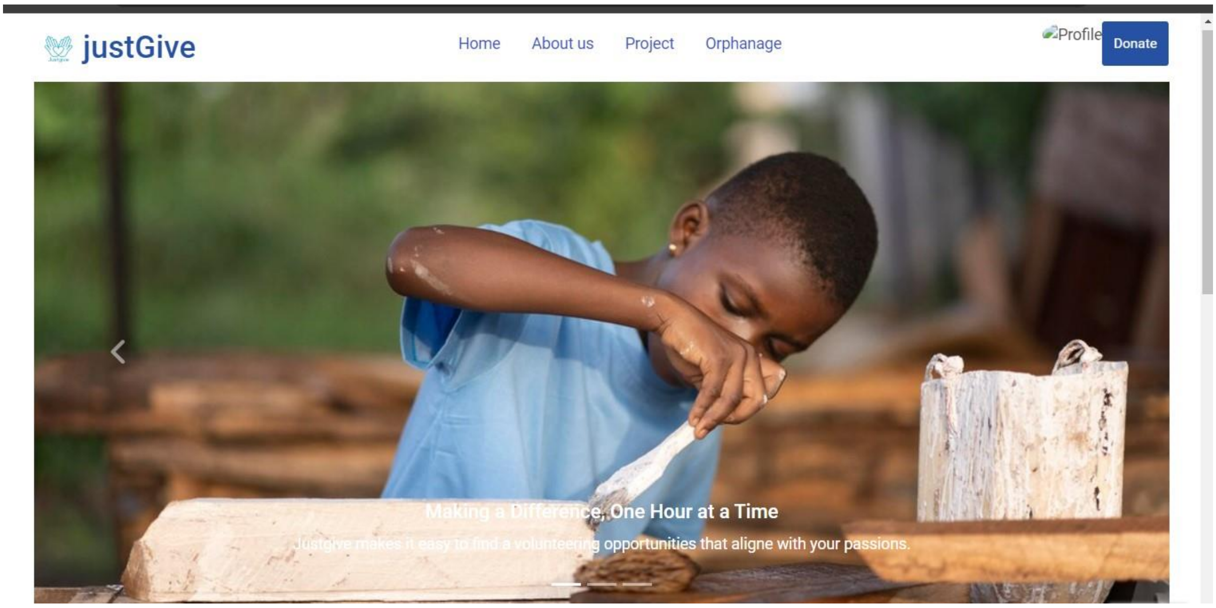



Figure 58 Orphanage page

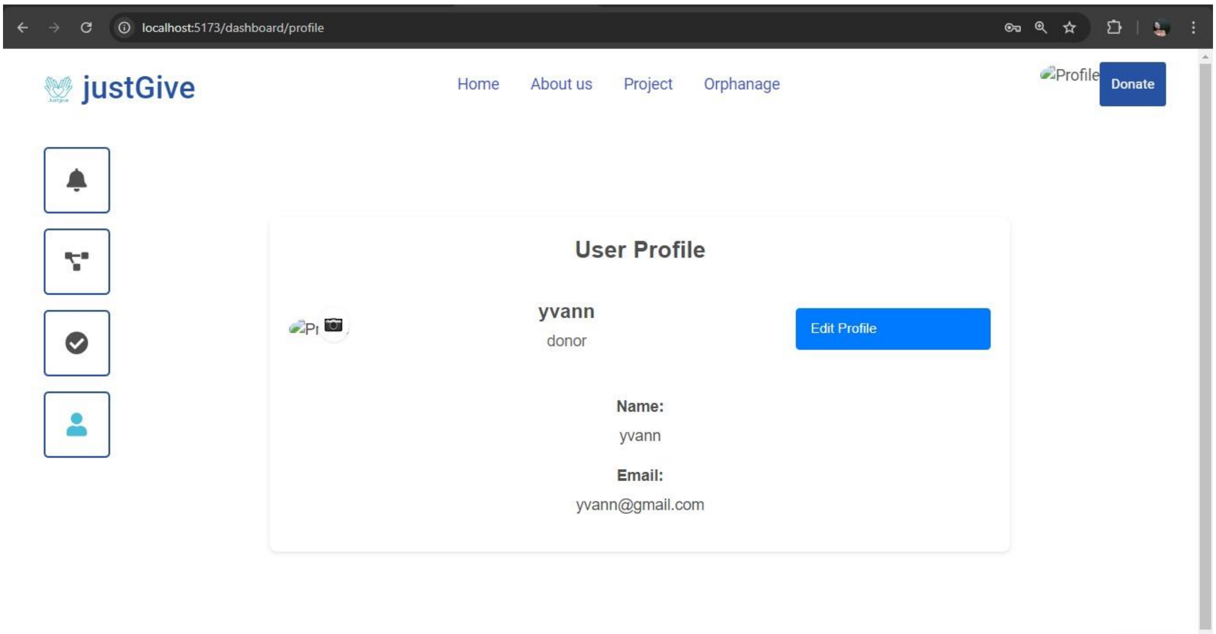



Figure 59 User dashboard

# CONCLUSION

This user guide has provided a comprehensive overview of our product's features, functionalities, and best practices for optimal use. By following the instructions and tips outlined in this document, you'll be well-equipped to maximize the potential of our product and streamline your workflow..

# GENERAL CONCLUSION

Having come to an end of our project and the end of our internship at WELLDONE PLANET, we have achieved a lot from our internship period, it was very challenging and demanding, we had to manage stress, failures and successes. Our stay in WELLDONE PLANET was very profitable; we learned how to adapt to the professional milieu and to work in team. For our project, we worked on the theme CONCEPTION AND IMPLIMENTATION OF A WEB APPLICATION FOR DONATION AND VOLUNTEERING FOR ORPHANAGES IN CAMEROON. Our greatest desire is to give a better life to those children, to whom life took one of the greatest treasure, their parents; by facilitating the donation procedure to orphanages and enabling an online volunteering system. We began by identifying the current system in place for donation and volunteering in orphanages of Cameroon, project constraints and requirements in the specification book, and then we went forth to analyze our system (Justgive) using UML-2TUP methodology. We made use of Visual Paradigm, a modelling software used to draw our various diagrams. In order to realize our project, we used React JS a java script library for our front-end, Django rest framework for our back end and Postgres sql for our database. Our system, Justgive will enable the donors to donate to orphanages (both material and financial donations) equally ,create projects to enable volunteering in these orphanages. In addition to that ,they will be able to find orphanages with respect to their locations and know the needs of that orphanage before making a donation. We won’t end here, updates and important improvements are going to be made in the nearest future in terms of functionality, security to make it more reliable.

# ANNEXES

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- https://www.youtube.com/watch?v=_uQrJ0TkZlc&list=PLTjRvDozrdlxj5wgH4qkvwSOdHLOCx10f Python tutorial August 15,2024 ,3pm
- https://www.youtube.com/watch?v=cJveiktaOSQ Django rest frame work tutorial on August 17,2024 3pm

# TABLE OF CONTENT